

Oracle® Real Application Clusters Installation Guide



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Preface

This guide explains how to install and configure Oracle Real Application Clusters (Oracle RAC).

Before you use this guide, you must first complete an installation of Oracle Clusterware, as described in the *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*.

- [Intended Audience](#)
- [Documentation Accessibility](#)
- [Diversity and Inclusion](#)
- [Set Up Java Access Bridge to Implement Java Accessibility](#)
Install Java Access Bridge so that assistive technologies on Microsoft Windows systems can use the Java Accessibility API.
- [Related Documents](#)
- [Conventions](#)

Intended Audience

Oracle Real Application Clusters Installation Guide for Microsoft Windows provides database installation information for database administrators (DBAs) who install and configure Oracle RAC.

Documentation Accessibility

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc>.

Access to Oracle Support

Oracle customers that have purchased support have access to electronic support through My Oracle Support. For information, visit <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=info> or visit <http://www.oracle.com/pls/topic/lookup?ctx=acc&id=trs> if you are hearing impaired.

Diversity and Inclusion

Oracle is fully committed to diversity and inclusion. Oracle respects and values having a diverse workforce that increases thought leadership and innovation. As part of our initiative to build a more inclusive culture that positively impacts our employees, customers, and partners, we are working to remove insensitive terms from our products and documentation. We are also mindful of the necessity to maintain compatibility with our customers' existing technologies and the need to ensure continuity of service as Oracle's offerings and industry standards evolve.

Because of these technical constraints, our effort to remove insensitive terms is ongoing and will take time and external cooperation.

Set Up Java Access Bridge to Implement Java Accessibility

Install Java Access Bridge so that assistive technologies on Microsoft Windows systems can use the Java Accessibility API.

Java Access Bridge is a technology that enables Java applications and applets that implement the Java Accessibility API to be visible to assistive technologies on Microsoft Windows systems.

Refer to *Java Platform, Standard Edition Accessibility Guide* for information about the minimum supported versions of assistive technologies required to use Java Access Bridge. Also refer to this guide to obtain installation and testing instructions, and instructions for how to use Java Access Bridge.

Related Topics

- *Java Platform, Standard Edition Java Accessibility Guide*

Related Documents

The related documentation for Oracle Database products includes the following manuals:

Installation Guides

- *Oracle Grid Infrastructure Installation Guide* for your platform
- *Oracle Database Installation Guide* for your platform
- *Oracle Database Client Installation Guide for Microsoft Windows*
- *Oracle Database Examples Installation Guide*

Operating System-Specific Administrative Guides

- *Oracle Database Administrator's Reference for Linux and UNIX-Based Operating Systems*
- *Oracle Database Platform Guide for Microsoft Windows*

Oracle Real Application Clusters Management

- *Oracle Clusterware Administration and Deployment Guide*
- *Oracle Real Application Clusters Administration and Deployment Guide*
- *Oracle Database 2 Day DBA*

Generic Documentation

- *Oracle Database New Features Guide*
- *Oracle Database Concepts*
- *Oracle Database Net Services Administrator's Guide*
- *Oracle Database Reference*
- *Oracle Database Sample Schemas* describes the sample schemas provided for Oracle Database. Many of the examples in the Oracle Database Documentation Library use these schemas.

Error Messages

Oracle error message documentation is available only in HTML. You can browse the error messages by range in *Oracle Database Error Messages Reference*. When you find a range, use your browser's "find in page" feature to locate a specific message. When connected to the Internet, you can search for a specific error message using the error message search feature of the Oracle online documentation.

Other Documentation

For any other documentation, you can access the documentation library at Oracle Help Center:

<http://docs.oracle.com>

Conventions

The following text conventions are used in this document:

Convention	Meaning
boldface	Boldface type indicates graphical user interface elements associated with an action, or terms defined in text or the glossary.
<i>italic</i>	Italic type indicates book titles, emphasis, or placeholder variables for which you supply particular values.
monospace	Monospace type indicates commands within a paragraph, URLs, code in examples, text that appears on the screen, or text that you enter.

Changes In This Release for Oracle Real Application Clusters

This preface lists changes in Oracle Real Application Clusters Installation Guide

- [Changes in Oracle Database 19c](#)
The following are changes in Oracle Real Application Clusters Installation Guide for Oracle Database 19c.

Changes in Oracle Database 19c

The following are changes in Oracle Real Application Clusters Installation Guide for Oracle Database 19c.

- [Deprecated Features for Oracle RAC 19c](#)
- [Desupported Features for Oracle RAC 19c](#)

Deprecated Features for Oracle RAC 19c

The following feature is deprecated in this release:

- **Deprecation of the `SERVICE_NAMES` Initialization Parameter**

Starting with Oracle Database 19c, customer use of the `SERVICE_NAMES` parameter is deprecated. It can be desupported in a future release.

The use of the `SERVICE_NAMES` parameter is no longer actively supported. It must not be used for high availability (HA) deployments. It is not supported to use service names parameter for any HA operations. This restriction includes FAN, load balancing, `FAILOVER_TYPE`, `FAILOVER_RESTORE`, `SESSION_STATE_CONSISTENCY`, and any other uses.

To manage your services, Oracle recommends that you use the `SRVCTL` or `GDSCTL` command line utilities, or the `DBMS_SERVICE` package.

Related Topics

- *Oracle Database Upgrade Guide*

Desupported Features for Oracle RAC 19c

The following feature is desupported in this release:

- **Desupport of Leaf Nodes in Flex Cluster Architecture**

Leaf nodes are no longer supported in the Oracle Flex Cluster Architecture in Oracle Grid Infrastructure 19c.

In Oracle Grid Infrastructure 19c (19.1) and later releases, all nodes in an Oracle Flex Cluster function as hub nodes. The capabilities offered by Leaf nodes in the original

implementation of the Oracle Flex Cluster architecture can as easily be served by hub nodes. Therefore, leaf nodes are no longer supported.

- **Desupport of Oracle Real Application Clusters for Standard Edition 2 (SE2) Database Edition**

Starting with Oracle Database 19c, Oracle Real Application Clusters (Oracle RAC) is not supported in Oracle Database Standard Edition 2 (SE2).

Upgrading Oracle Database Standard Edition databases that use Oracle Real Application Clusters (Oracle RAC) functionality from earlier releases to Oracle Database 19c is not possible. To upgrade those databases to Oracle Database 19c, either remove the Oracle RAC functionality before starting the upgrade, or upgrade from Oracle Database Standard Edition to Oracle Database Enterprise Edition. For more information about each step, including how to reconfigure your system after an upgrade, see My Oracle Support Note 2504078.1: "Desupport of Oracle Real Application Clusters (RAC) with Oracle Database Standard Edition 19c."

Related Topics

- *Oracle Database Upgrade Guide*

Related Topics

- [My Oracle Support Note 2504078.1](#)

1

Oracle RAC Installation Checklist

Review these checklists for installing Oracle Real Application Clusters (Oracle RAC).

- [Deployment Checklist for Oracle RAC Database](#)
Use the checklist to review the deployment methods for Oracle Real Application Clusters.
- [Server Hardware and Software Review Checklist for Oracle RAC Installation](#)
Use the checklist to check minimum hardware and software requirements for Oracle RAC.
- [Supported Storage Options for Oracle Database and Recovery Files](#)
The following table shows the storage options supported for Oracle Database and recovery files:
- [Installer Planning Checklist for Oracle Database Installation](#)
Use the checklist to assist you to be prepared before starting Oracle Universal Installer.
- [Upgrade Checklist for Oracle RAC](#)
Review the checklist for additional requirements related to upgrading an existing Oracle Real Application Clusters (Oracle RAC) installation to Oracle Database 19c.

Deployment Checklist for Oracle RAC Database

Use the checklist to review the deployment methods for Oracle Real Application Clusters.

Table 1-1 Deployment Checklist for Oracle RAC Database

Item	Task
To Deploy Oracle RAC software	Install Oracle RAC Database software using Oracle Universal Installer (OUI).
To Deploy Oracle Database software and create Oracle RAC databases	Install Oracle RAC Database software using Oracle Universal Installer (OUI) and choose to create a database.
To Create Oracle RAC database in an already-installed Oracle home.	Use Oracle Database Configuration Assistant (DBCA). See Creating Oracle RAC or Oracle RAC One Node Databases with Oracle DBCA for more information about creating database using DBCA.
Complete the installation by completing the post-installation tasks	See Oracle Real Application Clusters Postinstallation Procedures .

Server Hardware and Software Review Checklist for Oracle RAC Installation

Use the checklist to check minimum hardware and software requirements for Oracle RAC.

It is assumed that the servers were prepared as described in the Oracle Grid Infrastructure Installation Guide for your platform.

Table 1-2 Hardware and Software Checklist for Oracle RAC

Item	Task
Server Hardware on each node	• Use identical server hardware on each node, to simplify server maintenance. • Avoiding resource contention issues by not installing Oracle RAC on a primary domain controller or backup domain controller. • Review "Checking the Hardware Requirements" in <i>Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows</i> to ensure that your system has enough RAM. • Verify the <code>TEMP</code> environment variable points to a location that has enough available space for the installation. • For both the Enterprise and Standard Editions of Oracle RAC, the hard disk requirements for Oracle Database components include 1 GB required to install Java Runtime Environment (JRE) and Oracle Universal Installer (OUI) on the disk partition where the operating system is installed. If sufficient space is not detected, then the installation fails and an error message appears.

Table 1-2 (Cont.) Hardware and Software Checklist for Oracle RAC

Item	Task
Operating System General Requirements	• Install the supported operating system version and install the operating system packages and patches required for that version. – Windows Server 2025 x64 - Standard, Datacenter, and Essentials editions – Windows Server 2022 x64 - Standard, Datacenter, and Essentials editions (Supported starting Oracle Database 19c Release Update 19.13 or later) – Windows Server 2019 x64 - Standard, Datacenter, and Essentials editions – Windows Server 2016 x64 - Standard, Datacenter, and Essentials editions – Windows Server 2012 R2 x64 - Standard, Datacenter, Essentials, and Foundation editions For some operating systems, Oracle may require updates, such as service packs and individual patches. If such requirements exist, then they are stated in the Release Notes for a particular release. You can also apply other operating system patches as recommended by Microsoft, if there are no "certification exceptions" listed in the Release Notes. Refer to your operating system vendor for required operating system updates.  Note: You must use the same operating system on each node in the cluster. Oracle strongly recommends that you use the same software configurations on each node of the cluster. Oracle Clusterware and Oracle RAC do not support heterogeneous platforms (each server must run the same Oracle software binaries) in the same cluster. • (Optional) Stage all of the software on one node for installation (the "local node").
Virtualization	Oracle certifies the following virtualization technologies with Oracle Database in both Single Instance and RAC modes on Windows: • Oracle VM Server • Microsoft Hyper-V For more detailed information on certified Oracle VM Server combinations, check My Oracle Support note 464754.1. For more information on certified Hyper-V combinations, visit: https://www.oracle.com/database/technologies/virtualization-matrix.html

Table 1-2 (Cont.) Hardware and Software Checklist for Oracle RAC

Item	Task
Create the required users and configure the environments	- Review the users created during the Oracle Grid Infrastructure installation. - To install the Oracle Real Application Clusters software, you must use either a local or domain user. In either case, the Oracle Installation user must be an explicit member of the Administrators group on all nodes of the cluster. - To install Oracle RAC software on Oracle Automatic Storage Management Cluster File System (Oracle ACFS) you must use a Windows Domain user account for the Oracle Home user. - You can use a Local User to perform the installation provided the user has the same password on all nodes and is an explicit member of the Administrators group on all nodes of the cluster. - During installation, you can specify an Oracle Home user. The Oracle Home User may be a Windows Domain User Account or a Windows Group Managed service account (gMSA).
Configure the network interfaces	- Verify that each node in your cluster can communicate with the other nodes using the <code>net use</code> command, for example, on node1 you can use the following command: <pre>C:\> net use \\node2\c\$</pre> The command completed successfully. - Set up the domain name forwarding for Grid Naming Service (GNS) if you plan to deploy GNS or Multi-Cluster GNS, and set up the network addresses in the DNS and on the server as needed.
Set up the required shared storage.	- All Oracle RAC database instances share the control file, server parameter file (SPFILE), redo log files, and all data files. These files must be placed on shared storage, and all the cluster database instances on cluster nodes must have access to these files. Each instance also has its own set of redo log files. During failures, shared access to redo log files enables existing instances to perform recovery. - Oracle recommends that you choose Oracle ASM as the shared storage option for database and recovery files. - You can store shared files using Oracle ASM, Oracle ACFS, or on a Network File Server (NFS) using Direct NFS (DNFS). - Files supported by Oracle ACFS include database and application executable files, trace files, alert logs, application reports, BFILES, and configuration files. - For Enterprise Edition Oracle RAC installations, Oracle ASM is the only supported shared storage option for database or recovery files. - If you do not have a storage option that provides external file redundancy, then you must configure at least three voting file areas to provide voting file redundancy.
Time Zone Requirement	Upgrade the Time Zone File and TSTZ Data. As part of an installation of Oracle Database 19c, time zone version files from 1 to 12 are installed in the path <code>%ORACLE_HOME%\oracore\zoneinfo/</code>. You can continue to use the current time zone version or upgrade to the latest version. Oracle recommends that you upgrade the server to the latest time zone version. See <i>Oracle Database Globalization Support Guide</i> for information about how to upgrade the time zone file and TSTZ data

Table 1-2 (Cont.) Hardware and Software Checklist for Oracle RAC

Item	Task
Platform-Specific Server Configuration	Configure a Windows Domain user account to use when installing Oracle RAC on Oracle ACFS on Windows platforms.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*

Supported Storage Options for Oracle Database and Recovery Files

The following table shows the storage options supported for Oracle Database and recovery files:

Table 1-3 Supported Storage Options for Oracle Database and Recovery Files

Storage Option	File Types Supported / Database	File Types Supported / Recovery Area
Oracle ASM	Yes	Yes
Oracle ACFS	Yes (Oracle Database 12c Release 1, 12.1.0.2 and later)	Yes
Direct NFS	Yes	Yes
Local Storage	No	No
Shared unformatted partitions	No	No

Installer Planning Checklist for Oracle Database Installation

Use the checklist to assist you to be prepared before starting Oracle Universal Installer.

Table 1-4 Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Review the Documentation	• Review the Oracle Database Release Notes, which is available at the following location: <i>Oracle Database Release Notes</i> • Be familiar with the installation steps for Oracle RAC software and creating an Oracle RAC database.
Review the Licensing Information	You are permitted to use only those components in the Oracle Database media pack for which you have purchased licenses. For more information about licenses, refer to the following URL: <i>Oracle Database Licensing Information</i>
Obtain your My Oracle Support account information.	During installation, you require a My Oracle Support user name and password to configure security updates, download software updates, and other installation tasks. You can register for My Oracle Support at the following URL: https://support.oracle.com/

Table 1-4 (Cont.) Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Review Oracle Support Certification Matrix	New platforms and operating system software versions can be certified after this guide is published, review the certification matrix on the My Oracle Support website for the most up-to-date list of certified hardware platforms and operating system versions: https://support.oracle.com/ You must register online before using My Oracle Support. After logging in, from the menu options, select the Certifications tab. On the Certifications page, use the Certification Search options to search by Product, Release, and Platform. You can also search using the Certification Quick Link options such as Product Delivery, and Lifetime Support.
Review online information to assist with installation	- Log on to My Oracle Support to access certifications for your installation for your platform. - Refer to Oracle.com (http://www.oracle.com) for additional resources about planning for specific implementation scenarios, best practices, and other information that can help you with your installation plan. In particular, refer to the Oracle Real Application Clusters pages on the Oracle Technology Network at http://www.oracle.com/goto/rac
Run Oracle Universal Installer (OUI) with CVU and use fixup scripts	Oracle Universal Installer is fully integrated with Cluster Verification Utility (CVU), automating many CVU prerequisite checks. Oracle Universal Installer runs all prerequisite checks and creates fixup scripts when you run the installer. You can run OUI up to the Summary screen without starting the installation. - Obtain the latest version of CVU at the following URL: https://www.oracle.com/database/technologies/cvu-downloads.html - You can also run CVU commands manually to check that your system is prepared for installation before you start an Oracle RAC installation. If you have vendors performing hardware or operating system configuration steps, then ask the vendors to complete the relevant CVU checks of the cluster after they complete their work to ensure that your system is configured correctly. - Run OUI and DBCA from a node where an Oracle RAC Oracle database instance is located. - In case of an upgrade failure, follow common industry standards for data recovery planning, including backing up your existing database.

Table 1-4 (Cont.) Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Download and run Oracle ORAchk for runtime and upgrade checks, or runtime health checks	The Oracle ORAchk utility provides system checks that can help to prevent issues before and after installation. These checks include kernel requirements, operating system resource allocations, and other system requirements. Use the Oracle ORAchk Upgrade Readiness Assessment to obtain an automated upgrade-specific system health check for upgrades to 11.2.0.3, 11.2.0.4, 12.1.0.1, 12.2, 18c, and 19c. For example: - Before you perform a fresh database installation: <pre>%ORACLE-HOME%\suptools\orachk>orachk.bat -profile preinstall</pre> - To upgrade your existing database to a higher version or release: <pre>%ORACLE-HOME%\suptools\orachk>orachk.bat -o pre</pre> The Oracle ORAchk Upgrade Readiness Assessment automates many of the manual pre- and post-upgrade checks described in Oracle upgrade documentation. Oracle ORAchk is supported on Windows 2008 and Windows 2012 on a Cygwin environment only. For more information refer to the following URL: https://support.oracle.com/rs?type=doc&id=1268927.1
Verify if Oracle Grid Infrastructure is installed	- If you want to use Oracle ASM or Oracle Restart, then install Oracle Grid Infrastructure for a standalone server before you install and create the database. Otherwise, to use Oracle ASM, you must complete an Oracle Grid Infrastructure installation, and then manually register the database with Oracle Restart. - To install Oracle Real Applications Cluster (Oracle RAC), you must have Oracle Grid Infrastructure (Oracle Clusterware and Oracle ASM) installed on your cluster. The Oracle Clusterware version must be equal to or greater than the Oracle RAC version that you plan to install. - Currently, there are no supported clusterware products other than Oracle Clusterware for the Microsoft Windows platforms. If you intend to install Oracle RAC, then you must first install Oracle Grid Infrastructure for a cluster, which includes Oracle Clusterware.
Check running Oracle processes, and shut down if necessary	- On a standalone database not using Oracle ASM: You do not need to shut down the database while you install Oracle Grid Infrastructure. - On a standalone database using Oracle ASM: Stop the existing Oracle ASM instances. The Oracle ASM instances are restarted during installation. - On an Oracle RAC Database node: This installation requires an upgrade of Oracle Clusterware, as Oracle Clusterware is required to run Oracle RAC. As part of the upgrade, you must shut down the database one node at a time as the rolling upgrade proceeds from node to node.
Ensure Task Scheduler jobs do not run during installation	If the installer is running when daily scheduled jobs start, then you may encounter unexplained installation problems if your scheduled job is performing cleanup, and temporary files are deleted before the installation is finished. Oracle recommends that you complete installation before daily scheduled jobs are run, or disable daily scheduled jobs that perform cleanup until after the installation is completed.

Table 1-4 (Cont.) Oracle Universal Installer Planning Checklist for Oracle Database Installation

Check	Task
Decide on an Oracle Database management tool	By default, Oracle Database is managed by Oracle Enterprise Manager Database Express. If you have an existing Oracle Management Agent, and decide to use Oracle Enterprise Manager Cloud Control to centrally manage your database, then obtain the following information to enter during the database installation: • OMS host • OMS port • EM admin username • EM admin password • Specify password of ASMSNMP user You need a web browser to access documentation, to use Oracle Enterprise Manager Database Express, and to use Oracle Application Express. Web browsers must support JavaScript and the HTML 4.0 and Cascading Style Sheets (CSS) 1.0 standards.
Review memory allocation and Automatic Memory Management feature	You can enable automatic memory management either during, or after Oracle Database installation. If you enable automatic memory management after installation, then you must shut down and restart the database. With Automatic Memory Management, Oracle Database instances automatically manage and tune memory. You choose a memory target, and the instance automatically distributes memory between the system global area (SGA) and the instance program global area (instance PGA). As memory requirements change, the instance dynamically redistributes memory between the SGA and instance PGA.
Unzip utility	Unzip 6.0 or later. Unzip is required to extract the image files for Oracle Database and Oracle Grid Infrastructure installations.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*

Related Topics

- *Oracle Enterprise Manager Cloud Control Administrator's Guide*

Related Topics

- *Oracle Database Administrator's Guide*

Related Topics

- *Oracle Clusterware Administration and Deployment Guide*

Upgrade Checklist for Oracle RAC

Review the checklist for additional requirements related to upgrading an existing Oracle Real Application Clusters (Oracle RAC) installation to Oracle Database 19c.

The system must meet the following requirements:

- Each server must use the same Windows user account as the Oracle Home User across all Oracle homes.

- When upgrading, if the Oracle home from which you run the database upgrade uses a Windows Domain User as the Oracle Home User, then the Oracle Home User on the target version must use the same Windows Domain User.
- When upgrading, if the Oracle home from which you run the database upgrade uses the built-in account (LocalSystem) as the Oracle Home User, then the Oracle Home User on the target version may use the built-in account, a Windows Domain User account, or a Windows Group Managed service account (GMSA).

Table 1-5 Upgrade Checklist for Oracle RAC

Item	Task
Review existing Oracle installations and upgrade plans	To install Oracle RAC 19c, you must have Oracle Grid Infrastructure (Oracle Clusterware and Oracle ASM) 18c installed on your cluster. See Also: <i>Oracle Database Upgrade Guide</i> <i>Oracle Grid Infrastructure Installation Guide</i>
Ensure that Oracle RAC Databases you are installing are compatible with existing databases	• Oracle Clusterware and Oracle ASM are both 19c after you perform an Oracle Grid Infrastructure 19c installation or upgrade. • If you have an existing Oracle home, then you can create a new Oracle home and install Oracle Database 19c into the new Oracle home. Ensure that Oracle Grid Infrastructure is installed in a separate Oracle home. Do not install Oracle Grid Infrastructure in the same Oracle base directory that is used for Oracle Database homes.
Migrate files off of RAW devices	If you have any database data stored on RAW devices, then before you start Oracle Grid Infrastructure and Oracle RAC installation, you must use RMAN to copy that data to Oracle ASM or to another supported file system.
Prepare to upgrade all existing nodes	The Oracle RAC database instance is running on the <i>same</i> nodes that you intend to make members of the new cluster installation. For example, if you have an existing Oracle RAC database running on a three-node cluster, then you must select all three nodes when upgrading the database using Oracle Universal Installer. You cannot upgrade only two nodes of the cluster, removing the third instance in the upgrade.
Ensure that Oracle RAC database version is equal to or earlier than the version of Oracle Clusterware	You can have multiple Oracle homes for Oracle databases on your cluster. However, the Oracle RAC database software in these homes must be from a version that is equal to or before the version of Oracle Clusterware that is installed; you cannot have a version of Oracle Database running on Oracle Clusterware that was released after the version of Oracle Clusterware that you use. For example: • If your servers use Oracle Grid Infrastructure 19c, then you can have an Oracle Database 19c single-instance database running on one node, and separate Oracle Database 11g release 2 (11.2), Oracle RAC 12c release 1 (12.1), or Oracle RAC 19c databases also running on the cluster. • You cannot have Oracle Grid Infrastructure 11g release 2 (11.2) installed on your cluster, and install Oracle RAC 12c release 1 (12.1).

Related Topics

- *Oracle Database Upgrade Guide*

2

Installing Oracle RAC and Oracle RAC One Node

After installing Oracle Clusterware, as described in *Oracle Grid Infrastructure Installation Guide* for your platform, you can install Oracle RAC Database software.

- [About Image-Based Oracle Database Installation](#)
Starting with Oracle Database 18c, installation and configuration of Oracle Database software is simplified with image-based installation.
- [Setup Wizard Installation Options for Creating Images](#)
Before you start the setup wizards for your Oracle Database or Oracle Grid Infrastructure installation, decide if you want to use any of the available image-creation options.
- [Deciding Between Multitenant Container Databases and Non-CDBs in Oracle RAC](#)
Review the information to decide how to deploy your Oracle Database using the CDB or Non-CDB options.
- [Installing Oracle RAC and Oracle RAC One Node Databases](#)
Understand the process to install Oracle Real Application Clusters (Oracle RAC) and Oracle RAC One Node Databases.
- [Simplified Upgrade of TIMESTAMP WITH TIME ZONE Data](#)
Time zone files are upgraded when you install Oracle Real Application Clusters.
- [Overview of Installation Directories for Oracle RAC](#)
Both an Oracle Base directory and an Oracle Home directory are used for every installation of Oracle Database software.
- [Specify Oracle Home User Name and Password](#)
To provide enhanced security for your Oracle Database software, you can select to use an Oracle Home User.
- [Updating Environment Variables on Remote Nodes](#)
Changes made to registry and environment variables are not immediately visible to user sessions on remote nodes.
- [Creating an Oracle RAC Database on Direct NFS](#)
There are different configuration processes you must perform when installing and create an Oracle RAC database that uses Direct NFS (dNFS) for the database files.

About Image-Based Oracle Database Installation

Starting with Oracle Database 18c, installation and configuration of Oracle Database software is simplified with image-based installation.

To install Oracle Database, create the new Oracle home, extract the image file into the newly-created Oracle home, and run the setup wizard to register the Oracle Database product.

Using image-based installation, you can install and upgrade Oracle Database for single-instance and cluster configurations.

This installation feature streamlines the installation process and supports automation of large-scale custom deployments. You can also use this installation method for deployment of

customized images, after you patch the base-release software with the necessary Release Updates (Updates) or Release Update Revisions (Revisions).

**Note:**

You must extract the image software (`db_home.zip`) into the directory where you want your Oracle Database home to be located, and then run the Oracle Database Setup Wizard to start the Oracle Database installation and configuration. Oracle recommends that the Oracle home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations.

Setup Wizard Installation Options for Creating Images

Before you start the setup wizards for your Oracle Database or Oracle Grid Infrastructure installation, decide if you want to use any of the available image-creation options.

In image-based installations, you can start your Oracle Database installation or Oracle Grid Infrastructure installation by running the setup wizard `setup.exe`. This wizard comes with the following image-creation options:

**Note:**

`setup.exe` is the recommended setup wizard for installing both Oracle Database and Oracle Grid Infrastructure.

Table 2-1 Image-Creation Options for Setup Wizard

Option	Description
<code>-createGoldImage</code>	Creates a gold image from the current Oracle home.
<code>-destinationLocation</code>	Specify the complete path, or location, where the gold image will be created.
<code>-exclFiles</code>	Specify the complete paths to the files to be excluded from the newly created gold image.
<code>-help</code>	Displays help for all the available options.

For example:

```
setup.exe -createGoldImage -destinationLocation c:\my_images
```

Where:

`c:\my_images` is the file location where the image zip file is created.

Deciding Between Multitenant Container Databases and Non-CDBs in Oracle RAC

Review the information to decide how to deploy your Oracle Database using the CDB or Non-CDB options.

Starting in Oracle Database 12c Release 1 (12.1), you must create a database as either a multitenant container database (CDB) or as an Oracle database that is non-CDB. This also applies to Oracle RAC databases. The only difference to the installation process is to choose whether to create the Oracle RAC database as a CDB or non-CDB.

A pluggable database (PDB) is a portable collection of schemas, schema objects, and nonschema objects that appears to an Oracle Net client as a non-CDB. PDBs can be plugged into CDBs. A CDB can contain multiple PDBs. Each PDB appears on the network as a separate database.

If you create an Oracle RAC database as a CDB and plug one or more PDBs into the CDB, then, by default, a PDB is not started automatically on any instance of the Oracle RAC CDB. With the first dynamic database service assigned to the PDB (other than the default database service which has the same name as the database name), the PDB is made available on those instances on which the service runs.

Whether or not a PDB is available on more than one instance of an Oracle RAC, CDB is typically managed by the services running on the PDB. You can manually enable PDB access on each instance of an Oracle RAC CDB by starting the PDB manually on that instance.

See Also:

- *Oracle Database Concepts* for more information about PDB concepts
- *Oracle Database Administrator's Guide* for more information about managing PDBs
- *Oracle Real Application Clusters Administration and Deployment Guide* for information specific to the administration of Oracle RAC CDBs

Installing Oracle RAC and Oracle RAC One Node Databases

Understand the process to install Oracle Real Application Clusters (Oracle RAC) and Oracle RAC One Node Databases.

Installation of Oracle Real Application Clusters (Oracle RAC) and Oracle RAC One Node Databases is a two-step process:

1. Install Oracle RAC software by running the Oracle Database installer.
 2. Create and Configure Oracle RAC and Oracle RAC One Node Databases using Database Configuration Assistant (DBCA).
- [Installing Oracle RAC and Oracle RAC One Node Database Software](#)
Install Oracle RAC or Oracle RAC One Node software.

Installing Oracle RAC and Oracle RAC One Node Database Software

Install Oracle RAC or Oracle RAC One Node software.

Oracle Real Application Clusters Database installation is a two-step process. This procedure describes the first step — to install Oracle RAC software.

If you have an existing Oracle installation, then write down the version numbers, patches, and other configuration information, and review upgrade procedures for your existing installation. Review Oracle Database Upgrade Guide before proceeding with the installation.

1. Log in as an Administrator user.
2. Open a command prompt window to run the installer, and log in as the user account that you want to own the Oracle Database installation (for example, `oracle`).
3. Download the Oracle Database installation image files (`db_home.zip`) and extract the files into a new Oracle home directory.

 Note:

Oracle recommends that the Oracle home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations. Also, unzip the installation image files only in this Oracle home directory that you created.

4. From the Oracle home directory, start the Oracle Database software installation:

```
cd C:\app\oracle\product\19.0.0\dbhome_1
setup.exe
```

 Note:

Run the `setup.exe` command from the Oracle home directory only. Do not run `setup.exe` from any other location.

5. In the Select Configuration Option screen, select the **Setup Software Only** option. Click **Next**.
6. In the Select Database Installation Option screen, select the **Oracle Real Application Clusters Database Installation** option. Click **Next**.
7. In the Node Selection screen, select all the nodes where you want to install Oracle RAC or Oracle RAC One Node software. Click **Next**.
8. In the Select Database Edition screen, select the **Enterprise Edition** option. Click **Next**.
9. In the Specify Install Location screen, provide the location of Oracle base for Oracle RAC software installation.
The Oracle base directory must be different from the Grid home directory. Click **Next**.
10. The Perform Prerequisite Checks screen displays the results of the prerequisites checks. If any of the checks have a status of Failed and are not Fixable, then you must manually correct these issues. After you have fixed the issue, you can click the **Check Again** button

to have the installer recheck the requirement and update the status. Repeat as needed until all the checks have a status of Succeeded. Click **Next**.

11. Review the contents of the Summary screen and then click **Install.**

The installer displays a progress indicator enabling you to monitor the installation process.

After installing Oracle RAC software, run Database Configuration Assistant (DBCA) from the `ORACLE_HOME\bin\dbca` directory to create and configure Oracle RAC databases.

Related Topics

- [Oracle Real Application Clusters Postinstallation Procedures](#)
After you have installed the Oracle Database 19c with Oracle Real Application Clusters (Oracle RAC) software, there are postinstallation tasks to complete.
- [Creating Oracle RAC or Oracle RAC One Node Databases with DBCA](#)
Use Oracle Database Configuration Assistant (DBCA) in standalone mode to create and delete Oracle Real Application Clusters (Oracle RAC) databases.

Simplified Upgrade of TIMESTAMP WITH TIME ZONE Data

Time zone files are upgraded when you install Oracle Real Application Clusters.

As part of an installation of Oracle Database 19c, time zone files are installed in the path `Oracle_home\oracore\zoneinfo`. You can continue to use the current time zone file or upgrade to the latest version. Oracle recommends that you upgrade the server to the latest version of the time zone file. Upgrading to a new version of the time zone file may cause existing TIMESTAMP WITH TIME ZONE data to become stale. Using the newly provided DBMS_DST PL/SQL package, you can update the TIMESTAMP WITH TIME ZONE data transparently, with minimal manual procedures and system downtime.

All instances of an Oracle RAC database must use the same time zone. The Oracle RAC database time zone defaults to the time zone setting of the Grid user, unless an instance is started with SQL*Plus. When you use SQL*Plus, you must be sure to use the same time zone setting for the database instance that is used for Oracle Clusterware. You can change the time zone Oracle Clusterware uses for a database by using the following command, where *time_zone* is the time zone to which you want to change:

```
srvctl setenv database -env "TZ=time_zone"
```

Time zone version files are also installed with Oracle Client installations. You do not have to upgrade Oracle Client time zone files immediately. Upgrades can be done at a time when it is most convenient to the system administrator. However, there could be a small performance penalty when client and server use different time zone versions.



See Also:

- *Oracle Database Upgrade Guide* for information about preparing to upgrade TIMESTAMP WITH TIME ZONE data
- *Oracle Database Globalization Support Guide* for information about how to upgrade the time zone file and TIMESTAMP WITH TIME ZONE data
- *Oracle Call Interface Programmer's Guide* for information about performance effects of clients and servers operating with different versions of time zone files

Overview of Installation Directories for Oracle RAC

Both an Oracle Base directory and an Oracle Home directory are used for every installation of Oracle Database software.

Additionally, on the Windows operating system, Oracle provides a home name for each Oracle Home directory.

- [Overview of Oracle Base Directories](#)
The Oracle base directory is the location where Oracle software and configuration files are stored.
- [Overview of Oracle Home Directories](#)
The Oracle home directory is located under the Oracle base directory.

Overview of Oracle Base Directories

The Oracle base directory is the location where Oracle software and configuration files are stored.

By default, Oracle Universal Installer (OUI) installs the Oracle Database software binary files by version and Oracle Home Name in a subdirectory of the Oracle base directory. An Oracle base directory can be used for multiple installations of software by a given installation owner. A separate Oracle base directory is created for each Oracle Home user you specify during installing of the Oracle Database software.

The Oracle Home User has complete control over the Oracle base directory. For reasons of security, different Windows User Accounts used as Oracle Home Users for different Oracle home directories are not allowed to share the same Oracle base directory. However, to support Oracle Database upgrade, Oracle supports the sharing of an Oracle base directory between a Built-in Account and a Windows User Account. If you choose to reuse an Oracle base directory from an earlier release of Oracle Database in Oracle Database 19c Release, then the Oracle Home User of the Oracle Database 19c Release has complete control over the Oracle base directory of the earlier release.

The default Oracle base path contains the Oracle Home User name if an Oracle Home User is specified during installation of the Oracle Database software. In a default Windows installation, the Oracle base directory appears as follows, where *username* is the Oracle Installation user if you choose Windows Built-in Account as the Oracle Home User, or it is the Oracle Home user if one is specified:

DRIVE_LETTER: \app\username

If you have separate Oracle Home Users for the Oracle Grid Infrastructure installation and the Oracle RAC installation, then you have two Oracle base paths that are in accordance with Optimal Flexible Architecture (OFA) guidelines. For example, if the user `grid` is the Oracle Home User for the Oracle Grid infrastructure installation and the user `oracle` is the Oracle Home User for the Oracle Database installation, then you have two Oracle base directories. In the following examples, `X:` represents a mounted disk:

- `X:\app\grid`—This is the Oracle base for the Grid user (`grid` in this example), which is the Oracle Home User for the Oracle Grid Infrastructure installation.
- `X:\app\oracle`—This is the Oracle base for the Oracle user (`oracle` in this example), which is the Oracle Home User for the Oracle Database installation.

 Caution:

After installing Oracle Database 12c Release 1 (12.1) (or later) release with a Windows User Account as Oracle Home User, do not install older versions of Oracle Databases that share the same Oracle Base Directory. During installation of the software for older releases, the ACLs are reset and Oracle Database 12c Release 1 (12.1) (or later) services may not be able to access the Oracle Base directory and files.

Overview of Oracle Home Directories

The Oracle home directory is located under the Oracle base directory.

In a default Windows installation, if you name the Oracle home directory `dbhome_1`, it appears in the Oracle base directory as follows, where *username* is the Oracle Installation user if you do not choose Windows security, or it is the Oracle Home user if one is specified:

```
DRIVE_LETTER:\app\username\product\19.0.0\dbhome_1
```

Ensure that the paths that you select for Oracle software, such as Oracle home paths and the Oracle base path, use only ASCII characters. Because some Oracle software directory paths use installation user names by default, this ASCII character restriction applies to user names, file names, and directory names.

Specify Oracle Home User Name and Password

To provide enhanced security for your Oracle Database software, you can select to use an Oracle Home User.

An Oracle Home User is a standard Windows User Account (not an Administrator account), specified during installation, that runs the Windows services required by Oracle Database for the Oracle home.

The Oracle Home User is associated with an Oracle Home and it cannot be changed post installation. Different Oracle Homes on a system can share the same Oracle Home User or use different Oracle Home User names. For Oracle RAC databases, the Windows user account for the Oracle Home must be a domain account and it has to be an existing account.

For Administrator-managed databases, you can store the password for the Oracle Home User in a secure wallet in the Oracle Cluster Registry (OCR). If such a wallet exists in the OCR, then Oracle Database administration tools automatically use the password from the wallet and do not require you to enter the password for the Oracle Home User during administrative operations.

For Policy-managed databases, you must store the password for the Oracle Home User in a secure wallet in the OCR. Oracle Database Configuration Assistant automatically creates the wallet (if one does not exist) when a Policy-managed database is created.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*
- *Oracle Database Administrator's Reference for Microsoft Windows*

Updating Environment Variables on Remote Nodes

Changes made to registry and environment variables are not immediately visible to user sessions on remote nodes.

During the installation, the Windows registry and environment variables are modified on each node on which you installed Oracle RAC. The new registry entries and environment variable settings are visible on the node where the installation was performed (the local node). However, the new settings on the remote nodes are not immediately available to your user session. Attempting to run scripts or applications from the newly installed Oracle home can produce errors similar to the following:

- ORACONFIG.exe - Unable to Locate Component
- OCI.dll not found

To make the modified environment variables available on the remote nodes, you can do either of the following actions:

1. Close your current session on the remote nodes (log off) and then log on to the remote nodes to create a new session.
2. Make the environment variables available to the remote nodes:
 - a. From the Start menu, right-click **My Computer**, then select **Properties**
Alternatively, you can enter `SYSDM.CPL` in the Run window.
 - b. Select the **Advanced** tab.
 - c. Click **Environment Variables**.
The modified environment variables are now visible.
 - d. Click **OK** to close the System Properties window.

Creating an Oracle RAC Database on Direct NFS

There are different configuration processes you must perform when installing and create an Oracle RAC database that uses Direct NFS (dNFS) for the database files.

- [Performing a Software-Only Installation of Oracle Database](#)
In a software-only installation you install the Oracle Database software but do not create a database as part of the installation process.
- [Using Oracle ASMCA to Configure an ACFS Mount Point](#)
Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA) enables you to create an Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point which is used in the "common file location" step of Oracle Database Configuration Assistant (DBCA).
- [Using Oracle DBCA to Create and Configure the Oracle RAC Database](#)
Use Oracle Database Configuration Assistant (DBCA) to create an Oracle Real Application Clusters (Oracle RAC) database that uses Direct NFS for datafile storage.
- [Enabling and Configuring Direct NFS](#)
You must manually enable the Direct NFS option after installing the Oracle Database software.

- [Using Oracle ASMCA to Remove the ACFS Mount Point](#)

Now you have configured the mount point using Direct NFS, you can remove the Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point using Oracle Automatic Storage Management Configuration Assistant (ASMCA).

Performing a Software-Only Installation of Oracle Database

In a software-only installation you install the Oracle Database software but do not create a database as part of the installation process.

1. Start Oracle Universal Installer (OUI) by running `setup.exe` from the Oracle home directory.
2. On the Select Configuration Option screen select **Set Up Software Only**.
3. Select the nodes on which you want to install the database software.
4. Select the database edition to install.
5. Specify an Oracle Home user, or choose to use a Windows-built in user for the software installation owner.
6. On the Specify Installation Location screen, enter a path to the Oracle base directory and the software location (Oracle home directory).
7. On the Summary screen, verify your selections, then click **Install**.

Using Oracle ASMCA to Configure an ACFS Mount Point

Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA) enables you to create an Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point which is used in the "common file location" step of Oracle Database Configuration Assistant (DBCA).

When creating Oracle ACFS file systems on Windows, run ASMCA as a Windows domain user who is also an administrator of the computer.

1. From the `Grid_home/bin` directory, run `asmca.exe` to start the ASMCA.
2. Select the **Disk Groups** tab.
3. Right-click **Disk Group Name** and select **Create ACFS for Database use**.
4. In the Create ACFS for Database window specify the mount point location, volume name, and size, then click **OK**.

For example, you can specify the following:

- Mount Point: `C:\oradatamnt`
 - Volume Name: `dbnfs`
 - Size (GB): `70`
5. Click **OK** in the informational pop-up window that appears.
This window summarizes the actions performed by ASMCA.
 6. Select the **ASM Cluster File Systems** tab.
The mount point you just created is displayed on this page.

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Using Oracle DBCA to Create and Configure the Oracle RAC Database

Use Oracle Database Configuration Assistant (DBCA) to create an Oracle Real Application Clusters (Oracle RAC) database that uses Direct NFS for datafile storage.

1. From the `Oracle_home\bin` directory, run `dbca.exe` to start the Database Configuration Assistant.
2. On the Database Operation screen, select **Create Database**.
3. On the Creation Mode screen, select **Advanced Configuration**.
4. On the Database Template screen, select **Oracle Real Application Clusters (RAC) database** for the Database Type.

For the Configuration Type, you can choose either Policy-Managed or Administrator-Managed. Select the template most appropriate for the type of database you want to create.
5. For the next four screen, make selections and provide information that best meet your business requirements.
6. Perform the following steps on the Storage Locations screen:
 - a. Select **File System** for the Database files Storage type.
 - b. Select the option **Use Common Location for All Database Files**.
 - c. In the File location field, specify the location of the ACFS mount point, for example, `C:\oradatamnt`.
7. On the Specify Database Options screen, choose any additional configuration you want for your database.
8. On the Configurations Options screen, use the default settings, or provide customized values for the initialization parameters.
9. On the Creation Options screen, select the option **Generate Database Creation Scripts**. Specify a destination directory for the script file, or use the default value.
10. After the Prerequisite checks complete, on the Summary screen, minimize the installation window. **DO NOT** click Finish at this point.
11. Enable the Direct NFS option.
12. Remove the virtual mount point you created with Oracle Automatic Storage Management Configuration Assistant (ASMCA).
13. Create all the directories needed locally on each node as well as on the NFS server.

For this example, you can create the following, where `orcl` represents the database SID and `pdb1` represents the Pluggable Database (PDB) name:
 - On each node, create the directory `c:\oracle\oradatamnt\orcl\pdb1`
 - On the NFS server, create the directory `/export/abcd/orcl/pdb1`
14. Return to the DBCA window and click **Finish**.
15. Run the generated scripts on the cluster node to create the database.
16. Map a drive letter to a Common Internet File System (CIFS) share on the NFS server that represents the location of the database files.

Use a command similar to the following:

```
NET USE * \\filer\vol0\orcl
```

After you complete this step, both Oracle and Microsoft Windows operating system can access the location where the database files reside. Oracle is using DNFS, but Microsoft Windows operating system uses CIFS to access the same location on the NFS server.

17. Verify Direct NFS is configured for the database.
 - a. Start SQL*Plus.
 - b. Connect to the newly created database as a DBA user.
 - c. Run the following SQL command:

```
SELECT * FROM v$dtnfs_servers;
```

Enabling and Configuring Direct NFS

You must manually enable the Direct NFS option after installing the Oracle Database software.

1. Run the program `Oracle_home\bin\enable_dnfs.bat`.
2. Create an `oranfstab` file.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*

Using Oracle ASMCA to Remove the ACFS Mount Point

Now you have configured the mount point using Direct NFS, you can remove the Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point using Oracle Automatic Storage Management Configuration Assistant (ASMCA).

When creating Oracle ACFS file systems on Windows, run ASMCA as a Windows domain user who is also an administrator of the computer.

1. From the `Grid_home/bin` directory, run `asmca.exe` to start the ASMCA.
2. Select the **ASM Cluster File System** tab.
3. Select the mount point created earlier (`C:\oradatamnt`), then click **Dismount All**.
4. Select the **Volumes** tab.
5. Right-click the mount point and then select **Delete**.

3

Creating Oracle RAC or Oracle RAC One Node Databases with Oracle DBCA

Use Oracle Database Configuration Assistant (DBCA) in standalone mode to create and delete Oracle Real Application Clusters (Oracle RAC) databases.

- [Using Oracle DBCA with Oracle RAC or Oracle RAC One Node](#)
Oracle Database Configuration Assistant (DBCA) is a tool for creating and configuring an Oracle database.
- [About Oracle Database Configuration Assistant](#)
Understand the features Oracle Database Configuration Assistant (Oracle DBCA) offers for creating your Oracle RAC databases.
- [Selecting Installation Options for Oracle RAC](#)
Review the topics to select options for installing Oracle RAC.
- [Installing the Oracle Database Vault Option](#)
Installing and configuring Oracle Database Vault requires actions during and after installation.
- [Automatic Listener Migration from Earlier Releases](#)
Review this information for listener migration from earlier database releases.
- [Verifying Requirements for Oracle DBCA](#)
Use Cluster Verification Utility (CVU) to verify that your system is prepared for configuration changes.
- [Tasks to Complete Before Using DBCA to Create an Oracle RAC Database](#)
Before you can create an Oracle RAC database using Oracle Database Configuration Assistant, you must configure your system to meet the software requirements, if this was not done as part of the Oracle Grid Infrastructure installation.
- [Selecting DBCA Options to Create an Oracle RAC or Oracle RAC One Node Database](#)
Review this information to use Database Configuration Assistant (DBCA) to create Oracle RAC or Oracle RAC One Node Database.
- [Using DBCA to Create an Oracle RAC One Node Database](#)
If you have selected to install only the Oracle RAC software on cluster nodes, then you can use Oracle Database Configuration Assistant (DBCA) to configure Oracle RAC One Node.
- [Deleting an Oracle RAC Database Using DBCA](#)
Deleting an Oracle RAC database using Oracle Database Configuration Assistant (DBCA) involves deleting the database and database objects.
- [Creating an Oracle RAC Database on Direct NFS](#)

Using Oracle DBCA with Oracle RAC or Oracle RAC One Node

Oracle Database Configuration Assistant (DBCA) is a tool for creating and configuring an Oracle database.

Oracle DBCA has the following primary database functions:

- Create and delete databases

- Create database templates
- Create, plug, unplug, and delete pluggable databases (PDBs)
- Add and delete database instances
- Register databases in Oracle Enterprise Manager Cloud Control
- Configure and register database options (such as Oracle Database Vault) with the Directory Server



Note:

Cluster Managed Services are no longer managed through DBCA. Instead, use the Cluster Managed Services page in Oracle Enterprise Manager Cloud Control, if available, or SRVCTL. For more information, see *Oracle Real Application Clusters Administration and Deployment Guide*.

Related Topics

- [Selecting DBCA Options to Create an Oracle RAC or Oracle RAC One Node Database](#)
Review this information to use Database Configuration Assistant (DBCA) to create Oracle RAC or Oracle RAC One Node Database.

Related Topics

- *Oracle Database Net Services Administrator's Guide*

Related Topics

- *Oracle Real Application Clusters Administration and Deployment Guide*

About Oracle Database Configuration Assistant

Understand the features Oracle Database Configuration Assistant (Oracle DBCA) offers for creating your Oracle RAC databases.

Oracle DBCA enables you to create both policy-managed and administrator-managed databases. With Oracle DBCA, you can also create site-specific tablespaces as part of database creation. If you have data file requirements that differ from those offered by Oracle DBCA templates, then create your database with Oracle DBCA and modify the data files later. You can also run user-specified scripts as part of your database creation process.

Oracle DBCA also configures your Oracle RAC environment for various Oracle high availability features, such as cluster administration tools. Oracle DBCA also starts any database instances required to support your defined configuration.

You can use Oracle DBCA to create a database from templates supplied by Oracle, or from templates that you create. The templates contain settings optimized for a particular type of workload.

Oracle ships templates for the following two workload types:

- General purpose or transaction processing
- Data warehouse

For more complex environments, you can select the **Custom Database** option. This option does not use templates and results in a more extensive interview, which means that it takes longer to create your database.

Click **Show Details** to see the configuration for each type of database. Select the template suited to the type of workload your database supports. If you are not sure which to choose, then select the default **General Purpose or Transaction Processing** template.

Selecting Installation Options for Oracle RAC

Review the topics to select options for installing Oracle RAC.

- [Selecting a Security Notification Contact](#)
During installation, you are asked in the Configure Security Updates screen to provide a security contact.
- [Selecting an Installation Option](#)
You must choose one of the installation options for installing the software.
- [Selecting the Database Type for Oracle Grid Infrastructure Deployments](#)
During installation, Oracle Universal Installer (OUI) detects if you have Oracle Grid Infrastructure for a cluster installed. If you do, then you must specify the type of database you plan to create.
- [Choosing the Cluster Database Management Type](#)
When creating an Oracle RAC database, you can choose one of two types of databases to create.
- [Selecting an Installation Type](#)
When you run Oracle Universal Installer (OUI) to install Oracle RAC, you can select the **Typical** or the **Advanced** installation type.
- [Selecting a Database Name](#)
The database name is comprised of various strings and must contain only permitted characters. Review the following guidelines when selecting a database name.
- [Requirements for Database Passwords](#)
To secure your database, use passwords that satisfy the Oracle recommended password requirements, even the passwords for predefined user accounts.
- [About Automatic Memory Management Installation Options](#)
Decide if you want to configure Automatic Memory Management during installation.
- [About Character Set Selection During Installation](#)
Before you create the database, decide the character set that you want to use.
- [Managing Database Services After Installation](#)
Use the Server Control Utility (SRVCTL), or use Oracle Enterprise Manager Database Express, or Oracle Enterprise Manager Cloud Control for all administration and monitoring of database services for an Oracle RAC database.

Selecting a Security Notification Contact

During installation, you are asked in the Configure Security Updates screen to provide a security contact.

Oracle issues security alerts as needed for vulnerability fixes that are determined to be too critical to wait for distribution in the next Critical Patch Update.

1. Optional: Provide security contact information in one of the following forms:
 - An email address to receive security information for your installation.

- A My Oracle Support email address or account name to receive security information for your installation, and to enroll your system for Security Updates. You can receive information about alerts through My Oracle Support.

The information collected by Security Updates is limited to configuration information. The data collected does not include personally identifiable information (except a local contact name in case of transmission problems). You may still use all licensed Oracle functionality if you decline to enable Security Updates

If you provide your My Oracle Support credentials, then Security Updates automatically gathers configuration information regarding your installed Oracle products and uploads it to Oracle's support systems. You can access the information it collects through your My Oracle Support account, and review health check recommendations, patch recommendations and other recommendations for your system in addition to security alerts.

2. Optional: To choose not to receive security notifications, leave all fields in the Configure Security Updates screen blank.

You can choose not to provide this information, but Oracle strongly recommends that you configure a security notification contact.

3. Click **Next** to continue.



See Also:

The Oracle Security Policies page, which is available from the following URL:

<http://www.oracle.com/us/support/assurance/fixing-policies/index.html>

Selecting an Installation Option

You must choose one of the installation options for installing the software.

1. On the Select Installation Option page, select one of the following options:
 - **Create and Configure a Single Instance Database:** Provides you with the option to create a database using a preconfigured database template designed for particular system load demands, such as an online transaction processing (OLTP) database, or a decision support or data warehouse database.
 - **Set Up Software only:** Installs Oracle Database software; you must complete the database configuration after the installation completes using the installed utilities.
2. Optional: If you are installing Oracle Database software, then Oracle recommends that you use a preconfigured database option, or select the **Advanced** option on the Select Configuration page, and configure a custom starter database.
3. If you have an existing Oracle installation, then write down the version numbers, patches, and other configuration information, and review upgrade procedures for your existing installation.

**See Also:**

Oracle Database Upgrade Guide before proceeding with the installation, to decide how you want to proceed.

Selecting the Database Type for Oracle Grid Infrastructure Deployments

During installation, Oracle Universal Installer (OUI) detects if you have Oracle Grid Infrastructure for a cluster installed. If you do, then you must specify the type of database you plan to create.

- Determine which type of database you plan to create after installing the software:
 - A single-instance database
 - An Oracle RAC database
 - An Oracle RAC One Node database

If you plan to create databases of different types on this cluster, then choose the most advanced option.

For example, if you plan to create only single-instance and Oracle RAC One Node databases, then choose the Oracle RAC One Node database option. If you plan to create single-instance databases and Oracle RAC databases, then choose the Oracle RAC database option.

- If you plan to install Oracle RAC One Node, then you can install the Oracle RAC software on two or more nodes in the cluster.

An Oracle RAC One Node installation starts an instance on one of the nodes you select as an Oracle RAC One Node pool member. If that instance goes down, then the Oracle RAC One Node instance fails over to another pool member. This feature relocates database instances and connections to other cluster nodes for high availability.

**See Also:**

Oracle Real Application Clusters Administration and Deployment Guide for information about how to convert single-instance databases to Oracle RAC

Choosing the Cluster Database Management Type

When creating an Oracle RAC database, you can choose one of two types of databases to create.

- A **policy-managed database**: The database instances are automatically managed based on server pools for effective resource utilization.
- An **administrator-managed database**: The database instances are tied to specific servers in the cluster.

**See Also:**

Oracle Real Application Clusters Administration and Deployment Guide for more information about server pools and the different cluster database management types

Selecting an Installation Type

When you run Oracle Universal Installer (OUI) to install Oracle RAC, you can select the **Typical** or the **Advanced** installation type.

The Typical installation type installs a default configuration of Oracle Database, with basic configuration choices. Oracle recommends that most users select Typical as their installation type.

The Advanced installation type is for customized installations. Use Advanced installation only when you have a specific requirement for it, such as:

- Adding specific components to your installation
- Requiring different passwords for the SYS, SYSTEM and DBSNMP accounts
- Using a different database character set than is in use on your servers
- Changing product languages
- Other nonstandard configurations
- [Preconfigured Database Types Supplied with Oracle Database](#)
The General Purpose and Transaction Processing type and the Data Warehouse type use preconfigured database templates optimized for each type of database.
- [Using Advanced Database Configuration](#)
You use the Advanced Database Configuration option when you have special requirements for your Oracle Database.
- [About Installing Oracle Database with Other Languages](#)
To use languages other than the default (English), either for the database or for applications running on the database, you must use the Advanced Installation method.

Preconfigured Database Types Supplied with Oracle Database

The General Purpose and Transaction Processing type and the Data Warehouse type use preconfigured database templates optimized for each type of database.

During installation, Oracle Universal Installer (OUI) starts Oracle Net Configuration Assistant (NETCA) and Oracle Database Configuration Assistant (DBCA), and installs the preconfigured database without further input. During database installation, OUI displays a progress indicator.

DBCA processing for these two configuration types creates a starter database, and configures the Oracle network services.

Using Advanced Database Configuration

You use the Advanced Database Configuration option when you have special requirements for your Oracle Database.

Advanced configuration options available using this installation type include Oracle RAC, Automatic Storage Management, backup and recovery configuration, integration with Oracle Enterprise Manager Cloud Control, more fine-grained memory tuning, and other options.

About Installing Oracle Database with Other Languages

To use languages other than the default (English), either for the database or for applications running on the database, you must use the Advanced Installation method.

Tip:

By default, Oracle Universal Installer (OUI) configures the character set of a new database based on the language of the operating system.

See Also:

- *Oracle Database Globalization Support Guide* for detailed information about character set and language support
- *Oracle Database Installation Guide for Microsoft Windows* for information about running OUI in different languages

Selecting a Database Name

The database name is comprised of various strings and must contain only permitted characters. Review the following guidelines when selecting a database name.

The database name input field sets the following Oracle initialization parameter values:

- `DB_NAME`
- `DB_UNIQUE_NAME`
- `DB_DOMAIN`

In Oracle RAC environments, the database name (`DB_UNIQUE_NAME`) portion is a string of no more than 30 characters that can contain alphanumeric, underscore (`_`), dollar (`$`), and pound (`#`) characters, but must begin with an alphabetic character. No other special characters are permitted in a database name. The `DB_NAME` parameter for a database is set to the first 8 characters of the database name.

The domain portion of the global database name (`DB_DOMAIN`) can be no more than 128 characters. Domain names using underscores (`_`) are not allowed. The values for `DB_UNIQUE_NAME`.`DB_DOMAIN` in its entirety must be unique within the enterprise.

Note:

For Oracle Real Applications Cluster (Oracle RAC) databases, the pluggable database (PDB) name must be unique in the cluster.

Database Name and ORACLE_SID

The Oracle Service Identifier (SID) prefix is the first 8 characters of the database name. The SID prefix can contain only the characters a-z, A-Z, and 0-9. The SID prefix cannot contain operating system special characters, so if you use special characters in the first 8 characters of the database name, then these special characters are omitted in the SID prefix. There is a single SID prefix for every database. The SID prefix for a database must be unique within the cluster.

For an Oracle RAC database, each instance has a unique identifier, `ORACLE_SID`, which consists of the SID prefix and an instance number. The `ORACLE_SID` prefix can contain up to 12 characters. The `ORACLE_SID` for Oracle RAC database instances is generated differently, depending on how you choose to manage the database. If you select a policy-managed database, then Oracle generates the SID in the format *name_#*, where *name* is the first eight alphanumeric characters of `DB_UNIQUE_NAME`, and *#* is the instance number. If you select an administrator-managed database, then Oracle Database Configuration Assistant generates the default SID for the instance names, using the format *name#*, where *name* is the first eight alphanumeric characters of `DB_UNIQUE_NAME`, and *#* is the instance number. However, during installation or database creation you can specify a nondefault value for the SID. The instance number is automatically added to the end of this string for each instance.

For an Oracle RAC One Node database, the instance name is `ORACLE_SID_1`, which consists of `_1` appended to the SID prefix. During online relocation, a second instance `ORACLE_SID_2` is started, which becomes the only instance after the relocation completes. The next online relocation uses `ORACLE_SID_1` for the new instance.

Example 3-1 Global Database Name and Related Initialization Parameters

If your database has a global database name of `orl$racprod2551.example.com` which you supplied during installation, then the following values are used for initialization parameters:

Parameter	Value
<code>DB_UNIQUE_NAME</code>	<code>orl\$racprod2551</code>
<code>DB_DOMAIN</code>	<code>example.com</code>
<code>DB_NAME</code>	<code>orl\$racp</code>

Example 3-2 DB_UNIQUE_NAME and Related ORACLE_SID Values

If the `DB_UNIQUE_NAME` for a database is `orl$racprod2551`, then the following SID values are used:

Database or Instance Type	Value Used for ORACLE_SID
Single-instance Oracle database	<code>orl\$racpr</code>
Policy-managed Oracle RAC instance	<code>orl\$racpr_1</code>
Administrator-managed Oracle RAC instance	<code>orl\$racpr1</code>
Oracle RAC One Node database instance	<code>orl\$racpr_1</code>

Requirements for Database Passwords

To secure your database, use passwords that satisfy the Oracle recommended password requirements, even the passwords for predefined user accounts.

Oracle Database provides a set of predefined user accounts. Create passwords in a secure fashion. If you have default passwords, change these passwords to secure passwords.

You can manage the security for Oracle Database users in various ways:

- Enforce restrictions on the way that passwords are created
- Create user profiles
- Use user resource limits to further secure user accounts

Related Topics

- *Oracle Database Security Guide*

About Automatic Memory Management Installation Options

Decide if you want to configure Automatic Memory Management during installation.

During a **Typical** installation, you create your database with Oracle Database Configuration Assistant (DBCA), and automatic memory management is enabled. If you choose **Advanced** installation, then you can either specify memory allocation manually, or enable automatic memory management.

If the total physical memory of your database instance is greater than 4 GB, then you cannot select the Oracle Automatic Memory Management option during database installation and creation. Instead, use automatic shared memory management. Automatic shared memory management automatically distributes the available memory among the various components as required, allowing the system to maximize the use of all available SGA memory.

With automatic memory management, the Oracle Database instances automatically manage and tune memory for you. With automatic memory management, you choose a memory target, and the instance automatically distributes memory between the system global area (SGA) and the instance program global area (instance PGA). As memory requirements change, the instance dynamically redistributes memory between the SGA and instance PGA.

You can enable automatic memory management either during, or after the database installation. Enabling automatic memory management after installation involves a shutdown and restart of the database.



Note:

By default, automatic memory management is disabled when you perform typical installation on a node that has more than 4 GB of RAM.

Related Topics

- *Oracle Database Administrator's Guide*

About Character Set Selection During Installation

Before you create the database, decide the character set that you want to use.

After a database is created, changing its character set is usually very expensive in terms of time and resources. Such operations may require converting all character data by exporting the whole database and importing it back. Therefore, it is important that you carefully select the database character set at installation time.

Oracle Database uses character sets for the following:

- Data stored in SQL character data types (`CHAR`, `VARCHAR2`, `CLOB`, and `LONG`).
- Identifiers such as table names, column names, and PL/SQL variables.
- Stored SQL and PL/SQL source code, including text literals embedded in this code.

Starting with Oracle Database 12c Release 2 (12.2), the default database character set of a database created from the General Purpose/Transaction Processing or the Data Warehousing template is Unicode `AL32UTF8`.

Unicode is the universal character set that supports most of the currently spoken languages of the world. It also supports many historical scripts (alphabets). Unicode is the native encoding of many technologies, including Java, XML, XHTML, ECMAScript, and LDAP. Unicode is ideally suited for databases supporting the Internet and the global economy.

Because `AL32UTF8` is a multibyte character set, database operations on character data may be slightly slower when compared to single-byte database character sets, such as `WE8ISO8859P1` or `WE8MSWIN1252`. Storage space requirements for text in most languages that use characters outside of the ASCII repertoire are higher in `AL32UTF8` compared to legacy character sets supporting the language. English data may require more space only if stored in `CLOB` (character large object) columns. Storage for non-character data types, such as `NUMBER` or `DATE`, does not depend on a character set. The universality and flexibility of Unicode usually outweighs these additional costs.

Consider legacy character sets only when the database needs to support a single group of languages and the use of a legacy character set is critical for fulfilling compatibility, storage, or performance requirements. The database character set to be selected in this case is the character set of most clients connecting to this database.

The database character set of a multitenant container database (CDB) determines which databases can be plugged in later. Ensure that the character set you choose for the CDB is compatible with the database character sets of the databases to be plugged into this CDB. If you use Unicode `AL32UTF8` as your CDB character set, then you can plug in a pluggable database (PDB) in any database character set supported by Oracle Database (with the exception of EBCDIC-based character sets).



See Also:

Oracle Database Globalization Support Guide for more information about choosing a database character set for a multitenant container database (CDB)

Managing Database Services After Installation

Use the Server Control Utility (SRVCTL), or use Oracle Enterprise Manager Database Express, or Oracle Enterprise Manager Cloud Control for all administration and monitoring of database services for an Oracle RAC database.

You cannot use Oracle Database Configuration Assistant (DBCA) to manage database services for Oracle RAC databases.

Oracle Enterprise Manager Cloud Control is available separately with the Oracle Enterprise Manager Cloud Control installation, and on the Oracle Technology Network website at:

<https://www.oracle.com/downloads/>



See Also:

Oracle Enterprise Manager Online Help for service management using Oracle Enterprise Manager

Installing the Oracle Database Vault Option

Installing and configuring Oracle Database Vault requires actions during and after installation.

- [Starting the Listener with Oracle Database Vault Installations](#)
You must start the listener and database instance on all Oracle RAC nodes other than the one on which the installation is performed.
- [Configuring Oracle Database Vault Using DBCA](#)
You can configure Oracle Database Vault after installation using Oracle Database Configuration Assistant (DBCA), or choose not to configure Oracle Database Vault.
- [Perform Postinstallation Configuration for Oracle Database Vault](#)
After you install the Oracle Database Vault option, you can be required to make additional changes to your database.

Starting the Listener with Oracle Database Vault Installations

You must start the listener and database instance on all Oracle RAC nodes other than the one on which the installation is performed.

- Use Server Control (SRVCTL) to start and stop the Oracle RAC instances being configured for Oracle Database Vault.

Do not use SQL*Plus to start and stop Oracle RAC instances.

Configuring Oracle Database Vault Using DBCA

You can configure Oracle Database Vault after installation using Oracle Database Configuration Assistant (DBCA), or choose not to configure Oracle Database Vault.

To install Oracle Database Vault using DBCA:

1. After installing Oracle RAC, create the database.
2. Start DBCA and select the option **Configure Database**.
3. In the component list, select **Oracle Label Security** and **Oracle Database Vault**.
4. Provide the required Oracle Database Vault user IDs and passwords to proceed with configuration. To enable a separate Oracle Database Vault administrator, choose to configure the DV_ACCTMGR user.
5. After you have finished, restart each database instance to finish the software configuration.



See Also:

Oracle Database Vault Administrator's Guide for information about using Oracle Data Guard with Oracle Database Vault

Perform Postinstallation Configuration for Oracle Database Vault

After you install the Oracle Database Vault option, you can be required to make additional changes to your database.

1. Refer to *Oracle Database Vault Administrator's Guide* for required postinstallation steps.
2. If you use other Oracle Database products, then you will have to integrate Database Vault with other Oracle products, such as Transparent Data Encryption or Oracle Data Guard.

Related Topics

- [Configuring Oracle Database Vault](#)
Oracle Universal Installer (OUI) installs Oracle Database Vault by default when you install the Oracle RAC software, but requires additional configuration steps.
- *Oracle Database Vault Administrator's Guide*

Automatic Listener Migration from Earlier Releases

Review this information for listener migration from earlier database releases.

If your system has an Oracle Grid Infrastructure 11g Release 2 (11.2), 12c Release 1 (12.1), 12c Release 2 (12.2), or 18c installation, and you install Oracle Grid Infrastructure 19c either to coexist with or to upgrade the Oracle Grid Infrastructure 11.2, 12.1, 12.2, or 18c installation, then most installation types automatically migrate the existing listener to the 19c Oracle home. During migration, the upgrade process configures and starts a default Oracle Net Listener using the same TCP/IP port as the existing listener, with the IPC key value.

During the Oracle Clusterware upgrade, the default listener (`LISTENER_nodename`) is migrated to the Oracle Grid Infrastructure home (Grid home). Oracle Database Configuration Assistant always uses the default listener.



Note:

During migration, client applications may not be able to connect to any databases that are registered to the listener that is being migrated.

Verifying Requirements for Oracle DBCA

Use Cluster Verification Utility (CVU) to verify that your system is prepared for configuration changes.

- Prior to using Oracle Database Configuration Assistant (Oracle DBCA) to change the database configuration, run Cluster Verification Utility (CVU) to verify that your system is prepared for configuration changes using the following command syntax:

```
Grid_home\bin\cluvfy stage -pre dbcfg -n node_list -d Oracle_home [-verbose]
```

In the preceding syntax example, the variable *Grid_home* is the Oracle Grid Infrastructure home, the variable *node_list* is the list of nodes in your cluster, separated by commas, and the variable *Oracle_home* is the path for the Oracle home directory where Oracle Universal Installer (OUI) creates or modifies the database.

You can select the option `-verbose` to receive progress updates as CVU performs its system checks, and detailed reporting of the test results.

If the CVU summary indicates that the cluster verification check fails, then review and correct the relevant system configuration steps, and run the test again.

Example 3-3 Using CVU Prior to Verify Your System is Prepared for an Oracle RAC Installation

To verify that your system is prepared for an Oracle Database with Oracle RAC installation on a two-node cluster with nodes `node1` and `node2`, with the Grid home `C:\app\19.0.0\grid`, and with the Oracle home path `C:\app\oracle\product\19.0.0\dbhome_1`, enter the following command:

```
C:\app\19.0.0\grid\bin> cluvfy stage -pre dbcfg -n node1,node2 \
-d C:\app\oracle\product\19.0.0\dbhome_1
```

Tasks to Complete Before Using DBCA to Create an Oracle RAC Database

Before you can create an Oracle RAC database using Oracle Database Configuration Assistant, you must configure your system to meet the software requirements, if this was not done as part of the Oracle Grid Infrastructure installation.

- [Decide on a Naming Convention to Use for Your Oracle RAC Database](#)
The global database name for an Oracle RAC database must meet the naming requirements. The global database name consists of the database name and the domain name.
- [Configure Shared Storage for the Oracle RAC Database](#)
Before starting Oracle Database Configuration Assistant (DBCA) to configure an Oracle RAC database, you must install Oracle Grid Infrastructure for a cluster, and configure shared storage areas for Oracle RAC files.
- [Obtain the Password for the Oracle Home User](#)
If you use Oracle Database Configuration Assistant to create an Oracle Home User, then create an Oracle Home User password.

Decide on a Naming Convention to Use for Your Oracle RAC Database

The global database name for an Oracle RAC database must meet the naming requirements. The global database name consists of the database name and the domain name.

1. Choose a name for your database that has the following characteristics:

- a. Up to 30 characters in length
- b. Begins with an alphabetic character
2. Determine the domain name portion of the global database name, that satisfies these requirements:
 - a. Is up to 128 characters in length
 - b. Contains only alphabetic and numeric characters, and the period (.) character
3. Determine the `ORACLE_SID` values for each instance.

The maximum number of characters you can use for the SID prefix is 8 characters. Oracle Database Configuration Assistant uses the SID prefix to generate a unique value for the variable `ORACLE_SID` for each instance. The SID prefix must begin with an alphabetic character.

Configure Shared Storage for the Oracle RAC Database

Before starting Oracle Database Configuration Assistant (DBCA) to configure an Oracle RAC database, you must install Oracle Grid Infrastructure for a cluster, and configure shared storage areas for Oracle RAC files.

1. Log in as a user with SYSASM system privileges.
Storage administration tasks require the SYSASM system privileges, which are granted to members of the OSASM operating system group. This group may not be the same as the OSDBA group, whose members are granted the SYSDBA system privileges.
2. On Windows-based systems, if you plan to use Oracle ASM storage, then before you use DBCA to create a database, you must perform the following steps:
 - a. Create logical drives either on extended partitions or primary partitions.
 - b. Delete the drive letters for these partitions on all nodes.
 - c. Stamp these partitions with `asmtoolg`.
 - d. After you have configured the disks to be used by Oracle ASM, you must create the disk groups that will be used by the database.

You can create disk groups by using Oracle Automatic Storage Management Configuration Assistant (ASMCA), SQL*Plus, or Oracle Enterprise Manager.

See Also:

- *Oracle Database Installation Guide for Microsoft Windows* for more information about `asmtoolg`
- *Oracle Grid Infrastructure Installation Guide* for your platform more information on shared storage configuration requirements
- *Oracle Automatic Storage Management Administrator's Guide* for more information about creating disk groups

Obtain the Password for the Oracle Home User

If you use Oracle Database Configuration Assistant to create an Oracle Home User, then create an Oracle Home User password.

Log in as the user that installed the Oracle Database software and perform the following task:

- If an Oracle Home User was specified during installation, then obtain that user password.



See Also:

Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows for more information about the Oracle Home User.

Selecting DBCA Options to Create an Oracle RAC or Oracle RAC One Node Database

Review this information to use Database Configuration Assistant (DBCA) to create Oracle RAC or Oracle RAC One Node Database.



Note:

You can no longer set up email notification for Oracle RAC databases either from DBCA or Oracle Universal Installer (OUI).

- [Starting DBCA on Microsoft Windows Systems](#)
You can either start the Oracle Database Configuration Assistant (DBCA) utility from the command line or from the Windows Start menu.
- [Cluster Detection and Node Selection when Using DBCA](#)
When you start Oracle Database Configuration Assistant (DBCA), it automatically shows options for Oracle RAC if it detects from the central Oracle Inventory that the Oracle Home is enabled for Oracle RAC.
- [Using DBCA to Select Storage to Use With Oracle RAC Database](#)
You can choose to use either Oracle ASM Disk groups or a shared file system as storage for Oracle RAC database files.
- [Using DBCA to Specify Database Initialization Parameters for Oracle RAC](#)
Set the `CLUSTER_DATABASE_INSTANCES` parameter to the expected number of instances.
- [Actions Performed By DBCA for Oracle RAC Databases](#)
Review this information to understand about Oracle Database Configuration Assistant (DBCA) actions during Oracle RAC database creation.

Related Topics

- [Using Oracle DBCA with Oracle RAC or Oracle RAC One Node](#)
Oracle Database Configuration Assistant (DBCA) is a tool for creating and configuring an Oracle database.

Starting DBCA on Microsoft Windows Systems

You can either start the Oracle Database Configuration Assistant (DBCA) utility from the command line or from the Windows Start menu.



Note:

To run DBCA, you do not have to set operating system environment variables `ORACLE_HOME` to the Oracle RAC database home, or `ORACLE_UNQNAME` to the database unique name.

1. Log in as an Administrator user.

The user must also be a member of `ORA_DBA` or `ORA_Homename_DBA` group and must also be a member of `ORA_ASMDBA` if Oracle ASM is used as storage for the Oracle RAC database.

You are prompted to enter the password for the Oracle Home User if you are administering an Administrator-managed Oracle RAC database and chose not to store the password in an Oracle Wallet.

2. To start DBCA from the command line:

- a. Open a command prompt window.
- b. Navigate to the `Oracle_home\bin` directory.
- c. Enter the command `dbca`.

3. To start DBCA from the Start menu:

- a. Click **Start**.
- b. Select **Programs**.
- c. Under **Programs**, select **Oracle - Oracle_home name**.
- d. Select **Configuration and Migration Tools**.
- e. Select **Database Configuration Assistant**.

4. After you have started DBCA, to create an Oracle RAC database, you select the following:

- **Create Database** on the Database Operation/Welcome page
- **Advanced Configuration** on the Creation Mode page
- **Oracle RAC database** on the Deployment Type page

Related Topics

- [Deleting an Oracle RAC Database Using DBCA](#)
Deleting an Oracle RAC database using Oracle Database Configuration Assistant (DBCA) involves deleting the database and database objects.

Cluster Detection and Node Selection when Using DBCA

When you start Oracle Database Configuration Assistant (DBCA), it automatically shows options for Oracle RAC if it detects from the central Oracle Inventory that the Oracle Home is enabled for Oracle RAC.

If DBCA does not detect the Oracle home as an Oracle RAC home, check that the Oracle Universal Installer (OUI) inventory is correctly located in the directory `C:\Program Files\Oracle\Inventory`, and that the `oraInventory` file is not corrupted. Also, perform clusterware diagnostics by using the following CVU command syntax:

```
Grid_home\bin\cluvfy\cluvfy.bat stage -post crsinst -n nodelist
```

When using DBCA, if nodes that are part of your cluster installation do not appear on the Node Selection page, then run the `Opatch lsinventory` command to perform inventory diagnostics. Also use CVU to perform clusterware diagnostics.

Example 3-4 Performing Clusterware Diagnostics If DCBA Fails To Detect A Two-Node Cluster

If the Grid Home is `D:\app\19.0.0\grid`, and the nodes are named `node1` and `node2`, then run the following command to perform clusterware diagnostics:

```
D:\app\19.0.0\grid\bin> cluvfy stage -post crsinst -n node1,node2
```

Using DBCA to Select Storage to Use With Oracle RAC Database

You can choose to use either Oracle ASM Disk groups or a shared file system as storage for Oracle RAC database files.

- On the Specify Database Storage Options page, if you do not see the diskgroups in Oracle Database Configuration Assistant (DBCA), then either Oracle ASM is not configured, or the diskgroups are not mounted.
You can create diskgroups using Oracle Automatic Storage Management Configuration Assistant (ASMCA) in the Grid Infrastructure home before starting DBCA.
- If you are using Oracle ASM, then you can select the Fast Recovery Area and size on the Specify Database Storage Options page.

When using Oracle ASM, the Fast Recovery Area defaults to the Oracle ASM Disk Group.

Using DBCA to Specify Database Initialization Parameters for Oracle RAC

Set the `CLUSTER_DATABASE_INSTANCES` parameter to the expected number of instances.

- On the Initialization Parameters page, if you intend to add more nodes in your cluster than you have during the current Oracle Database Configuration Assistant session, then click **All Initialization Parameters**, and change the parameter `CLUSTER_DATABASE_INSTANCES` to the total number of nodes that you plan to add to the cluster.
- In addition, if you click **All Initialization Parameters**, note that if your global database name is longer than 8 characters, then the database name value (in the `DB_NAME` parameter) is truncated to the first 8 characters, and the `DB_UNIQUE_NAME` parameter value is set to the global name.



See Also:

Oracle Database Administrator's Guide for information about initialization parameters

Actions Performed By DBCA for Oracle RAC Databases

Review this information to understand about Oracle Database Configuration Assistant (DBCA) actions during Oracle RAC database creation.

After you respond to DBCA prompts, review the Summary dialog information and click **OK**, DBCA performs several actions.

- Creates an Oracle RAC database, and its instances
- Creates the Oracle RAC data dictionary views
- Starts the Oracle services if you are on a Windows-based platform
- Starts the Oracle Clusterware high availability services
- Starts the database instances across cluster nodes

Caution:

After you have created the Oracle RAC database, if you decide to install additional Oracle Database products in the Oracle RAC database you have created, then before you attempt to install the products, you must stop all processes running in the Oracle RAC database homes.

You must stop all processes running in the Oracle RAC homes so that Oracle Universal Installer can relink certain executables and libraries. Refer to "[Preparing to Upgrade an Existing Oracle RAC Database](#)" for additional information.

Using DBCA to Create an Oracle RAC One Node Database

If you have selected to install only the Oracle RAC software on cluster nodes, then you can use Oracle Database Configuration Assistant (DBCA) to configure Oracle RAC One Node.

After installation of Oracle Real Application Clusters (Oracle RAC) software, start Oracle Database Configuration Assistant (DBCA).

1. From the Database Operation page, select the option **Create Database**.
2. On the Creation Mode page, select **Advanced Configuration**.
3. On the Deployment Type page, select **Oracle RAC One Node database**.

Selecting one node deploys Oracle RAC One Node on a single node. Oracle recommends that you select all nodes in the cluster to which you want Oracle RAC One Node to be able to fail over.

When you create an administrator-managed Oracle RAC One Node database, note that while the database is started on only one of the pool of nodes you installed the binaries, all the candidate servers are placed into the Generic server pool. If the servers are not already in Generic or Free, then this may result in stopping resources that are running on candidate servers.

When you use DBCA to create an Oracle RAC One Node database, a failover service is automatically configured.

Deleting an Oracle RAC Database Using DBCA

Deleting an Oracle RAC database using Oracle Database Configuration Assistant (DBCA) involves deleting the database and database objects.

DBCA first deletes the database, and then removes the database's initialization parameter files, instances, Optimal Flexible Architecture (OFA) structure, and the Oracle network configuration for the database.

1. Start DBCA on one of your cluster nodes.
DBCA displays the Operations page, displaying different database deployment options.
2. Select **Delete a database**, and click **Next**.
DBCA displays a list of all Oracle RAC and single-instance databases running from the Oracle home where DBCA is run.
3. If your user ID and password are not operating-system authenticated, then the List of Cluster Databases page displays the user name and password fields. If these fields appear, then enter a user ID and password for a user account that has SYSDBA privileges.
4. Select the database to delete, and click **Finish**.
After you click Finish, DBCA displays a dialog box to confirm the database and instances that you have configured DBCA to delete.
5. Click **OK** to begin the deletion of the database and its associated files, services, and environment settings, or click **Cancel** to stop the operation.

When you click **OK**, DBCA continues the operation and deletes all the associated instances for this database. DBCA also removes the parameter files and password files.

At this point, you have accomplished the following:

- Deleted the selected Oracle RAC database from the cluster
- Deleted the selected Oracle RAC Database Oracle services on Windows-based platforms
- Deleted high availability services assigned to the Oracle RAC database
- Deleted the Oracle Net configuration for the Oracle RAC database
- Deconfigured Oracle Enterprise Manager for the Oracle RAC database
- Deleted the OFA directory structure for that Oracle RAC database from the cluster
- Deleted the Oracle RAC database data files

Related Topics

- [Starting DBCA on Microsoft Windows Systems](#)
You can either start the Oracle Database Configuration Assistant (DBCA) utility from the command line or from the Windows Start menu.

Creating an Oracle RAC Database on Direct NFS

There are different configuration processes you must perform when installing and create an Oracle RAC database that uses Direct NFS (dNFS) for the database files.

- [Performing a Software-Only Installation of Oracle Database](#)
In a software-only installation you install the Oracle Database software but do not create a database as part of the installation process.

- [Using Oracle ASMCA to Configure an ACFS Mount Point](#)
Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA) enables you to create an Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point which is used in the "common file location" step of Oracle Database Configuration Assistant (DBCA).
- [Using Oracle DBCA to Create and Configure the Oracle RAC Database](#)
Use Oracle Database Configuration Assistant (DBCA) to create an Oracle Real Application Clusters (Oracle RAC) database that uses Direct NFS for datafile storage.
- [Enabling and Configuring Direct NFS](#)
You must manually enable the Direct NFS option after installing the Oracle Database software.
- [Using Oracle ASMCA to Remove the ACFS Mount Point](#)
Now you have configured the mount point using Direct NFS, you can remove the Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point using Oracle Automatic Storage Management Configuration Assistant (ASMCA).

Performing a Software-Only Installation of Oracle Database

In a software-only installation you install the Oracle Database software but do not create a database as part of the installation process.

1. Start Oracle Universal Installer (OUI) by running `setup.exe` from the Oracle home directory.
2. On the Select Configuration Option screen select **Set Up Software Only**.
3. Select the nodes on which you want to install the database software.
4. Select the database edition to install.
5. Specify an Oracle Home user, or choose to use a Windows-built in user for the software installation owner.
6. On the Specify Installation Location screen, enter a path to the Oracle base directory and the software location (Oracle home directory).
7. On the Summary screen, verify your selections, then click **Install**.

Using Oracle ASMCA to Configure an ACFS Mount Point

Oracle Automatic Storage Management Configuration Assistant (Oracle ASMCA) enables you to create an Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point which is used in the "common file location" step of Oracle Database Configuration Assistant (DBCA).

When creating Oracle ACFS file systems on Windows, run ASMCA as a Windows domain user who is also an administrator of the computer.

1. From the `Grid_home/bin` directory, run `asmca.exe` to start the ASMCA.
2. Select the **Disk Groups** tab.
3. Right-click **Disk Group Name** and select **Create ACFS for Database use**.
4. In the Create ACFS for Database window specify the mount point location, volume name, and size, then click **OK**.

For example, you can specify the following:

- Mount Point: `C:\oradatamnt`

- Volume Name: `dbnfs`
 - Size (GB): `70`
5. Click **OK** in the informational pop-up window that appears.
This window summarizes the actions performed by ASMCA.
 6. Select the **ASM Cluster File Systems** tab.
The mount point you just created is displayed on this page.

Related Topics

- *Oracle Automatic Storage Management Administrator's Guide*

Using Oracle DBCA to Create and Configure the Oracle RAC Database

Use Oracle Database Configuration Assistant (DBCA) to create an Oracle Real Application Clusters (Oracle RAC) database that uses Direct NFS for datafile storage.

1. From the `Oracle_home\bin` directory, run `dbca.exe` to start the Database Configuration Assistant.
2. On the Database Operation screen, select **Create Database**.
3. On the Creation Mode screen, select **Advanced Configuration**.
4. On the Database Template screen, select **Oracle Real Application Clusters (RAC) database** for the Database Type.

For the Configuration Type, you can choose either Policy-Managed or Administrator-Managed. Select the template most appropriate for the type of database you want to create.
5. For the next four screen, make selections and provide information that best meet your business requirements.
6. Perform the following steps on the Storage Locations screen:
 - a. Select **File System** for the Database files Storage type.
 - b. Select the option **Use Common Location for All Database Files**.
 - c. In the File location field, specify the location of the ACFS mount point, for example, `C:\oradatamnt`.
7. On the Specify Database Options screen, choose any additional configuration you want for your database.
8. On the Configurations Options screen, use the default settings, or provide customized values for the initialization parameters.
9. On the Creation Options screen, select the option **Generate Database Creation Scripts**. Specify a destination directory for the script file, or use the default value.
10. After the Prerequisite checks complete, on the Summary screen, minimize the installation window. **DO NOT** click Finish at this point.
11. Enable the Direct NFS option.
12. Remove the virtual mount point you created with Oracle Automatic Storage Management Configuration Assistant (ASMCA).
13. Create all the directories needed locally on each node as well as on the NFS server.

For this example, you can create the following, where `orcl` represents the database SID and `pdb1` represents the Pluggable Database (PDB) name:

- On each node, create the directory `c:\oracle\oradatamnt\orcl\pdb1`
- On the NFS server, create the directory `/export/abcd/orcl/pdb1`

14. Return to the DBCA window and click **Finish**.
15. Run the generated scripts on the cluster node to create the database.
16. Map a drive letter to a Common Internet File System (CIFS) share on the NFS server that represents the location of the database files.

Use a command similar to the following:

```
NET USE * \\filer\vol0\orcl
```

After you complete this step, both Oracle and Microsoft Windows operating system can access the location where the database files reside. Oracle is using DNFS, but Microsoft Windows operating system uses CIFS to access the same location on the NFS server.

17. Verify Direct NFS is configured for the database.
 - a. Start SQL*Plus.
 - b. Connect to the newly created database as a DBA user.
 - c. Run the following SQL command:

```
SELECT * FROM v$dnfs_servers;
```

Enabling and Configuring Direct NFS

You must manually enable the Direct NFS option after installing the Oracle Database software.

1. Run the program `Oracle_home\bin\enable_dnfs.bat`.
2. Create an `oranfstab` file.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*

Using Oracle ASMCA to Remove the ACFS Mount Point

Now you have configured the mount point using Direct NFS, you can remove the Oracle Automatic Storage Management Cluster File System (Oracle ACFS) mount point using Oracle Automatic Storage Management Configuration Assistant (ASMCA).

When creating Oracle ACFS file systems on Windows, run ASMCA as a Windows domain user who is also an administrator of the computer.

1. From the `Grid_home/bin` directory, run `asmca.exe` to start the ASMCA.
2. Select the **ASM Cluster File System** tab.
3. Select the mount point created earlier (`C:\oradatamnt`), then click **Dismount All**.
4. Select the **Volumes** tab.
5. Right-click the mount point and then select **Delete**.

4

Oracle Real Application Clusters Postinstallation Procedures

After you have installed the Oracle Database 19c with Oracle Real Application Clusters (Oracle RAC) software, there are postinstallation tasks to complete.

- [Required Postinstallation Tasks](#)
Perform these tasks after completing your installation.
- [Recommended Postinstallation Tasks](#)
Oracle recommends that you complete these tasks after completing an Oracle RAC installation.
- [Product-Specific Postinstallation Tasks](#)
Many Oracle products and options must be configured before you use them for the first time.
- [Configuring the Oracle Home User](#)
Under certain circumstances you may have to perform additional configuration steps for the Oracle Home user.
- [Oracle Configuration Manager Postinstallation Configuration for Oracle RAC](#)
If you have installed Oracle Configuration Manager, then you must run a script to create a database account to collect database configuration collections.
- [Enabling and Disabling Database Options After Installation](#)
When you install Oracle Database, some options are enabled and the others disabled. You can view the enabled Oracle Database options by querying the `V$OPTION` view using `SQL*Plus`.

Required Postinstallation Tasks

Perform these tasks after completing your installation.

- [Determine If Any Patches Are Required For Your New Software](#)
If you did not choose to download software options during installation, then after installing Oracle RAC, verify if there are any patches needed for your system.
- [Configure Exceptions for the Windows Firewall](#)
If the Windows Firewall feature is enabled on one or more nodes of your Oracle RAC cluster, then you must create exceptions for Oracle RAC applications and ports.
- [Create the OraMTS Service for Microsoft Transaction Server](#)
To enable client components to use Oracle databases as resource managers in Microsoft application coordinated transactions, create the Oracle Service for Microsoft Transaction Server (OraMTS).
- [Recompiling All Invalid Objects](#)
Identify and recompile invalid objects on the CDB and PDBs using the `catcon` utility to run `utlrp.sql` after you install, patch, or upgrade a database.

- [Configuring Services on Oracle RAC with CDBs](#)
During installation, if you select a multitenant container database (CDB), and configure pluggable databases (PDBs), then Oracle recommends that you add services to the PDBs after installation.

Determine If Any Patches Are Required For Your New Software

If you did not choose to download software options during installation, then after installing Oracle RAC, verify if there are any patches needed for your system.

- To determine if any patches are required for your system, review the *Oracle Database Release Notes*.

Related Topics

- *Oracle Database Release Notes*

Configure Exceptions for the Windows Firewall

If the Windows Firewall feature is enabled on one or more nodes of your Oracle RAC cluster, then you must create exceptions for Oracle RAC applications and ports.

Enabling the Windows Firewall feature on one or more nodes in your cluster blocks virtually all TCP network ports to incoming connections. As a result, any Oracle product that listens for incoming connections on a TCP port does not receive any of those connection requests and the clients making those connection requests report errors.

Review the instructions in *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows* for details on how to configure exceptions for the Windows Firewall, if you have not done so already as part of the Oracle Grid Infrastructure installation.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows*

Create the OraMTS Service for Microsoft Transaction Server

To enable client components to use Oracle databases as resource managers in Microsoft application coordinated transactions, create the Oracle Service for Microsoft Transaction Server (OraMTS).

OraMTS acts as a proxy for the Oracle database to the Microsoft Distributed Transaction Coordinator (MSDTC). As a result, OraMTS provides client-side connection pooling and allows client components that leverage Oracle to participate in promotable and distributed transactions. In addition, OraMTS can operate with Oracle databases running on any operating system, given that the services themselves are run on Windows.

On releases earlier than Oracle Database 12c, the OraMTS service was created as part of a software-only installation. Starting with Oracle Database 12c, you must use a configuration tool to create this service.

To create the OraMTS service after performing a software-only installation of Oracle RAC or after adding a node to an existing cluster, perform the following steps:

1. Open a command window.
2. Change directories to `%ORACLE_HOME%\bin`.

3. Run the `OraMTSctl` utility to create the OraMTS Service, where `host_name` is a list of nodes on which you want to create the service:

```
C:\..\bin> oramtsctl.exe -new -host host_name
```

**See Also:**

Oracle Services for Microsoft Transaction Server Developer's Guide for Microsoft Windows for more information about installing OraMTS

Recompiling All Invalid Objects

Identify and recompile invalid objects on the CDB and PDBs using the `catcon` utility to run `utlrlp.sql` after you install, patch, or upgrade a database.

**Note:**

If you upgraded using the AutoUpgrade utility, then AutoUpgrade automatically takes care of this task during the upgrade. You do not need to perform this task.

Oracle recommends that you use the `catcon.pl` utility to run `utlrlp.sql` on all containers in your container database (CDB). The `utlrlp.sql` script recompiles all invalid objects. Run the script immediately after installation, to ensure that users do not encounter invalid objects.

1. Change directory to `Oracle_home/rdbms/admin`. For example

```
$ cd $ORACLE_HOME/rdbms/admin
```

2. Use the `catcon.pl` script in the Oracle home to run `utlrlp.sql`. For example:

```
$ORACLE_HOME/perl/bin/perl catcon.pl --n 1 --e --b utlrlp --d ''.'''  
utlrlp.sql
```

Note the following conditions of this use case:

- `--n` parameter: is set to 1, so the script runs each PDB recompilation in sequence.
- `--e` parameter: turns echo on.
- `--b` parameter: Sets the log file base name. It is set to `utlrlp`.

Expect a time delay for the serial recompilation of PDBs to complete. Depending on the number of PDBs that you are upgrading, the recompilation can extend significantly beyond the time required for the upgrade scripts to complete.

The `utlrlp.sql` script automatically recompiles invalid objects in either serial or parallel recompilation, based on both the number of invalid objects, and on the number of CPUs available. CPUs are calculated using the number of CPUs (`cpu_count`) multiplied by the number of threads for each CPU (`parallel_threads_per_cpu`). On Oracle Real Application Clusters (Oracle RAC), this number is added across all Oracle RAC nodes.

For more information about catcon utility syntax and options, refer to *Oracle Multitenant Administrator's Guide*.

Related Topics

- Syntax and Parameters for catcon.pl

Configuring Services on Oracle RAC with CDBs

During installation, if you select a multitenant container database (CDB), and configure pluggable databases (PDBs), then Oracle recommends that you add services to the PDBs after installation.

If you do not add services to PDBs, and then the Oracle RAC One Node CDB fails over to another node, or you manually relocate the CDB to another node, then by default, all PDBs associated with that CDB that do not have registered services are restarted in `MOUNTED` state. PDBs are opened in Read Write mode after failover or relocation only after you have configured the PDBs to have associated services. If you have not associated services to PDBs, then the PDBs remains in `MOUNTED` state when the CDB instance restarts.

- Use the following `srvctl` command syntax, where *cdbname* is the name of the CDB, *service_name* is the name of the service, and *pdbname* is the name of the PDB:

```
srvctl add service -d cdbname -s service_name -pdb pdbname
```

After you add services to your PDBs, if you relocate the CDB with which the PDBs are associated, or the CDB fails over, then the PDBs associated with that CDB automatically open in `Read Write` state.

Recommended Postinstallation Tasks

Oracle recommends that you complete these tasks after completing an Oracle RAC installation.

- [Setting Up Additional User Accounts](#)
You can set up additional user accounts to manage your database.
- [Setting the Oracle User Environment Variables](#)
Unlike on other platforms, do not set `ORACLE_HOME` as a fixed environment variable when running Oracle Database on Windows operating systems. This is because the Oracle software determines where executable files reside at run time.
- [About Installing Oracle Autonomous Health Framework](#)
Install the latest version of Oracle Autonomous Health Framework to perform proactive health checks and collect diagnostics data for the Oracle software stack.
- [About Using CVU Cluster Healthchecks After Installation](#)
You can use the `CVU healthcheck` command to check your Oracle Clusterware and Oracle Database installations for their compliance with mandatory requirements and best practices guidelines, and to ensure that they are functioning properly.

Setting Up Additional User Accounts

You can set up additional user accounts to manage your database.

Refer to *Oracle Database Administrator's Reference for Microsoft Windows* and *Oracle Database Security Guide* for information about setting up additional optional user accounts.

Setting the Oracle User Environment Variables

Unlike on other platforms, do not set `ORACLE_HOME` as a fixed environment variable when running Oracle Database on Windows operating systems. This is because the Oracle software determines where executable files reside at run time.

When you invoke an Oracle executable program on Windows, for example `sqlplus.exe`, the `ORACLE_HOME`, `ORACLE_BASE`, and `ORACLE_SID` variables are determined by the `PATH` environment variable and the location of the executable program (which Oracle home it resides in). To use SQL*Plus to manage a different database or Oracle ASM instance, click the Windows Start button, select the correct Oracle Home for the instance you want to manage, and then select the SQL*Plus utility.

You can use Oracle Universal Installer (OUI) to specify an Oracle home as the default Oracle home and update the `PATH` environment variable to point to that Oracle home.



See Also:

"[Multiple Oracle Home Directories on Windows](#)" for detailed instructions on how to change the default Oracle home.

About Installing Oracle Autonomous Health Framework

Install the latest version of Oracle Autonomous Health Framework to perform proactive health checks and collect diagnostics data for the Oracle software stack.

Oracle Autonomous Health Framework includes the functionality from Oracle ORAchk, Oracle EXAchk, and Oracle Trace File Analyzer (TFA). Oracle Autonomous Health Framework extends health check coverage to the entire Oracle software stack, based on critical and reoccurring problems. Oracle Autonomous Health Framework proactively scans for known problems with Oracle products and deployments, including the following:

- Standalone Oracle Database
- Oracle Grid Infrastructure
- Oracle Real Application Clusters
- Maximum Availability Architecture (MAA) Validation
- Upgrade Readiness Validations
- Oracle GoldenGate

Oracle Autonomous Health Framework is pre-installed with Oracle Database. However, Oracle recommends that you update to the latest version of Oracle Autonomous Health Framework by downloading and installing it from My Oracle Support Note 2550798.1.

<https://support.oracle.com/epmos/faces/DocContentDisplay?id=2550798.1&parent=DOCUMENTATION&sourceId=USERGUIDE>

About Using CVU Cluster Healthchecks After Installation

You can use the `CVU healthcheck` command to check your Oracle Clusterware and Oracle Database installations for their compliance with mandatory requirements and best practices guidelines, and to ensure that they are functioning properly.

Syntax

```
cluvfy comp baseline -collect {all|cluster|database|asm} [-n node_list|-allnodes] [-d oracle_home|-db db_unique_name]
[-bestpractice|-mandatory] [-binlibfilesonly]
[-reportname reportname] [-savedir directory_path]
```

Options

Option	Description
<code>-collect [cluster database]</code>	Use this option to specify that you want to perform checks for Oracle Clusterware (cluster) or Oracle Database (database). If you do not use the collect flag with the healthcheck command, then <code>cluvfy comp healthcheck</code> performs checks for both Oracle Clusterware and Oracle Database.
<code>-db db_unique_name</code>	Use this flag to specify checks on the database unique name that you enter after the <code>-db</code> option. CVU uses JDBC to connect to the database as the user <code>CVUSYS</code> to verify various database parameters. For this reason, if you want checks to be performed for the database you specify with the <code>-db</code> option, then you must first create the <code>CVUSYS</code> user on that database, and grant that user the CVU-specific role, <code>CVUSAPP</code>. You must also grant members of the <code>CVUSAPP</code> role <code>SELECT</code> permissions on system tables. The SQL script <code>cvusys.sql</code> is included in the <code>CVU_home\cv\admin</code> directory to facilitate the creation of this user. Use this SQL script to create the <code>CVUSYS</code> user on all the databases that you want to verify using CVU. If you use the <code>-db</code> option but do not provide a database unique name, then CVU discovers all the Oracle databases on the cluster. To perform best practices checks on these databases, you must create the <code>CVUSYS</code> user on each database, and grant that user the <code>CVUSAPP</code> role with the <code>SELECT</code> privileges needed to perform the best practice checks.

Option	Description
<code>[-bestpractice -mandatory] [-deviations]</code>	• Use the <code>-bestpractice</code> option to specify best practice checks • Use the <code>-mandatory</code> option to specify mandatory checks • Add the <code>-deviations</code> option to specify that you want to see only the deviations from either the best practice recommendations or the mandatory requirements • If you specify neither <code>-bestpractice</code> or <code>-mandatory</code>, then both best practices and mandatory requirements are displayed. You can specify either the <code>-bestpractice</code> or <code>-mandatory</code> option, but not both options.
<code>-html</code>	Use the <code>-html</code> option to generate a detailed report in HTML format. If you specify the <code>-html</code> option, and a browser that is recognized by CVU is available on the system, then the browser is started and the report is displayed on the browser when the checks are complete. If you do not specify the <code>-html</code> option, then the detailed report is generated in a text file.
<code>-save [-savedir dir_path]</code>	Use the <code>-save</code> or <code>-save -savedir</code> options to save validation reports (<code>cvuchecdkreport_timestamp.txt</code> and <code>cvuchecdkreport_timestamp.htm</code>), where <i>timestamp</i> is the time and date of the validation report. If you use the <code>-save</code> option by itself, then the reports are saved in the path <code>CVU_home\cv\report</code>, where <i>CVU_home</i> is the location of the CVU executable files. If you use the options <code>-save -savedir</code>, and enter a path where you want the CVU reports saved, then the CVU reports are saved in the path you specify.

Example 4-1 Running a Cluster Healthcheck After the Software Installation

To run a healthcheck for your Oracle Grid Infrastructure cluster, to check for any deviations from best practices, and display the results in HTML format, use the following command:

```
C:\> cd app\19.0.0\grid\bin
C:\..bin> cluvfy comp healthcheck -html -bestpractice -deviations
```

Example 4-2 Running a Healthcheck for Oracle RAC Database

To run a healthcheck for your Oracle RAC cluster, to check best practices recommendations and mandatory requirements, and display the results in HTML format, use the following command:

```
C:\> cd app\19.0.0\grid\bin
C:\..bin> cluvfy comp healthcheck -html
```

Product-Specific Postinstallation Tasks

Many Oracle products and options must be configured before you use them for the first time.

Before using individual Oracle Database 19c products or options, refer to the appropriate manual in the product documentation library, which is available online at: <http://docs.oracle.com>.

Refer to the following topics for information about configuring various products and features after installation.

- [Configuring Oracle Database Vault](#)
Oracle Universal Installer (OUI) installs Oracle Database Vault by default when you install the Oracle RAC software, but requires additional configuration steps.
- [Configuring Oracle Label Security](#)
After installation, you must configure Oracle Label Security in a database before you use it.
- [Configuring the OraClrAgnt Service for Oracle Database Extensions for .NET](#)
Oracle Database Extensions (ODE) for .NET depends on a Windows service to operate properly. This service is called the OraClrAgnt service.
- [Configuring Oracle XML DB](#)
Oracle XML DB is a required component of the Oracle Database installation. However, you must manually configure the FTP and HTTP ports for Oracle XML DB.
- [Configure Storage for External Tables, Shared Files, or Directory Objects](#)
If your Oracle RAC database uses files that are external to the database, then the external files must be located on shared storage that is accessible to all nodes.

Configuring Oracle Database Vault

Oracle Universal Installer (OUI) installs Oracle Database Vault by default when you install the Oracle RAC software, but requires additional configuration steps.

1. Register Oracle Database Vault with the Oracle RAC database.
2. Create the Database Vault Owner user and, optionally, the Database Vault Account Manager administrative user accounts.

Related Topics

- [Perform Postinstallation Configuration for Oracle Database Vault](#)
After you install the Oracle Database Vault option, you can be required to make additional changes to your database.

Configuring Oracle Label Security

After installation, you must configure Oracle Label Security in a database before you use it.

You can configure Oracle Label Security in two ways: with Oracle Internet Directory integration and without Oracle Internet Directory integration.

Table 4-1 Configuration Options and Requirements for Oracle Label Security

Configuration	Requirements
With Oracle Internet Directory integration	To configure Oracle Label Security with Oracle Internet Directory integration, Oracle Internet Directory must be installed in your environment and the Oracle database must be registered in the directory.
Without Oracle Internet Directory integration	If you configure Oracle Label Security (OLS) without Oracle Internet Directory integration, then you cannot configure it to use Oracle Internet Directory at a later stage. To configure Oracle Label Security with Oracle Internet Directory on your database at a later time, you must remove the OLS option on the database and configure the OLS with Oracle Internet Directory integration option.



See Also:

Oracle Label Security Administrator's Guide for information about configuring Oracle Label Security.

Configuring the OraClrAgnt Service for Oracle Database Extensions for .NET

Oracle Database Extensions (ODE) for .NET depends on a Windows service to operate properly. This service is called the OraClrAgnt service.

In versions of Oracle Database prior to Oracle Database 12c, this CLR service was created automatically by the installer.

- After installation you use the `OraClrCtl.exe` utility to create, start, stop, and delete the OraClrAgnt service.

When you use the `OraClrCtl.exe` utility to create the service, a new service is created named `OracleHomenamClrAgent`, where *Homenam* represents an Oracle Home name. The OraClrAgnt service is configured by this tool using the Oracle Home User account specified during the Oracle Database installation.

 See Also:

Oracle Database Extensions for .NET Developer's Guide for Microsoft Windows for more information about using the `OraClrCtl.exe` tool and installing and configuring the `OraClrAgnt` service

Configuring Oracle XML DB

Oracle XML DB is a required component of the Oracle Database installation. However, you must manually configure the FTP and HTTP ports for Oracle XML DB.

- Refer to *Oracle XML DB Developer's Guide* for information on configuring the FTP and HTTP protocols for Oracle XML DB.

Related Topics

- *Oracle XML DB Developer's Guide*

Configure Storage for External Tables, Shared Files, or Directory Objects

If your Oracle RAC database uses files that are external to the database, then the external files must be located on shared storage that is accessible to all nodes.

- Each node must use the same mount point to access the file.
Acceptable shared file systems include Database File System (DBFS), Oracle Automatic Storage Management Cluster File System (Oracle ACFS), or a supported network file system (NFS) using the Direct NFS Client.
- The database directory object used to write and read files external to the database must point to a shared storage location.
- Each node must use the same mount point for the same shared storage location.

For example, each node can have a directory object called `DPUMP` for the mount point `C:\app\acfsmounts\dpump`, which accesses Oracle ACFS shared storage.

 Note:

There is no checking of the contents of the external files or directory object specified as part of the external table to ensure that the directory contents are consistent on each node. To avoid unpredictable results, you must ensure that the same file is accessed from all nodes, or that the same file is used on all nodes.

Configuring the Oracle Home User

Under certain circumstances you may have to perform additional configuration steps for the Oracle Home user.

The additional configuration steps you might need to perform for the Oracle Home user include:

- Creating an OCR wallet for Oracle Home user
- Changing the password for the Oracle Home user

Related Topics

- *Oracle Database Administrator's Reference for Microsoft Windows*

Oracle Configuration Manager Postinstallation Configuration for Oracle RAC

If you have installed Oracle Configuration Manager, then you must run a script to create a database account to collect database configuration collections.

You must create this database account in both *Connected* and *Disconnected* modes. The database account stores the PL/SQL procedures that collect the configuration information, and the account owns the database management system (DBMS) job that performs the collection. After the account has been set up, the account is locked because login privileges are no longer required.

- Run the script `installCCRSQL.exe`.

The script `installCCRSQL.exe` creates an Oracle Configuration Manager user and loads the PL/SQL procedure into the database defined by the `ORACLE_SID` environment variable. For Oracle RAC, you must run the database script against only one instance, such as the local instance on which you performed the installation. However, Oracle Configuration Manager must be installed in all instance homes.

You can also specify the database SID by using the `-s` option in the command line, as in the following example, where the SID is `orcl`:

```
%ORACLE_HOME%/ccr/admin/scripts/installCCRSQL.exe collectconfig -s orcl
```

By default, the connection to the database is through OS authentication, `"/as sysdba."` To specify a different SYSDBA user and password, you can use these options:

`-r SYSDBA-USER`: The login name of the **SYSDBA** user

`-p SYSDBA-PASSWORD`: The password for the **SYSDBA** user

Note:

- If you specify the user name without specifying the password (with the `-p` parameter), then the script prompts you to enter the password.
- If you specify only the password without specifying the user name, then the script uses the user `SYS` by default.

Enabling and Disabling Database Options After Installation

When you install Oracle Database, some options are enabled and the others disabled. You can view the enabled Oracle Database options by querying the `V$OPTION` view using SQL*Plus.

If you need to enable or disable a particular database feature for an Oracle home, then use the `chopt` tool. The `chopt` tool is a command-line utility that is located in the `ORACLE_HOME\bin` directory. The syntax for `chopt` is as follows:

```
chopt [ enable | disable] db_option
```

The possible values for *db_option* are described in the following table:

Table 4-2 Database Options for Chopt Tool Command

Value	Description
olap	Oracle OLAP
rat	Oracle Real Application Testing



Note:

The Oracle Advanced Analytics (OAA) feature is enabled by default for Oracle Database. You cannot disable it using the `chopt` tool.

Example 4-3 Running the Chopt Tool

To enable the Oracle Real Application Testing option in your Oracle binary files:

1. Shut down the database with `srvctl` or `SQL*Plus`:

```
srvctl stop database -d myDb
```

2. Stop the database service, `OracleServiceSID`, using the Services program in Control Panel.
3. Run the following commands:

```
cd ORACLE_HOME/bin
chopt enable rat
```

4. Start the database service, `OracleServiceSID`, using the Services program in Control Panel.
5. Start up the database:

```
srvctl start database -d myDb
```

5

Using Server Pools with Oracle RAC

Understand the server pool concepts in Oracle Real Application Clusters (Oracle RAC) environments.

- [Policy-Managed Clusters and Capacity Management](#)
Starting with Oracle Clusterware 11g release 2, resources that Oracle Clusterware manages are contained in logical groups of servers called **server pools**.
- [Oracle RAC Database and Server Pools](#)
Oracle RAC databases support two different management styles and deployment models.
- [About Creating Server Pools for Oracle RAC Databases](#)
You can create a server pool with Oracle Database Configuration Assistant while creating an Oracle RAC database, but Oracle recommends that you create server pools before you deploy database software and databases.
- [Oracle RAC One Node and Server Pools](#)
Oracle RAC One Node supports the use of server pools, with some limitations.

Policy-Managed Clusters and Capacity Management

Starting with Oracle Clusterware 11g release 2, resources that Oracle Clusterware manages are contained in logical groups of servers called **server pools**.

Resources are hosted on a shared infrastructure and are contained within server pools. Resources are no longer defined as belonging to a specific instance or node. Instead, the priority of resource requirements is defined. You can use a cluster configuration policy set to provide dynamic management of cluster policies across the cluster.

- [Server Pools and Server Categorization](#)
You can manage servers dynamically using server pools by identifying servers distinguished by particular attributes, a process called **server categorization**.
- [Server Pools and Policy-Based Management](#)
With policy-based management, database administrators specify the server pool (excluding Generic or Free) in which the database resource runs.
- [How Server Pools Work](#)
Server pools divide the cluster into groups of servers hosting singleton and uniform database services and applications.
- [Default Server Pools](#)
When Oracle Clusterware is installed, two server pools are created automatically: *Generic* and *Free*.



See Also:

Oracle Real Application Clusters Administration and Deployment Guide

Server Pools and Server Categorization

You can manage servers dynamically using server pools by identifying servers distinguished by particular attributes, a process called **server categorization**.

In this way, you can manage clusters made up of heterogeneous nodes.

Server Pools and Policy-Based Management

With policy-based management, database administrators specify the server pool (excluding Generic or Free) in which the database resource runs.

Policy-based management:

- Enables dynamic capacity assignment when needed to provide server capacity in accordance with the priorities you set with policies
- Enables allocation of resources by importance, so that applications obtain the required minimum resources, whenever possible, and so that lower priority applications do not take resources from more important applications
- Ensures isolation where necessary, so that you can provide dedicated servers in a cluster for applications and databases
- Enables policies to be configured to change pools in accordance with business needs or application demand, so that pools provide the right service at the right time

Applications and databases running in server pools do not share resources. Because server pools do not share resources, they isolate resources where necessary, but enable dynamic capacity assignments as required. Together with role-separated management, this capability addresses the needs of organizations that have standardized cluster environments, but allow multiple administrator groups to share the common cluster infrastructure.

Oracle Clusterware efficiently allocates different resources in the cluster. You need only to provide the minimum and maximum number of nodes on which a resource can run, combined with a level of importance for each resource that is running on these nodes.

See Also:

- The Oracle Clusterware resource reference in *Oracle Clusterware Administration and Deployment Guide*
- *Oracle Clusterware Administration and Deployment Guide* for details about managing server pools to respond to business or application demand

How Server Pools Work

Server pools divide the cluster into groups of servers hosting singleton and uniform database services and applications.

Server pools distribute a uniform workload (a set of Oracle Clusterware resources) over several servers in the cluster. For example, you can restrict Oracle databases to run only in certain server pools. When you enable role-separated management, you can grant permission to operating system users to use server pools.

You manage server pools that contain Oracle RAC databases with the Server Control (SRVCTL) utility. Use the Oracle Clusterware Control (CRSCTL) utility to manage all other server pools. Only cluster administrators have permission to create top-level server pools.

Top-level server pools:

- Logically divide the cluster
- Are always exclusive, meaning that one server can only reside in one particular server pool at a certain point in time

Default Server Pools

When Oracle Clusterware is installed, two server pools are created automatically: *Generic* and *Free*.

All servers in a new installation are assigned to the Free server pool, initially. Servers move from Free to newly defined server pools automatically.

- [The Free Server Pool](#)
The Free server pool contains servers that are not assigned to any other server pools.
- [The Generic Server Pool](#)
The Generic server pool stores any Oracle Database that is not policy-managed.

The Free Server Pool

The Free server pool contains servers that are not assigned to any other server pools.

The attributes of the Free server pool are restricted, as follows:

- `SERVER_NAMES`, `MIN_SIZE`, and `MAX_SIZE` cannot be edited by the user
- `IMPORTANCE` and `ACL` can be edited by the user

The Generic Server Pool

The Generic server pool stores any Oracle Database that is not policy-managed.

Additionally, the Generic server pool contains servers with names you specified in the `SERVER_NAMES` attribute of the server pools that list the Generic server pool as a parent server pool.

The Generic server pool's attributes are restricted, as follows:

- No one can modify configuration attributes of the Generic server pool (all attributes are read-only)
- When Oracle Database Configuration Assistant (DBCA) or SRVCTL specifies a server name in the `HOSTING_MEMBERS` resource attribute, Oracle Clusterware only allows it if the server is one of the following:
 - Online and exists in the Generic server pool
 - Online and exists in the Free server pool, in which case Oracle Clusterware moves the server into the Generic server pool
 - Online and exists in any other server pool *and* the user is either a cluster administrator or is allowed to use the server pool's servers, in which case, the server is moved into the Generic server pool
 - Offline and the user is a cluster administrator

Oracle RAC Database and Server Pools

Oracle RAC databases support two different management styles and deployment models.

Policy-Managed

Deployment is based on server pools, where database services run within a server pool as *singleton* or *uniform* across all of the servers in the server pool. Databases are deployed in one or more server pools and the size of the server pools determine the number of database instances in the deployment. Policy management allows clusters and databases to expand or shrink as requirements change.

A policy-managed database is defined by **cardinality**, which is the number of database instances you want running during normal operations. A policy-managed database runs in one or more database server pools that the cluster administrator creates in the cluster, and it can run on different servers at different times. A database instance starts on all servers that are in the server pools defined for the database.

Clients can connect to a policy-managed database using the same SCAN-based connect string no matter which servers they happen to be running on at the time.

Administrator-Managed

Deployment is based on the Oracle RAC deployment types that existed before Oracle Database 11g release 2 (11.2) and requires that you statically configure each database instance to run on a specific node in the cluster, and that you configure database services to run on specific instances belonging to a certain database using the preferred and available designation.

When you review the database resource for an administrator-managed database, you see a server pool defined with the same name as the Oracle database. This server pool is part of a special Oracle-defined server pool called Generic. Oracle RAC manages the Generic server pool to support administrator-managed databases. When you add or remove an administrator-managed database using either SRVCTL or Oracle Database Configuration Assistant (DBCA), Oracle RAC creates or removes the server pools that are members of the Generic server pool.

See Also:

- "Overview of Server Pools and Policy-Based Management" in *Oracle Clusterware Administration and Deployment Guide*
- "Overview of Cluster Configuration Policies and the Policy Set" in *Oracle Clusterware Administration and Deployment Guide*

About Creating Server Pools for Oracle RAC Databases

You can create a server pool with Oracle Database Configuration Assistant while creating an Oracle RAC database, but Oracle recommends that you create server pools before you deploy database software and databases.

Oracle also recommends that you perform the following steps:

- Enable role separation before you create the first server pool in the cluster.

- Create and manage server pools using configuration policies and a respective policy set.

You can implement role-separated management in one of two ways: Vertical or Horizontal.

Vertical Implementation (Between Layers)

Vertical implementation describes a role separation approach based on different operating system users and groups used for various layers in the technology stack. Permissions on server pools and resources are granted to different users (and groups) for each layer in the stack using access control lists. Oracle Automatic Storage Management (Oracle ASM) offers setting up role separation as part of the Oracle Grid Infrastructure installation based on a granular assignment of operating system groups for specific roles.

Horizontal Implementation (Within One Layer)

Horizontal implementation describes a role separation approach that restricts resource access within one layer using access permissions for resources that are granted using access control lists assigned to server pools and policy-managed databases or applications.

For example, consider an operating system user called `grid`, that installs Oracle Grid Infrastructure and creates two database server pools. The operating system users `ouser1` and `ouser2` must be able to operate within a server pool, but must not be able to modify those server pools and withdraw hardware resources from other server pools either accidentally or intentionally.



See Also:

- "Overview of Cluster Configuration Policies and the Policy Set" in *Oracle Clusterware Administration and Deployment Guide*
- "Role-Separated Management" in *Oracle Clusterware Administration and Deployment Guide*

Oracle RAC One Node and Server Pools

Oracle RAC One Node supports the use of server pools, with some limitations.

Note the following about Oracle RAC One Node and server pools:

- Oracle RAC One Node runs only in one server pool. This server pool is treated the same as any other server pool.
- Online relocation of an Oracle RAC One Node database instance permits planned migrations of an Oracle RAC One Node database from one node to another node. Relocations must always be within a server pool.

6

Understanding the Oracle RAC Installed Configuration

There are many ways in which the Oracle Real Application Clusters (Oracle RAC) database is different from a single-instance Oracle database.

- [Understanding the Configured Environment in Oracle RAC](#)
Oracle Net Configuration Assistant (NETCA) and Database Configuration Assistant (DBCA) configure your environment to meet the requirements for database creation and Oracle Enterprise Manager discovery of Oracle RAC databases.
- [Understanding Operating System Privileges Groups](#)
As an administrator, you often perform special operations such as shutting down or starting up a database, or configuring storage.
- [Understanding Time Zone Settings on Cluster Nodes](#)
Oracle RAC requires that all cluster nodes have the same time zone setting.
- [Understanding the Server Parameter File for Oracle RAC](#)
When you create the database, Oracle Database creates an SPFILE in the file location that you specify. This location can be either an Oracle ASM disk group or a cluster file system.
- [Multiple Oracle Home Directories on Windows](#)
Install each Oracle product in its own Oracle home.
- [About Pluggable Databases in Oracle RAC](#)
A pluggable database (PDB) is a portable collection of schemas, schema objects, and nonschema objects that appears to an Oracle Net client as a non-CDB.
- [Database Components Created Using Database Configuration Assistant](#)
Database Configuration Assistant (DBCA) create various database components.
- [About Managing Undo Tablespaces in Oracle RAC](#)
Oracle Database stores rollback or undo information in undo tablespaces.
- [About Initialization Parameter Files](#)
Oracle recommends using the server parameter file (SPFILE) for storing Oracle Database initialization parameters.
- [Oracle Net Services Configuration for Oracle RAC Databases](#)
When connecting to an Oracle Database, you can use a connect descriptor or a net service name.
- [Performance Features of Oracle Net Services and Oracle RAC](#)
Oracle RAC databases provide the important benefits of connection load balancing and failover.
- [Oracle Net Services Configuration Files and Parameters](#)
Networking elements for the Oracle Database server and clients are preconfigured for most environments.

Understanding the Configured Environment in Oracle RAC

Oracle Net Configuration Assistant (NETCA) and Database Configuration Assistant (DBCA) configure your environment to meet the requirements for database creation and Oracle Enterprise Manager discovery of Oracle RAC databases.



Note:

Configuration files are created on each node of your cluster database.

Avoid changing host names after you complete your Oracle RAC installation, including adding or deleting domain qualifications. Node names are created from the host names during an Oracle Clusterware installation and are used extensively with database processes. Nodes with changed host names must be deleted from the cluster and added back with the new host names.

Understanding Operating System Privileges Groups

As an administrator, you often perform special operations such as shutting down or starting up a database, or configuring storage.

Only an administrator, responsible for these administration decisions, should perform these operations. System privileges for Oracle Database or Oracle Automatic Storage Management (Oracle ASM) administration require a secure authentication scheme.

Membership in special operating system groups enables administrators to authenticate to Oracle Database or Oracle ASM through the operating system rather than with a user name and password. This is known as **operating system authentication**. Each Oracle Database in a cluster can have its own operating system privileges groups, so that operating system authentication can be separated for each Oracle Database on a cluster. Because there can be only one Oracle Grid Infrastructure installation on a cluster, there can be only one set of operating system privileges groups for Oracle ASM.

During installation of Oracle Grid Infrastructure and Oracle Database, the installer creates operating system groups. These operating system groups are designated with the logical role of granting operating system authentication for administration system privileges for Oracle Database and Oracle ASM. Oracle Grid Infrastructure uses operating system authentication to manage Oracle Database. To enable this access, you must set the `AUTHENTICATION_SERVICES` parameter in the `sqlnet.ora` file to contain the value `NTS`.

You can use a single operating system group as the logical group whose members are granted all system privileges for Oracle Database and Oracle ASM, or you can delegate system privileges to two or more operating system groups. Oracle recommends that you designate separate operating system groups for each logical system privilege. Using separate operating system groups enables you to grant one or more subsets of administrator system privileges to database administrators. These database administrators can then perform standard database administration tasks without requiring the `SYSDBA` system privileges.

**See Also:**

- "Net Services Profile File (sqlnet.ora)" for more information on how to set the `AUTHENTICATION_SERVICES` parameter.
- *Oracle Grid Infrastructure Installation and Upgrade Guide for Microsoft Windows* for more information about operating system groups and Oracle Database system privileges.
- *Oracle Automatic Storage Management Administrator's Guide* for more information about operating system groups and Oracle ASM system privileges.

Understanding Time Zone Settings on Cluster Nodes

Oracle RAC requires that all cluster nodes have the same time zone setting.

During an Oracle Grid Infrastructure for a cluster installation, the installation process determines the time zone setting of the Oracle Installation user on the node where Oracle Universal Installer (OUI) runs. OUI uses that time zone value on all of the nodes as the default time zone setting for all processes that Oracle Clusterware manages. This default setting is used for databases, Oracle ASM, and any other managed processes.

However, if you start an instance with SQL*Plus, you must ensure that the time zone value that Oracle RAC uses is the same as the Oracle Clusterware time zone. You can change the time zone that Oracle Clusterware uses for a database by running the command:

```
srvctl setenv database -envs 'TZ=time zone'
```

Understanding the Server Parameter File for Oracle RAC

When you create the database, Oracle Database creates an SPFILE in the file location that you specify. This location can be either an Oracle ASM disk group or a cluster file system.

All instances in the cluster database use the same SPFILE at startup. Because the SPFILE is a binary file, do not directly edit the SPFILE with an editor. Instead, change SPFILE parameter settings using Oracle Enterprise Manager or `ALTER SYSTEM SQL` statements.

**See Also:**

Oracle Real Application Clusters Administration and Deployment Guide for information about creating and modifying SPFILE

Multiple Oracle Home Directories on Windows

Install each Oracle product in its own Oracle home.

The value for `%ORACLE_BASE%` is stored in the registry (for example, in `HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE\HOME0`). The values for `ORACLE_HOME` and `ORACLE_SID` are also stored in the registry. Symbolic links for these directories, like those used on UNIX platforms, are not supported on Windows platforms.

Oracle Universal Installer (OUI) records the values for environment variables such as `ORACLE_BASE`, `ORACLE_HOME`, and `ORACLE_SID` in the registry and also updates the value for the `PATH` environment variable for the user performing the installation. In Linux and UNIX systems, you must manually set these environment variables in the user session or user profile.

- [Changing the Current Setting for Oracle Home](#)
Use Oracle Universal Installer (OUI) to change the current Oracle home.

Related Topics

- [Running Oracle Net Configuration Assistant Using Response Files](#)
You can run Oracle Net Configuration Assistant (NETCA) in silent mode to configure and start an Oracle Net listener on the system, configure naming methods, and configure Oracle Net service names.
- *Oracle Database Administrator's Reference for Microsoft Windows*

Changing the Current Setting for Oracle Home

Use Oracle Universal Installer (OUI) to change the current Oracle home.

This procedure changes the value of the default `ORACLE_HOME` variable in the registry to the value you selected. It also ensures that the `%ORACLE_HOME%\bin` directories for each product are listed in the correct order in your `PATH` environment variable.

1. Start the Oracle Universal Installer.
2. Click the **Installed Products** button.
3. Click the **Environment** tab at the top of the window.
4. Move the Oracle home directory that you want as your default to the top of the list.
5. Apply the changes, and exit the installer.

Related Topics

- [Running Oracle Net Configuration Assistant Using Response Files](#)
You can run Oracle Net Configuration Assistant (NETCA) in silent mode to configure and start an Oracle Net listener on the system, configure naming methods, and configure Oracle Net service names.

About Pluggable Databases in Oracle RAC

A pluggable database (PDB) is a portable collection of schemas, schema objects, and nonschema objects that appears to an Oracle Net client as a non-CDB.

PDBs can be plugged into CDBs. A CDB can contain multiple PDBs. Each PDB appears on the network as a separate database.

Starting in Oracle Database 12c, you must create a database as either a multitenant container database (CDB) or as an Oracle database that is non-CDB. This also applies to Oracle RAC databases. The only difference to the installation process is to choose whether to create the Oracle RAC database as a CDB or a non-CDB.

If you create an Oracle RAC database as a CDB and plug one or more PDBs into the CDB, then, by default, a PDB is not started automatically on any instance of the Oracle RAC CDB. With the first dynamic database service assigned to the PDB (other than the default database service which has the same name as the database name), the PDB is made available on those instances on which the service runs.

Whether a PDB is available on more than one instance of an Oracle RAC CDB is typically managed by the services running on the PDB. You can manually enable PDB access on each instance of an Oracle RAC CDB by starting the PDB manually on that instance.



See Also:

- *Oracle Database Concepts* for more information about PDBs
- *Oracle Database Administrator's Guide* for more information about managing PDBs
- *Oracle Real Application Clusters Administration and Deployment Guide* for information specific to the administration of Oracle RAC CDBs

Database Components Created Using Database Configuration Assistant

Database Configuration Assistant (DBCA) create various database components.

- [About Tablespaces and Data Files](#)
For both single-instance and cluster database environments, Oracle Database is divided into smaller logical areas of space known as **tablespaces**.
- [About Control Files](#)
The database is configured with two control files that must be stored on shared storage.
- [About Online Redo Log Files](#)
Each database instance must have at least two online redo log files. The online redo log files for a database instance are called the **redo thread**.

About Tablespaces and Data Files

For both single-instance and cluster database environments, Oracle Database is divided into smaller logical areas of space known as **tablespaces**.

Each tablespace corresponds to one or more data files on the shared storage.

Table 6-1 Tablespace Names Used with Oracle Real Application Clusters Databases

Tablespace Name	Contents
SYSAUX	An auxiliary system tablespace that contains the DRSYS (contains data for Oracle Text), CWMLITE (contains the OLAP schemas), XDB (for XML features), ODM (for Oracle Data Mining), and INDEX schemas
SYSTEM	Consists of the data dictionary, including definitions of tables, views, and stored procedures needed by the database. Oracle Database automatically maintains information in this tablespace.
TEMP	Contains temporary tables and indexes created during SQL statement processing. You may need to expand this tablespace if you run a SQL statement that involves significant sorting, such as ANALYZE COMPUTE STATISTICS on a very large table, or the constructs GROUP BY, ORDER BY, or DISTINCT.

Table 6-1 (Cont.) Tablespace Names Used with Oracle Real Application Clusters Databases

Tablespace Name	Contents
UNDOTBS <i>n</i>	Contains undo tablespaces for each instance that Oracle Database Configuration Assistant creates for automatic undo management
USERS	Consists of application data. As you create and enter data into tables, Oracle Database fills this space with your data.

You cannot alter these tablespace names when using the preconfigured database configuration option from Oracle Universal Installer (OUI). However, you can change the names of the tablespaces if you use the advanced database creation method.

As mentioned, each tablespace has one or more data files on shared file systems. The data file names created by the preconfigured database configuration options vary by storage type such as Oracle ASM or a cluster file system.

**See Also:**

Oracle Database Administrator's Guide for more information about the `SYSTEM`, `SYSAUX`, and other tablespaces

About Control Files

The database is configured with two control files that must be stored on shared storage.

Every database must have one unique control file; any additional control files configured for the database are identical copies of the original control file. If a control file becomes unusable, then the database instance fails when it attempts to access the damaged control file. By multiplexing (creating multiple copies of) a control file on different disks, the database can achieve redundancy and thereby avoid a single point of failure.

**See Also:**

- "Overview of Control Files" in *Oracle Database Concepts*
- "Managing Control Files" in *Oracle Database Administrator's Guide*

About Online Redo Log Files

Each database instance must have at least two online redo log files. The online redo log files for a database instance are called the **redo thread**.

Each Oracle RAC database instance has its own redo thread to avoid contention for a single set of online redo log files. In case of instance failure, the online redo log files must be accessible by the surviving instances. Therefore, the online redo log files for an Oracle RAC

database must be placed on shared storage or Oracle ASM. If you use a file system for storage, then the file system must be a shared or cluster file system.

The file names of the redo log files that are created with the preconfigured database configuration option vary by storage type.

 See Also:

- *Oracle Database Concepts* for more information about the online redo log files
- *Oracle Real Application Clusters Administration and Deployment Guide* for more information about storage for online redo log files

About Managing Undo Tablespaces in Oracle RAC

Oracle Database stores rollback or undo information in undo tablespaces.

To manage undo tablespaces, Oracle recommends that you use Automatic Undo Management. Automatic Undo Management is an automated management mode for undo tablespaces that is easier to administer than manual undo management.

When Oracle ASM and Oracle Managed Files are used along with Automatic Undo Management, an instance that is started for the first time, and thus does not have an undo tablespace, has its undo tablespace created for it by another instance automatically. The same is also true for online redo logs.

 See Also:

- *Oracle Database Administrator's Guide* for more information about automatic undo management
- *Oracle Real Application Clusters Administration and Deployment Guide* for more information about managing undo tablespaces

About Initialization Parameter Files

Oracle recommends using the server parameter file (SPFILE) for storing Oracle Database initialization parameters.

Oracle recommends that you store all SPFILES on Oracle ASM, including the Oracle ASM SPFILE. SPFILES must be located on shared storage; all instances in a cluster database can access this parameter file.

 See Also:

Oracle Real Application Clusters Administration and Deployment Guide for more information about the creation and use of parameter files

Oracle Net Services Configuration for Oracle RAC Databases

When connecting to an Oracle Database, you can use a connect descriptor or a net service name.

For Oracle RAC databases, you can also use the single client access name (SCAN) to connect to any available instance of the Oracle RAC database. Users can access an Oracle RAC database using a client/server configuration or through one or more middle tiers, with or without connection pooling.

- [Database Services for an Oracle RAC Database](#)
Each database is represented by one or more services. A service is identified by a service name, such as `sales.example.com`.
- [Naming Methods and Connect Descriptors](#)
Each net service name is associated with a connect descriptor. A **connect descriptor** provides the location of the database and the name of the database service.
- [Easy Connect Naming Method](#)
The Easy Connect naming method eliminates the need to look up service names in the `tnsnames.ora` file or other repository for TCP/IP environments.
- [Understanding SCANS](#)
The SCAN is a domain name registered to at least one and up to three IP addresses, either in domain name service (DNS) or in Grid Naming Service (GNS).
- [About Connecting to an Oracle RAC Database Using SCANS](#)
Oracle recommends that you configure Oracle RAC database clients to use the SCAN to connect to the database instead of configuring the `tnsnames.ora` file.
- [About Listener Configuration for an Oracle RAC Database](#)
An Oracle RAC database uses multiple listeners to direct client requests to the available instances.
- [About Service Registration for an Oracle RAC Database](#)
An Oracle Database 19c database service automatically registers with the listeners specified in the database initialization parameters `LOCAL_LISTENER` and `REMOTE_LISTENER`.
- [How Database Connections are Created When Using SCANS](#)
Based on the environment, the following actions occur when you use a SCAN to connect to an Oracle RAC database using a service name.

Database Services for an Oracle RAC Database

Each database is represented by one or more services. A service is identified by a service name, such as `sales.example.com`.

A client uses a service name to identify the database it must access. During installation, Oracle RAC databases are configured with a default database service that has the same name as the database. This service can be used for performing database management tasks. Additional services must be created for client and application connections to the database.

A service name can be associated with multiple database instances, and an instance can be associated with multiple services. The listener acts as a mediator between the client and database instances and routes the connection request to the appropriate instance. Clients connecting to a service do not have to specify which instance they want to connect to.

Naming Methods and Connect Descriptors

Each net service name is associated with a connect descriptor. A **connect descriptor** provides the location of the database and the name of the database service.

A connect descriptor includes one or more protocol addresses of the listener and the connect information for the destination service.

The information needed to use a service name to create a database connection can be stored in a repository, used by one or more naming methods. A **naming method** is a resolution method used by a client application to resolve a service name to a connect descriptor. Oracle Net Services offers several types of naming methods that support localized configuration on each client, or centralized configuration that can be accessed by all clients in the network.

Easy Connect Naming Method

The Easy Connect naming method eliminates the need to look up service names in the tnsnames.ora file or other repository for TCP/IP environments.

With Easy Connect, clients use a connect string for a simple TCP/IP address, which consists of a host name, and an optional port and service name. If you use this method, then no naming or directory system is required. See ["Example 6-1"](#) for an example.

Networking elements for the Oracle Database server and clients are preconfigured for most environments. The Easy Connect naming method is enabled by default and does not require a repository. If you use a naming method other than Easy Connect, then additional configuration of Oracle Net Services may be required.

Understanding SCANS

The SCAN is a domain name registered to at least one and up to three IP addresses, either in domain name service (DNS) or in Grid Naming Service (GNS).

- [About the SCAN](#)
During the installation of Oracle Grid Infrastructure, several Oracle Clusterware resources are created for the SCAN.
- [About SCAN VIP Addresses](#)
SCAN virtual IP addresses (VIPs) function like node VIPs. However, unlike node VIPs, SCAN VIPs can run on any node in the cluster.
- [About SCAN Listeners](#)
During Oracle Grid Infrastructure installation, SCAN listeners are created for as many IP addresses as there are SCAN VIP addresses assigned to resolve to the SCAN.

About the SCAN

During the installation of Oracle Grid Infrastructure, several Oracle Clusterware resources are created for the SCAN.

- A SCAN virtual IP (VIP) is created for each IP address that Oracle Single Client Access Name (SCAN) resolves to
- A SCAN listener is created for each SCAN VIP
- A dependency on the SCAN VIP is configured for the SCAN listener

SCANS are defined using one of two options:

- The SCAN is defined in DNS

If you configure a SCAN manually, and use DNS for name resolution, then your network administrator should create a single name for the SCAN that resolves to three IP addresses on the same network as the public network for the cluster. The SCAN name must be resolvable without the domain suffix (for example, the address `sales1-scan.example.com` must be resolvable using `sales1-scan`). The SCAN must not be assigned to a network interface, because Oracle Clusterware resolves the SCAN.

The default SCAN is `cluster_name-scan.domain_name`. For example, in a cluster that does not use GNS, if your cluster name is `sales1`, and your domain is `example.com`, then the default SCAN address is `sales1-scan.example.com:1521`.

- The SCAN is defined in GNS

When using GNS and DHCP, Oracle Clusterware configures the VIP addresses for the SCAN name that is provided during cluster configuration. The node VIP and the three SCAN VIPs are obtained from the DHCP server when using GNS. If a new server joins the cluster, then Oracle Clusterware dynamically obtains the required VIP address from the DHCP server, updates the cluster resource, and makes the server accessible through GNS.

Oracle recommends that you configure clients connecting to the cluster to use the SCAN name, rather than node VIPs used in releases before Oracle Grid Infrastructure 11g Release 2 (11.2). Clients connecting to Oracle RAC databases using SCANS do not have to be configured with addresses of each node that hosts a particular database or database instance. For example, if you configure policy-managed server pools for a cluster, then connecting to the database using a SCAN enables connections to server pools in that database, regardless of which nodes are allocated to the server pool. You can add or remove nodes from the database without having to reconfigure clients connecting to the database.

Related Topics

- *Oracle Grid Infrastructure Installation and Upgrade Guide*

About SCAN VIP Addresses

SCAN virtual IP addresses (VIPs) function like node VIPs. However, unlike node VIPs, SCAN VIPs can run on any node in the cluster.

Clients (users or applications) that connect using a SCAN instead of a node VIP name or address do not have to update the list of node names or addresses in their local `tnsnames.ora` file when nodes are added to or removed from the cluster, or when a database instance runs on a different node.

Note:

Configuring three SCAN VIPs in DNS does not by itself ensure failover of connections. Instead, the Oracle Client uses the returned SCAN VIPs to failover the connection request to a different SCAN listener. If the connection attempt to a SCAN VIP fails, then the client uses the next returned SCAN VIP address to connect. For this reason, Oracle recommends that you use Oracle Client 11g Release 2 or later clients for connections that use the SCAN.

If you use GNS for name resolution, then you only provide the SCAN name during installation (for example, `sales1-scan`). GNS obtains DHCP address leases for three IP addresses and

resolves these addresses to the SCAN. The GNS daemon listens for registrations. When a SCAN VIP starts on a node, it registers its addresses with GNS.

Service requests to the cluster domain that GNS manages are routed to the GNS VIP address, which routes these requests to the GNS daemon for the cluster. When GNS receives a request from a DNS for the SCAN, it returns the registered addresses of the SCAN listeners to the DNS. The DNS then returns the three SCAN VIP addresses to the client.



See Also:

Oracle Clusterware Administration and Deployment Guide for more information about SCAN names, listeners, and client service requests

About SCAN Listeners

During Oracle Grid Infrastructure installation, SCAN listeners are created for as many IP addresses as there are SCAN VIP addresses assigned to resolve to the SCAN.

Oracle recommends that the SCAN resolves to three VIP addresses, to provide high availability and scalability. If the SCAN resolves to three addresses, then three SCAN VIPs and three SCAN listeners are created.

Each SCAN listener depends on its corresponding SCAN VIP. The SCAN listeners cannot start until the SCAN VIP is available on a node.

The addresses for the SCAN listeners resolve either through an external domain name service (DNS), or through the Grid Naming Service (GNS) within the cluster. SCAN listeners and SCAN VIPs can run on any node in the cluster. If a node where a SCAN VIP is running fails, then the SCAN VIP and its associated listener fails over to another node in the cluster. If the number of available nodes within the cluster falls to less than three, then one server hosts two SCAN VIPs and SCAN listeners. The SCAN listener also supports HTTP protocol for communication with Oracle XML Database (XDB).



See Also:

Oracle Clusterware Administration and Deployment Guide for more information about SCAN listeners

About Connecting to an Oracle RAC Database Using SCANS

Oracle recommends that you configure Oracle RAC database clients to use the SCAN to connect to the database instead of configuring the `tnsnames.ora` file.

Clients configured to connect to the cluster using node VIP addresses for Oracle RAC releases earlier than Oracle Database 11g Release 2 can continue to use their existing connection addresses. Using the SCAN is not required. When an earlier release of Oracle Database is upgraded, the database is not only registered with the local listeners, but is also registered with the SCAN listeners, allowing clients to start using the SCAN to connect to that database.

If the SCAN is resolved by DNS, then DNS returns all three SCAN VIP addresses to the client. If the SCAN is resolved by GNS, then DNS zone delegation sends the lookup request to GNS, which then returns all three SCAN VIP addresses to the client.

Oracle Database 19c database clients use SCAN to connect to the database. Oracle recommends against using the easy connect method with SCAN because the easy connect method does not have the ability to specify timeouts and retries for connection establishment. Instead, applications must use an Oracle Net connect descriptor with the following format:

```
(DESCRIPTION =
  (CONNECT_TIMEOUT=90) (RETRY_COUNT=20) (RETRY_DELAY=3)
  (TRANSPORT_CONNECT_TIMEOUT=3)
    ( ADDRESS = (PROTOCOL = TCP) (HOST=scan) (PORT=1521))
      (CONNECT_DATA=(SERVICE_NAME=service_name)))
```

scan represents the SCAN for your cluster. If you do not specify a port number, then the default value of 1521 is used for the TCP port identifier. The *service_name* is the name of a dynamic database service.

The client then uses one of the returned SCAN VIP addresses to contact a SCAN listener. When a SCAN listener receives a connection request from a client, the SCAN listener identifies the least loaded instance in the cluster that provides the requested service. It then redirects the connection request to the local listener on the node where the least loaded instance is running, and the client is given the local listener address. The local listener then creates the connection to the database instance.

Example 6-1 Connecting to Oracle RAC Using an Oracle Net Connect Descriptor

If the Oracle RAC database runs on a cluster for which the SCAN is *sales1-scan.mycluster.example.com*, then you can submit a connection request for the database service *oltp.example.com* by using a connect descriptor similar to the following:

```
(DESCRIPTION =
  (CONNECT_TIMEOUT=90) (RETRY_COUNT=20) (RETRY_DELAY=3)
  (TRANSPORT_CONNECT_TIMEOUT=3)
    ( ADDRESS = (PROTOCOL = TCP) (HOST=sales1-
scan.mycluster.example.com) (PORT=1521))
      (CONNECT_DATA=(SERVICE_NAME=oltp.example.com)))
```

If the SCAN is resolved by DNS, then DNS returns all three SCAN VIP addresses to the client. If the SCAN is resolved by GNS, then DNS zone delegation sends the lookup request to GNS, which then returns all three SCAN VIP addresses to the client. The client then uses one of the returned SCAN VIP addresses to contact a SCAN listener.

When a SCAN listener receives a connection request from a client, the SCAN listener identifies the least loaded instance in the cluster that provides the requested service. It then redirects the connection request to the local listener on the node where the least loaded instance is running, and the client is given the local listener address. The local listener then creates the connection to the database instance.

About Listener Configuration for an Oracle RAC Database

An Oracle RAC database uses multiple listeners to direct client requests to the available instances.

An Oracle database receives connection requests through the local listener. The local listener brokers a client request, handing off the request to the server. The listener is configured with a protocol address, and clients configured with the same protocol address can send connection requests to the listener. When a connection is established, the client and Oracle database communicate directly with one another.

The local listener, or default listener, is located in the Grid home when you have Oracle Grid Infrastructure installed. Local listeners are configured to respond to database connection requests, and to non-database connection requests, such as external procedures or Oracle XML Database (XDB) requests. When the database starts, the Database Agent process (`oraagent.exe`, previously known as `racgimon`) sets the `LOCAL_LISTENER` parameter to a connect descriptor that does not require an Oracle Net service name. The value for `LOCAL_LISTENER` is computed to be the endpoints of the Grid home listeners.

You can configure multiple Oracle Database listeners, each with a unique name, in one `listener.ora` file. Multiple listener configurations for database listeners are possible because each of the top-level configuration parameters has a suffix of the listener name or is the listener name itself. To configure a database to register with multiple local listeners, you must manually modify the `LOCAL_LISTENER` parameter.

**Note:**

Oracle recommends running only one listener for each node in most customer environments.

For an Oracle RAC database, the database parameter `REMOTE_LISTENER` identifies the SCAN listeners. The database registers with the local and SCAN listeners by using the connect description information contained in these parameters. Oracle Database 11g Release 2 and later instances only register with SCAN listeners as remote listeners. Upgraded databases register with SCAN listeners as remote listeners, and also continue to register with all node listeners.

The `REMOTE_LISTENER` parameter for an Oracle RAC database is always set to the SCAN address. For example, if the SCAN for the cluster is `myscan`, and the GNS subdomain for the cluster is `mycluster.example.com`, then the `REMOTE_LISTENER` parameter has the following value:

```
myscan.mycluster.example.com:1521
```

**Note:**

Do not set the `REMOTE_LISTENER` parameter for an Oracle RAC database to an Oracle Net alias that has a single address that uses the SCAN for the host name (`HOST=scan`).

About Service Registration for an Oracle RAC Database

An Oracle Database 19c database service automatically registers with the listeners specified in the database initialization parameters `LOCAL_LISTENER` and `REMOTE_LISTENER`.

During registration, the listener registration (LREG) process sends information such as the service name, instance names, and workload information to the listeners. This feature is called **service registration**.

When a listener starts after the Oracle instance starts, and the listener is available for service registration, registration does not occur until the next time the Oracle Database LREG process starts its discovery routine. By default, the LREG discovery routine is started every 60 seconds. To override the 60-second delay, use the SQL statement `ALTER SYSTEM REGISTER`. This statement forces LREG to register the service immediately

 Note:

Oracle recommends that you create a script to run the `ALTER SYSTEM REGISTER` statement immediately after starting the listener. If you run this statement when the instance is registered and all services are currently registered, or while the listener is down, then the statement has no effect.

 See Also:

Oracle Database Net Services Administrator's Guide for more information about service registration

How Database Connections are Created When Using SCANS

Based on the environment, the following actions occur when you use a SCAN to connect to an Oracle RAC database using a service name.

The numbered actions correspond to the arrows shown in the figure displayed after the steps.

1. The LREG process of each instance registers the database services with the default listener on the local node and with each SCAN listener, which is specified by the `REMOTE_LISTENER` database parameter. The listeners are dynamically updated on the amount of work being handled by the instances and dispatchers.
2. The client issues a database connection request using a connect descriptor of the form:

```
orausr/@scan_name:1521/webapp
```

 Note:

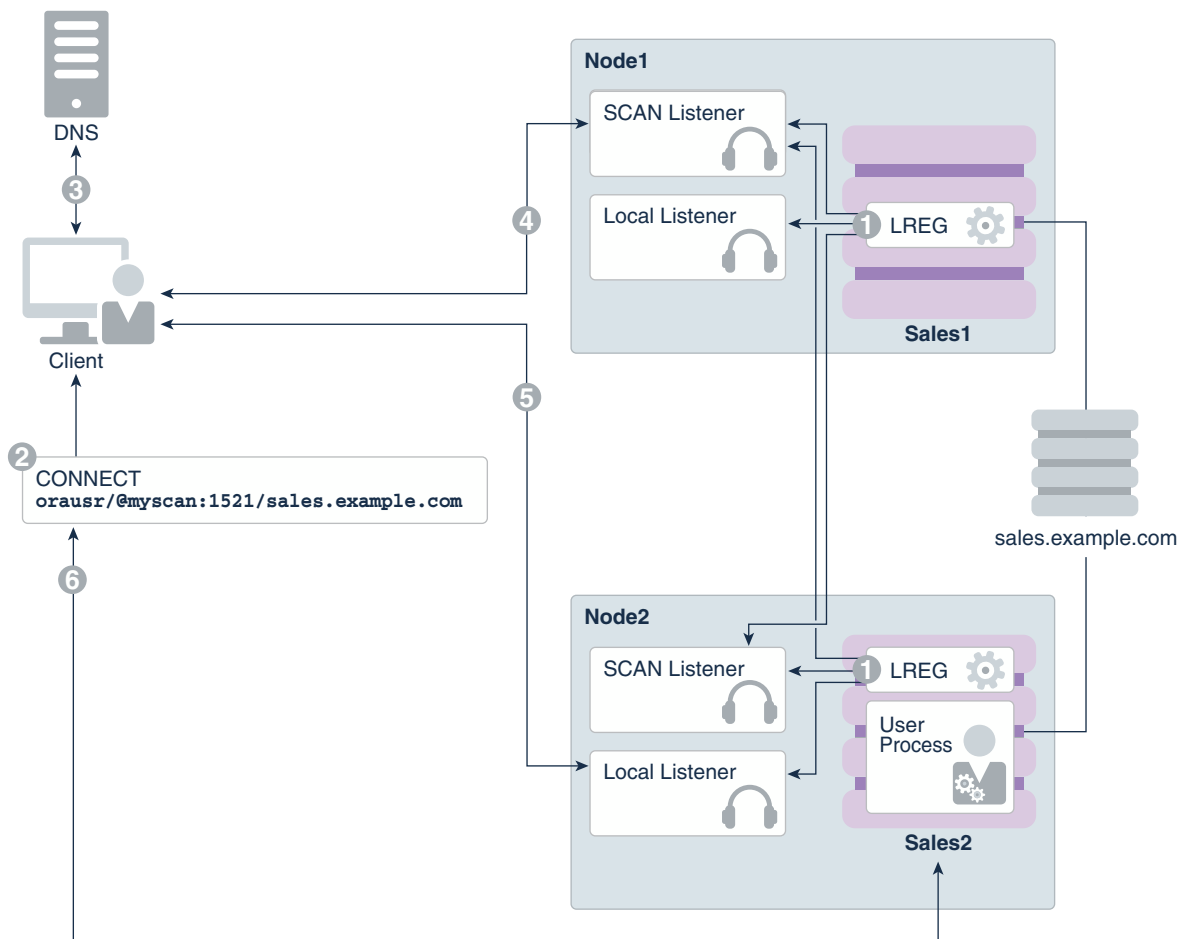
If you use the Easy Connect naming method, then ensure that the `sqlnet.ora` file on the client contains `EZCONNECT` in the list of naming methods specified by the `NAMES.DIRECTORY_PATH` parameter.

3. The client uses DNS to resolve `scan_name`. After DNS returns the three addresses assigned to the SCAN, the client sends a connect request to the first IP address. If the connect request fails, then the client attempts to connect using the next IP address.
4. When the connect request is successful, the client connects to a SCAN listener for the cluster that hosts the `sales` database and has an instance offering the `webapp` service,

which in this example is `sales1` and `sales2`. The SCAN listener compares the workload of the instances `sales1` and `sales2` and the workload of the nodes on which they run. If the SCAN listener determines that `node2` is less loaded than `node1`, then the SCAN listener selects `node2` and sends the address for the local listener on that node back to the client.

5. The client connects to the local listener on `node2`. The local listener starts a dedicated server process for the connection to the database.
6. The client connects directly to the dedicated server process on `node2` and accesses the `sales2` database instance.

Figure 6-1 Load Balancing Actions for Oracle RAC Connections That Use SCAN



Performance Features of Oracle Net Services and Oracle RAC

Oracle RAC databases provide the important benefits of connection load balancing and failover.

- [Load Balancing of Connections to Oracle RAC Databases](#)
There are two types of load balancing that you can implement for an Oracle RAC database: client-side and server-side load balancing.

- [Connection Failover for Oracle RAC Databases](#)
When a client issues a connection request using SCAN, the three SCAN addresses are returned to the client.
- [Shared Server Configuration for an Oracle RAC Database](#)
Standalone Oracle databases perform load balancing by distributing connections among the shared server dispatcher processes.

Load Balancing of Connections to Oracle RAC Databases

There are two types of load balancing that you can implement for an Oracle RAC database: client-side and server-side load balancing.

Services coordinate their sessions by registering their **workload**, or the amount of work they are currently handling, with the local listener and the SCAN listeners. Clients are redirected by the SCAN listener to a local listener on the least-loaded node that is running the instance for a particular service. This feature is called **load balancing**. The local listener either directs the client to a dispatcher process (if you configured the database to use shared servers), or directs the client to a dedicated server process.

Client-side load balancing balances the connection requests across the listeners. With server-side load balancing, the SCAN listener directs a connection request to the best instance currently providing the service by using the load balancing advisory.

See Also:

- *Oracle Grid Infrastructure Installation Guide* for more information about SCAN and its configuration
- *Oracle Real Application Clusters Administration and Deployment Guide* for more information about failover, load balancing, and the load balancing advisory

Connection Failover for Oracle RAC Databases

When a client issues a connection request using SCAN, the three SCAN addresses are returned to the client.

If the first address fails, then the connection request to the SCAN fails over to the next address. Using multiple addresses allows a client to connect to an instance of the database even if the initial instance has failed.

Oracle RAC provides failover with the node VIP addresses by configuring multiple listeners on multiple nodes to manage client connection requests for the same database service. If a node fails, then the service connecting to the VIP is relocated transparently to a surviving node, enabling fast notification of the failure to the clients connecting through the VIP. If the application and client are configured with transparent application failover options, then the client is reconnected to the surviving node.

Shared Server Configuration for an Oracle RAC Database

Standalone Oracle databases perform load balancing by distributing connections among the shared server dispatcher processes.

By default, Oracle Database Configuration Assistant (DBCA) configures your Oracle RAC database with dedicated servers, not shared servers. However, if you select the shared server option when using DBCA, then DBCA configures shared servers. Oracle RAC uses both dedicated and shared server processing when shared servers are configured.

**See Also:**

Oracle Database Net Services Administrator's Guide for more information about shared server configurations

Oracle Net Services Configuration Files and Parameters

Networking elements for the Oracle Database server and clients are preconfigured for most environments.

The Easy Connect naming method is enabled by default and does not require a repository. If you use a naming method other than Easy Connect, then additional configuration of Oracle Net Services may be required.

- [Database Initialization Parameters for Database Service Registration](#)
An Oracle Database 19c database service automatically registers with the listeners specified in the `LOCAL_LISTENER` and `REMOTE_LISTENER` parameters.
- [Net Service Names and the `tnsnames.ora` File](#)
The installation process creates a `tnsnames.ora` file on each node.
- [Net Service Names Created by DBCA](#)
Oracle Database Configuration Assistant creates net service names for connecting to the database instances.
- [Listener Configuration and the `listener.ora` File](#)
In Oracle RAC environments, Oracle recommends that you let the Oracle Agent manage Oracle listeners for Oracle Databases.
- [Net Services Profile File \(`sqlnet.ora`\)](#)
Oracle Universal Installer starts Oracle Net Configuration Assistant (NETCA) after the database software is installed. NETCA creates the Oracle Net Services profile, or the `sqlnet.ora` file.

Database Initialization Parameters for Database Service Registration

An Oracle Database 19c database service automatically registers with the listeners specified in the `LOCAL_LISTENER` and `REMOTE_LISTENER` parameters.

During registration, the listener registration (LREG) process sends information such as the service name, instance names, and workload information to the listeners.

When a listener starts after the Oracle instance starts, and the listener is available for service registration, registration does not occur until the next time the Oracle Database LREG process starts its discovery routine. By default, the LREG discovery routine is started every 60 seconds. To override the 60-second delay, use the SQL statement `ALTER SYSTEM REGISTER`. This statement forces LREG to register the service immediately.

**Note:**

Oracle recommends that you create a script to run the `ALTER SYSTEM REGISTER` statement immediately after starting the listener. If you run this statement when the instance is registered and all services are currently registered, or while the listener is down, then the statement has no effect.

**See Also:**

Oracle Database Net Services Administrator's Guide for more information about service registration

Net Service Names and the tnsnames.ora File

The installation process creates a `tnsnames.ora` file on each node.

The `tnsnames.ora` file acts as a repository of net service names. Each net service name is associated with a connect identifier. A **connect identifier** is an identifier that maps a user-defined name to a connect descriptor. A connect descriptor contains the following information:

- The network route to the service, including the location of the listener through a protocol address
- The `SERVICE_NAME` parameter, with the value set to the name of a database service

**Note:**

The `SERVICE_NAME` parameter you use in the `tnsnames.ora` file is singular, because you can specify only one service name.

The `tnsnames.ora` file is located in both the `Grid_home\network\admin` and `Oracle_home\network\admin` directories. By default, the `tnsnames.ora` file is read from the Grid home when Oracle Grid Infrastructure is installed.

With Oracle Clusterware 11g Release 2 and later, the listener association no longer requires `tnsnames.ora` file entries. The listener associations are configured as follows:

- Oracle Database Configuration Assistant (DBCA) no longer sets the `LOCAL_LISTENER` parameter. The Oracle Clusterware agent that starts the database sets the `LOCAL_LISTENER` parameter dynamically, and it sets it to the actual value, not an alias. So `listener_alias` entries are no longer needed in the `tnsnames.ora` file.
- The `REMOTE_LISTENER` parameter is configured by DBCA to reference the SCAN and SCAN port, without any need for a `tnsnames.ora` entry. Oracle Clusterware uses the Easy Connect naming method with `scanname:scanport`, so no listener associations for the `REMOTE_LISTENER` parameter are needed in the `tnsnames.ora` file.

Example 6-2 Adding a Second Listener to an Oracle RAC Database

If you created a database named `orc11`, to add a second listener, listening on port 2012, use a command similar to the following command to have the database register with both listeners on startup:

```
SQL> alter system set local_listener='(DESCRIPTION=(  
ADDRESS_LIST=(ADDRESS=(PROTOCOL=TCP) (HOST=192.168.0.61) (PORT=1521))  
(ADDRESS=(PROTOCOL=TCP) (HOST=192.168.0.61) (PORT=2012)))) '  
scope=BOTH SID='OCRL1';
```

See Also:

- *Oracle Database Administrator's Guide*
- *Oracle Database Net Services Administrator's Guide* for more information about the `tnsnames.ora` file

Net Service Names Created by DBCA

Oracle Database Configuration Assistant creates net service names for connecting to the database instances.

- [Net Service Names for Database Connections](#)
Clients that connect to any instance of Oracle RAC use the SCAN in the connect descriptor.
- [Net Service Names for Instance Connections](#)
Clients that connect to a particular instance of the database use the net service name for the instance.

Net Service Names for Database Connections

Clients that connect to any instance of Oracle RAC use the SCAN in the connect descriptor.

You can also use net service names to connect to Oracle RAC. The default database service created by Oracle Database Configuration Assistant (DBCA) enables Oracle Enterprise Manager to discover an Oracle RAC database, and must not be used for client connections.

If you use DBCA to create an Oracle RAC database that is a multitenant container database (CDB), then DBCA creates a database service that has the same name as the database. Clients that use this database service can connect to any database instance for the Oracle RAC CDB. However, if you use DBCA to add a pluggable database (PDB) to an existing CDB, then DBCA does not create a database service for the new PDB.

The net service name does not require a fully qualified domain name for the server on which a database, database instance, or listener runs. SCANS are resolved by the DNS or GNS, which returns three addresses to the client. The client then submits connection requests to each address in succession until a connection is made.

Example 6-3 Net Service Name Entry for a Database Connection

This example shows a connect descriptor that is used in a `tnsnames.ora` file. The connect identifier in this case is the same as the cluster domain, `mycluster.example.com`. Instead of

specifying the address for an individual server, a virtual Internet Protocol (VIP) address, or a cluster node name, the connect descriptor uses the SCAN, which is `myscan.mycluster.example.com`.

```
mycluster.example.com =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = myscan.mycluster.example.com)
      (PORT = 1522))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = myApp)
    )
  )
```

Oracle Clusterware resolves connection requests that use the net service name `mycluster.example.com` to any of the database instances of the `mycluster` database that run the `myApp` database service. The specific cluster node on which the instance is running is invisible to the client.

Net Service Names for Instance Connections

Clients that connect to a particular instance of the database use the net service name for the instance.

Example 6-4 Example Net Service Name Entry for an Instance Connection

In this example, the connect identifier is the same as the instance name, `mycluster1.example.com`. The connect descriptor uses the SCAN to locate the instance within the cluster. Clients connecting to the net service name `mycluster1.example.com` are connected to the `mycluster1` database instance of the `mycluster` database. Oracle Clusterware resolves that connection to the cluster node on which the instance is running. The specific cluster node on which the instance is running is invisible to the client.

```
mycluster1.example.com=
  (DESCRIPTION=
    (ADDRESS=(PROTOCOL=TCP) (HOST=myscan.mycluster.example.com) (PORT=1521))
    (CONNECT_DATA=
      (SERVICE_NAME=mycluster.example.com)
      (INSTANCE_NAME=mycluster1)
    )
  )
```

Listener Configuration and the listener.ora File

In Oracle RAC environments, Oracle recommends that you let the Oracle Agent manage Oracle listeners for Oracle Databases.



Note:

If you enable GNS, then you do not have to manually configure the listener.

- [Local Listener for an Oracle RAC Database](#)
The local listener, or default listener, is located in the Grid home when you have Oracle Grid Infrastructure installed.
- [Remote Listeners for an Oracle RAC Database](#)
A **remote listener** is a listener residing on one computer that redirects connections to a database instance on another computer.
- [Managing Multiple Listeners for an Oracle RAC Database](#)
In Oracle RAC environments, Oracle recommends that you let the Oracle Agent manage the Oracle listeners for the databases.
- [How Oracle Database Uses the Listener File \(listener.ora\)](#)
The `listener.ora` file is the configuration file for a listener.

Local Listener for an Oracle RAC Database

The local listener, or default listener, is located in the Grid home when you have Oracle Grid Infrastructure installed.

The `listener.ora` file is located in the `Grid_home\network\admin` directory. If needed, you can edit the `listener.ora` file for the Grid home listeners to define listener parameters for node and SCAN listeners. Do not modify the endpoints because these are automatically managed by the listener agent.

During Oracle Database creation, the `LOCAL_LISTENER` parameter is automatically configured to point to the local listener for the database. You can set a value manually for `LOCAL_LISTENER`. If you modify the value of the `LOCAL_LISTENER` parameter, then the Database Agent process does not automatically update this value. Oracle recommends that you leave the parameter unset so that the Database Agent process can maintain it automatically. If you do not set `LOCAL_LISTENER`, then the Database Agent process automatically updates the database associated with the local listener in the Grid home, even when the ports or IP address of that listener are changed.

See Also:

- ["Net Service Names and the tnsnames.ora File"](#) for more information about listener associations defined in the `tnsnames.ora` file
- [Oracle Database Net Services Reference](#) for more information about the `listener.ora` file
- [Oracle Database Net Services Administrator's Guide](#) for information about registering information with a local listener

Remote Listeners for an Oracle RAC Database

A **remote listener** is a listener residing on one computer that redirects connections to a database instance on another computer.

For example, SCAN listeners are remote listeners. In Oracle RAC environments, Oracle recommends that you let the Oracle Agent manage the Oracle listeners for the databases.

 See Also:

- ["Net Service Names and the tnsnames.ora File"](#) for more information about listener associations defined in the `tnsnames.ora` file
- *Oracle Database Net Services Reference* for more information about the `listener.ora` file
- *Oracle Database Net Services Administrator's Guide* for information about registering information with a remote listener

Managing Multiple Listeners for an Oracle RAC Database

In Oracle RAC environments, Oracle recommends that you let the Oracle Agent manage the Oracle listeners for the databases.

- Use the `lsnrctl` executable located in the Grid home to administer Oracle Database 19c local and SCAN listeners.

Do not attempt to use the `lsnrctl` executables from Oracle home locations for earlier releases, because they cannot be used with Oracle Database 19c.

- Use `SRVCTL` and the `setenv` command to modify the value of `TNS_ADMIN` for each listener.

For listeners not managed by Oracle Clusterware, you can use a nondefault location for the `listener.ora` file by setting the `TNS_ADMIN` environment variable or registry value to point to the directory that contains the Oracle Net Services configuration files.

How Oracle Database Uses the Listener File (listener.ora)

The `listener.ora` file is the configuration file for a listener.

The `listener.ora` file can include the protocol addresses it is accepting connection requests on, a list of the database and other services it is listening for, and control parameters used by the listener. You can modify the configuration of the listeners used by Oracle Clusterware and Oracle Real Application Clusters (Oracle RAC) with Server Control Utility (`SRVCTL`) commands, or by using Oracle Net Configuration Assistant (NETCA). Manual editing of the `listener.ora` file is not required.

Each listener is configured with one or more protocol addresses that specify its listening endpoints. The listener agent dynamically updates endpoints with the listener. Starting with Oracle Database 11g Release 2, the `listener.ora` file now only contains an IPC key and the following information:

```
(ADDRESS = (PROTOCOL=TCP) (HOST=) (PORT=1521))
```

In the previous example, the protocol `ADDRESS` refers implicitly to the `HOST` endpoint of the local node. The `listener.ora` file is the same on every node for an Oracle RAC database. Listening endpoints, such as the port numbers, are dynamically registered with the listener.

Before you install Oracle RAC, during the Oracle Grid Infrastructure installation, NETCA creates and starts a default listener in the Grid home called `LISTENER`. The listener is configured with default protocol listening addresses. The listener is configured to respond to connection requests that are directed to one protocol address specified during installation.

During the Oracle RAC installation, the Oracle RAC database uses the listener in the Grid home, and configures service information about the Oracle RAC database. The database services automatically register their information with the listener, such as the service name, instance names, and load information. Dynamic service registration eliminates the need for static configuration of database services. However, static service configuration is required if you plan to use Oracle Enterprise Manager.

Example 6-5 Example listener.ora File for an Oracle RAC Node

The following is an example `listener.ora` file for the `mycluster` cluster as it appears after installation, with an entry for a node named `node1` and a SCAN listener.

```
LISTENER_SCAN1=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)
(KEY=LISTENER_
SCAN1))))          # line added by Agent
LISTENER_NODE1=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=IPC)
(KEY=LISTENER))))
                # line added by Agent
# listener.ora.mycluster Network Configuration File:
C:\app\oracle\product\12.2.0\dbhome_1\network\admin\listener.ora.mycluster
# Generated by Oracle configuration tools.

LISTENER_NODE1 =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = IPC) (KEY = EXTPROC1521))
    )
  )

ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_NODE1=ON # line added by Agent
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN2=ON # line added by Agent
ENABLE_GLOBAL_DYNAMIC_ENDPOINT_LISTENER_SCAN1=ON # line added by Agent
```

Net Services Profile File (sqlnet.ora)

Oracle Universal Installer starts Oracle Net Configuration Assistant (NETCA) after the database software is installed. NETCA creates the Oracle Net Services profile, or the `sqlnet.ora` file.

In an Oracle Grid Infrastructure installation, the `sqlnet.ora` file is located in the `Grid_home\network\admin` directory by default.

For the local listener for the Oracle RAC database instance, the default location of the `sqlnet.ora` file is `%ORACLE_HOME%\network\admin` directory. In this directory there is a default `sqlnet.ora` file. Also, you can find a sample `sqlnet.ora` file in the subdirectory `sample`.

During installation of the Oracle RAC software, NETCA creates the following entries in the `sqlnet.ora` file, where `%ORACLE_BASE%` is the path to the Oracle base directory for the Oracle RAC installation:

```
NAMES.DIRECTORY_PATH=(TNSNAMES, EZCONNECT)
ADR_BASE =%ORACLE_BASE%
```

The `AUTHENTICATION_SERVICES` parameter (not shown in the above example) specifies the method by which users are authenticated for database access. The value `NTS` indicates that Microsoft Windows native authentication should be used to authorize access to the database. Oracle Databases that use Oracle Automatic Storage Management (Oracle ASM) and the databases that are managed by Oracle Grid infrastructure must use Windows native authentication, which is enabled by default.

The parameter `NAMES.DIRECTORY_PATH` specifies the priority order of the naming methods to use to resolve connect identifiers to connect descriptors. The `ADR_BASE` parameter specifies the base directory into which tracing and logging incidents are stored when Automatic Diagnostic Repository (ADR) is enabled for the database.



See Also:

- *Oracle Database Net Services Administrator's Guide* for more information about the `sqlnet.ora` file
- *Oracle Database Concepts* for more information about authentication
- *Oracle Database Client Installation Guide for Microsoft Windows* for more information about configuring clients for database connectivity
- *Oracle Database Administrator's Guide* for more information about ADR

7

Removing Oracle RAC Software

The `deinstall.bat` command removes Oracle Clusterware and Oracle Automatic Storage Management (Oracle ASM) from your server, and Oracle Database installations, for both Oracle RAC and single-instance databases.

- [Overview of Deinstallation Procedures](#)
There are several tasks to perform when completely removing all Oracle databases, instances, and software from an Oracle home directory.
- [About Oracle Deinstallation Options](#)
The `deinstall.bat` command stops Oracle software, and removes Oracle software and configuration files on the operating system.
- [Files Deleted by the deinstall Command](#)
The `deinstall` command removes Oracle software and files from your system.
- [Identifying All Instances On a Cluster](#)
You can identify the database instances on your cluster using either SRVCTL or the Windows Services control interface.
- [deinstall Command Reference](#)
You can run the `deinstall` command to remove Oracle software. You can run this command from an Oracle home directory after installation.
- [Using the Deinstallation Tool to Remove Oracle RAC](#)
You can run the deinstallation tool in multiple ways.
- [Cleaning Up After a Failed Installation](#)
If an installation fails, then you must remove the Oracle home directory and remove all files that Oracle Universal Installer (OUI) created during the attempted installation.

Overview of Deinstallation Procedures

There are several tasks to perform when completely removing all Oracle databases, instances, and software from an Oracle home directory.

- Identify all instances associated with the Oracle home
- Shut down processes
- Remove listeners installed in the Oracle Database home
- Remove database instances
- Remove Oracle Automatic Storage Management (Oracle ASM) release 11.1 or earlier
- Remove Oracle Clusterware and Oracle ASM (Oracle Grid Infrastructure)

 Note:

- For information on removing Oracle Database Vault, see *Oracle Database Vault Administrator's Guide*.
- For information on removing Oracle Configuration Manager, see *Oracle Configuration Manager Installation and Administration Guide*.
- With Oracle Grid Infrastructure 11g release 2 (11.2) and later, Oracle ASM and Oracle Clusterware comprise the Oracle Grid Infrastructure installation. These components are installed and removed together.

 Caution:

If any cluster member node has more than one database with the same global database name (GDN) on a server, then you cannot use the deinstallation tool to remove only one database.

For example, if you have a standalone database on one of your cluster nodes with the GDN `mydb.example.com`, and your Oracle RAC database GDN is also `mydb.example.com`, then both databases on that node are removed by the deinstallation tool.

About Oracle Deinstallation Options

The `deinstall.bat` command stops Oracle software, and removes Oracle software and configuration files on the operating system.

The `deinstall` command is available in Oracle home directories after installation. It is located in the `%ORACLE_HOME%\deinstall` directory.

`deinstall` creates a response file by using information in the Oracle home and using the information you provide. You can use a response file that you generated previously by running the `deinstall` command using the `-checkonly` option. You can also edit the response file template.

 Note:

- You must run the `deinstall` command from the same release to remove Oracle software. Do not run the `deinstall` command from a later release to remove Oracle software from an earlier release. For example, do not run the `deinstall` command from the 19c Oracle home to remove Oracle software from an existing 11.2.0.4 Oracle home.
- Starting with Oracle Database 12c Release 1 (12.1.0.2), the `roothas.bat` script replaces the `roothas.pl` script in the Oracle Grid Infrastructure home for Oracle Restart, and the `rootcrs.bat` script replaces the `rootcrs.pl` script in the Grid home for Oracle Grid Infrastructure for a cluster.

If the software in the Oracle home is not running (for example, after an unsuccessful installation), then the `deinstall` cannot determine the configuration and you must provide all the configuration details either interactively or in a response file.

Files Deleted by the deinstall Command

The `deinstall` command removes Oracle software and files from your system.

When you run `deinstall`, if the central inventory (`Inventory`) contains no other registered homes besides the home that you are deconfiguring and removing, then the `deinstall` removes the following files and directory contents in the Oracle base directory of the Oracle Database installation owner:

- `admin`
- `cfgtoollogs`
- `checkpoints`
- `diag`
- `oradata`
- `fast_recovery_area`

Oracle strongly recommends that you configure your installations using an Optimal Flexible Architecture (OFA) configuration, and that you reserve Oracle base and Oracle home paths for exclusive use of Oracle software. If you have any user data in these locations in the Oracle base that is owned by the user account that owns the Oracle software, then `deinstall` deletes this data.

Caution:

The `deinstall` command deletes Oracle Database configuration files, user data, and fast recovery area (FRA) files even if they are located outside of the Oracle base directory path.

Identifying All Instances On a Cluster

You can identify the database instances on your cluster using either SRVCTL or the Windows Services control interface.

- [Identifying All Instances On a Cluster Using SRVCTL](#)
You can use SRVCTL to identify all database instances associated with an Oracle home.
- [Identifying All Instances On a Cluster Using the Windows Services Control Manager](#)
You can use Windows Services Control Manager to identify all database instances and services associated with an Oracle home.

Identifying All Instances On a Cluster Using SRVCTL

You can use SRVCTL to identify all database instances associated with an Oracle home.

- Enter the following command, where `dbname` is the name of the database:

```
C:\> srvctl status database -db dbname
```

Identifying All Instances On a Cluster Using the Windows Services Control Manager

You can use Windows Services Control Manager to identify all database instances and services associated with an Oracle home.

1. Log in to a cluster node as a user with Administrator privileges.
2. Use the Windows Services Control Manager to locate the Oracle services related to the Oracle home.

Look for any Oracle services (their names begin with `Ora`) that access the Oracle home you are removing and have the status `Started`.

To determine which Oracle home a service is associated with, check "Path to Executable" for a service to see the directory where the executable for the service is located.

deinstall Command Reference

You can run the `deinstall` command to remove Oracle software. You can run this command from an Oracle home directory after installation.

Purpose

`deinstall` stops Oracle software, and removes Oracle software and configuration files on the operating system for a specific Oracle home.

File Path

`%ORACLE_HOME%\deinstall`

Prerequisites

Before you run the `deinstall` command for Oracle Grid Infrastructure installations:

- Dismount Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and disable Oracle Automatic Storage Management Dynamic Volume Manager (Oracle ADVM).
- If Grid Naming Service (GNS) is in use, then notify your DNS administrator to delete the subdomain entry from the DNS.

Syntax When Using the `deinstall.bat` Program

`deinstall.bat`

```
[-silent] [-checkonly] [-paramfile complete path on input
parameter properties file]
[-checkonly]
[-local]
[-paramfile complete path of input parameter properties
file]
[-params name1=value [name2=value . . .]]
[-o complete path of directory for saving files]
[-tmpdir complete path of temporary directory to use]
[-logdir complete path of log directory to use]
[-skipLocalHomeDeletion]
```

```
[-skipRemoteHomeDeletion]
[-help]
```

Options

Table 7-1 Options for the Deinstallation Tool

Command Option	Description
<code>home</code> <i>complete path of Oracle home</i>	Specify this option to indicate the home path of the Oracle home to check or deinstall. To deinstall Oracle software using the <code>deinstall.bat</code> command located within the Oracle home being removed, provide a response file in a location outside the Oracle home, and do not use the <code>-home</code> option. If you run <code>deinstall.bat</code> from the <code>%ORACLE_HOME%\deinstall</code> path, then the <code>-home</code> option is not required because the command knows from which home it is being run.
<code>silent</code>	Specify this option to run the <code>deinstall</code> in noninteractive mode. This option requires one of the following: - A working system that it can access to determine the installation and configuration information; the <code>-silent</code> option does not work with failed installations. - A response file that contains the configuration values for the Oracle home that is being deinstalled or deconfigured.
<code>checkonly</code>	Specify this option to check the status of the Oracle software home configuration. Running the <code>deinstall</code> command with the <code>-checkonly</code> option does not remove the Oracle configuration. This option generates a response file that you can use with the <code>deinstall.bat</code> command. When you use the <code>-checkonly</code> option to generate a response file, you are prompted to provide information about your system. You can accept the default value the <code>deinstall</code> command has obtained from your Oracle installation, indicated inside brackets (<code>[]</code>), or you can provide different values. To accept the defaults, press <code>Enter</code> at each prompt.
<code>local</code>	Specify this option on a multinode environment to deconfigure Oracle software in a cluster. When you run <code>deinstall.bat</code> with this option, it deconfigures and deinstalls the Oracle software only on the local node (the node on which you run <code>deinstall.bat</code>) for non-shared Oracle home directories. The <code>deinstall</code> command does not deinstall or deconfigure Oracle software on remote nodes.

Table 7-1 (Cont.) Options for the Deinstallation Tool

Command Option	Description
<code>paramfile</code> <i>complete path of input parameter properties file</i>	(Optional) You can specify this option to run <code>deinstall.bat</code> with a response file in a location other than the default. When you use this option, provide the complete path where the response file is located. If you run the <code>deinstall.bat</code> command from the Oracle home that you plan to deinstall, then you do not need to specify the <code>-paramfile</code> option. The default location of the response file is <code>%ORACLE_HOME%\deinstall\response</code> .
<code>params</code> <i>name1=value[name2=value name3=value...]</i>	Use this option with a response file to override one or more values in a response file that you created.
<code>o</code> <i>complete path of directory for saving response files</i>	Use this option to provide a path other than the default location where the response file (<code>deinstall.rsp.tmp1</code>) is saved. The default location of the response file is <code>%ORACLE_HOME%\deinstall\response</code> .
<code>tmpdir</code> <i>complete path of temporary directory to use</i>	Specifies a non-default location where the <code>deinstall</code> command writes the temporary files for the deinstallation.
<code>logdir</code> <i>complete path of log directory to use</i>	Specifies a non-default location where the <code>deinstall</code> command writes the log files for the deinstallation.
<code>skipLocalHomeDeletion</code>	Specify this option in Oracle Grid Infrastructure installations on a multinode environment to deconfigure a local Grid home without deleting the Grid home.
<code>skipRemoteHomeDeletion</code>	Specify this option in Oracle Grid Infrastructure installations on a multinode environment to deconfigure a remote Grid home without deleting the Grid home.
<code>help</code>	Use the <code>-help</code> option to obtain additional information about the command option flags.

Location of Log Files for the Deinstallation Tool

If you use the `deinstall.bat` command located in an Oracle home, then the `deinstall` writes log files in the `C:\Program Files\Oracle\Inventory\logs` directory.

If you are using the `deinstall.bat` command to remove the last Oracle home installed on the server, then the log files are written to the current user's home directory. For example, if you are logged in as the domain user `RACDBA\dba1`, then the log files are stored in the directory `C:\Users\dba1.RACDBA\logs`.

Using the Deinstallation Tool to Remove Oracle RAC

You can run the deinstallation tool in multiple ways.

- [Running the deinstall Command From an Oracle Home](#)
You can run the `deinstall` command from an Oracle home.

- [Generating a Response File For Use With the deinstall Command](#)
To use a response file with the `deinstall` command, you must first create the response file.

Running the deinstall Command From an Oracle Home

You can run the `deinstall` command from an Oracle home.

1. The default method for running the `deinstall` command is from the `deinstall` directory in the Oracle home as the Oracle Installation user:

```
C:\> %ORACLE_HOME%\deinstall\deinstall.bat
```

2. Provide information about your servers as prompted or accept the defaults.

The `deinstall` command stops Oracle software, and removes Oracle software and configuration files on the operating system.

Example 7-1 Running deinstall.bat From Within the Oracle Home

The most common method of running the `deinstall` command is to use the version installed in the Oracle home being removed. The `deinstall` command determines the software configuration for the local Oracle home, and then provides default values at each prompt. You can either accept the default value, or override it with a different value. If the software in the Oracle home is not running (for example, after an unsuccessful installation), then the `deinstall` command cannot determine the configuration, and you must provide all the configuration details either interactively or in a response file. To use the `deinstall` command located in the current Oracle home directory, issue the following commands while logged in as a member of the Administrators group:

```
C:\> C:\app\oracle\product\19.0.0\dbhome_1\deinstall\deinstall.bat
```

Provide additional information as prompted.



Note:

When using the `deinstall` command from a location other than within the Oracle home being removed, you must specify the `-home` option on the command line.

Generating a Response File For Use With the deinstall Command

To use a response file with the `deinstall` command, you must first create the response file.

You can generate the a response file by running the `deinstall.bat` command with the `-checkonly` and `-o` options before you run the command to deinstall the Oracle home, or you can use the response file template and manually edit it to create the response file.

Alternatively, you can use the response file template located at `%ORACLE_HOME%\deinstall\response\deinstall.rsp.tmpl`.

- To generate the response file `deinstall_dbhome_1.rsp` using the `deinstall.bat` command located in the Oracle home and the `-checkonly` option, enter a command similar to the following, where `C:\app\oracle\product\19.0.0\dbhome_1` is the location of

the Oracle home and `C:\Users\oracle` is the directory in which the generated response file is created:

```
C:\> app\oracle\product\19.0.0\dbhome_1\deinstall\deinstall.bat -checkonly  
-o C:\Users\oracle\
```

For example, to use the response file with the `deinstall` command, run the following command:

```
%ORACLE_HOME%\deinstall\deinstall.bat -paramfile response_file
```

Cleaning Up After a Failed Installation

If an installation fails, then you must remove the Oracle home directory and remove all files that Oracle Universal Installer (OUI) created during the attempted installation.

1. Run OUI to deinstall Oracle RAC.
2. Manually remove the directory that was used as the Oracle home directory during the installation.
3. Remove the following Windows Registry keys created by OUI during the previous installation attempt:

```
HKEY_LOCAL_MACHINE\SOFTWARE\ODBC\ODBCINST.INI  
HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE\KEY_ORACLE_HOME_NAME
```

After you have completed these steps, you can start the installation again.



See Also:

“Removing Oracle Real Application Clusters Software” in this guide for more information about removing Oracle RAC software.

A

Using Scripts or Response Files to Create Oracle RAC Databases

During noninteractive installations you can create Oracle Real Application Clusters (Oracle RAC) databases using scripts.



Note:

The scripts generated by Database Configuration Assistant (DBCA) are for reference only. Oracle strongly recommends that you use DBCA to create a database.

- [Using DBCA to Generate Installation Scripts for Oracle RAC](#)
You can generate scripts that create an Oracle RAC database and prepare the database for use.
- [About DBCA Noninteractive \(Silent\) Configuration for Oracle RAC](#)
You can perform a noninteractive, or *silent* configuration of Oracle RAC using Oracle Database Configuration Assistant (DBCA).
- [Using DBCA Commands for Noninteractive \(Silent\) Configuration of Oracle RAC](#)
You can use Oracle Database Configuration Assistant (DBCA) in non-interactive mode to create an Oracle RAC database.
- [How Response Files Work](#)
Response files can assist you with installing an Oracle product multiple times on multiple computers.
- [Preparing Response Files](#)
There are two methods you can use to prepare response files for silent mode or response file mode installations.
- [Running Oracle Universal Installer Using a Response File](#)
After creating the response file, run Oracle Universal Installer at the command line, specifying the response file you created, to perform the installation.
- [Running Configuration Assistants Using Response Files](#)
You can run configuration assistants in response file mode to configure and start Oracle software after it is installed on your system.
- [Postinstallation Configuration Using Response File Created During Installation](#)
Use response files to configure Oracle software after installation. You can use the same response file created during installation to also complete postinstallation configuration.
- [Postinstallation Configuration Using the ConfigToolAllCommands Script](#)
You can create and run a response file configuration after installing Oracle software.

Using DBCA to Generate Installation Scripts for Oracle RAC

You can generate scripts that create an Oracle RAC database and prepare the database for use.

1. Start Oracle Database Configuration Assistant (DBCA) and select your preferred options to build the Oracle RAC database.
 - a. On the Creation Options page of your DBCA session, deselect Create Database.
 - b. Select **Generate Database Creation Scripts**.
 - c. Click **Finish**.

You can accept the default destination directory for the scripts, or browse for a different location. In either case, note the path name for use in the next step.
2. Go to the directory where DBCA created the scripts, and review the SQL scripts to ensure that they contain the statements to build a database with the characteristics you require.

If the scripts do not contain the statements for the specific database characteristics you need, then Oracle recommends that you rerun DBCA to create scripts with the desired configuration rather than editing the scripts yourself.
3. On each cluster node you identified during your DBCA session, run the script `sid.bat`, where `sid` is the SID prefix that you entered on the DBCA Database Name page.
4. Set the initialization parameter, `cluster_database`, to the value `TRUE` in your SPFILE by entering an `ALTER SYSTEM` statement in SQL*Plus, or by uncommenting the parameter in the PFILE for each instance.
5. Configure Oracle Net Services to support your new database and instances.
6. Set the `REMOTE_LISTENER` parameter to the SCAN (using the Easy Connect Naming syntax `scanname:scanport`) in your SPFILE by entering an `ALTER SYSTEM` statement in SQL*Plus, or by uncommenting the parameter in the PFILE for each instance.
7. Run the Server Control Utility (SRVCTL) to configure and start database and instance applications as described in *Oracle Real Application Clusters Administration and Deployment Guide*.



See Also:

- ["Understanding the Oracle RAC Installed Configuration"](#)
- ["Selecting DBCA Options to Create an Oracle RAC or Oracle RAC One Node Database"](#) for additional information about running a DBCA session.

About DBCA Noninteractive (Silent) Configuration for Oracle RAC

You can perform a noninteractive, or *silent* configuration of Oracle RAC using Oracle Database Configuration Assistant (DBCA).

To perform a silent configuration, you must have completed an Oracle Grid Infrastructure (Oracle Clusterware and Oracle Automatic Storage Management (Oracle ASM)) installation.

You can use DBCA to create a database from templates supplied by Oracle, or from templates that you create. The templates contain settings optimized for a particular type of workload.

Oracle provides templates for the following two workload types:

- General purpose or transaction processing
- Data warehouse

For more complex environments, you can select the **Custom Database** option. This option does not use templates and results in a more extensive installation interview, which means that it takes longer to create your database.

Using DBCA Commands for Noninteractive (Silent) Configuration of Oracle RAC

You can use Oracle Database Configuration Assistant (DBCA) in non-interactive mode to create an Oracle RAC database.

Example A-1 Using DBCA in Silent Mode to Create an Oracle RAC Database

You can use the following command syntax to create an Oracle RAC database using the general purpose template, placing the data files in an existing Oracle ASM disk group. Nodes `node1` and `node2` are the cluster nodes on which Oracle RAC database instances are created. The disk group name is `+ASMgrp1`. An Oracle Home User was specified for this installation, and is indicated with the `-serviceUserPassword` option:

```
%ORACLE_HOME%\bin\dbca -silent -createDatabase -templateName
General_Purpose.dbc
-gdbName %DBNAME% -sid %ORACLE_SID% -sysPassword -systemPassword
-sysmanPassword -dbSNMPPassword -serviceUserPassword
-emConfiguration LOCAL -storageType ASM -diskGroupName +ASMgrp1
-datafileJarLocation %ORACLE_HOME%\assistants\dbca\templates
-nodeinfo node1,node2 -characterSet WE8MSWIN1252
-obfuscatedPasswords false -sampleSchema false -asmSysPassword
```

After you run this command, if you did not include the passwords as values in the above command, then DBCA prompts you for the passwords for the SYS, SYSTEM, SYSMAN, DBSNMP, Oracle Home (or Oracle Service) and SYSASM users, for example:

```
Enter SYS user password:
password
Enter SYSTEM user password:
password
...
```



See Also:

Oracle Database Administrator's Guide for a complete description of the `dbca` commands and options

How Response Files Work

Response files can assist you with installing an Oracle product multiple times on multiple computers.

When you start the installer, you can use a response file to automate the installation and configuration of Oracle software, either fully or partially. The installer uses the values contained in the response file to provide answers to some or all installation prompts.

Typically, the installer runs in interactive mode, which means that it prompts you to provide information in graphical user interface (GUI) screens. When you use response files to provide this information, you run the installer from a command prompt using either of the following modes:

- Silent mode

If you include responses for all of the prompts in the response file and specify the `-silent` option when starting the installer, then it runs in silent mode. During a silent mode installation, the installer does not display any screens. Instead, it displays progress information in the terminal that you used to start it.

- Response file mode

If you include responses for some or all of the prompts in the response file and omit the `-silent` option, then the installer runs in response file mode. During a response file mode installation, the installer displays all the screens, screens for which you specify information in the response file, and also screens for which you did not specify the required information in the response file.

You define the settings for a silent or response file installation by entering values for the variables listed in the response file. For example, to specify the Oracle home name, supply the appropriate value for the `ORACLE_HOME` variable:

```
ORACLE_HOME=C:\app\oracle\product\19.0.0\dbhome_1
```

Another way of specifying the response file variable settings is to pass them as command line arguments when you run the installer. For example:

```
-silent directory_path
```

In this command, *directory_path* is the path of the database directory on the DVD, or the path of the directory on the hard drive.

Ensure that you enclose the variable and its setting in double-quotes.

- [Deciding to Use Silent Mode or Response File Mode](#)

There are several reasons for running the installer in silent mode or response file mode.

- [Creating a Database Using Oracle ASM for Database Files in Silent Mode](#)

Creating an Oracle Real Application Clusters (Oracle RAC) database that uses Oracle Automatic Storage Management (Oracle ASM) for storage is a multi-step process.

- [Using Response Files](#)

Use these general steps for installing and configuring Oracle products using the installer in silent or response file mode.

Deciding to Use Silent Mode or Response File Mode

There are several reasons for running the installer in silent mode or response file mode.

Table A-1 Reasons for Using Silent Mode or Response File Mode

Mode	Reasons to Use
Silent	Use silent mode for the following installations: • To complete an unattended installation, which you schedule using operating system utilities • To complete several similar installations on multiple systems without user interaction • Install the software on a system that cannot display the Oracle Universal Installer (OUI) graphical user interface OUI displays progress information on the terminal that you used to start it, but it does not display any of the installer screens.
Response file	Use response file mode to complete similar Oracle software installations on more than one system, providing default answers to some, but not all the installer prompts. If you do not specify information required for a particular OUI screen in the response file, then the installer displays that screen. OUI suppresses screens for which you have provided all of the required information.

Creating a Database Using Oracle ASM for Database Files in Silent Mode

Creating an Oracle Real Application Clusters (Oracle RAC) database that uses Oracle Automatic Storage Management (Oracle ASM) for storage is a multi-step process.

Before you create a database that uses Oracle ASM, you must run the `root.bat` script. For this reason, you cannot create a database using Oracle ASM as the storage option for database files during a silent-mode installation.

Note:

This limitation applies only to databases that use Oracle Automatic Storage Management as the storage option for database files. You can create a database that uses the file system option during a silent-mode installation.

1. Complete a software-only installation of Oracle RAC using silent-mode.
2. Run Oracle Database Configuration Assistant (DBCA) in silent mode.

See Also:

["Performing a Software-Only Installation of Oracle Database"](#)

Using Response Files

Use these general steps for installing and configuring Oracle products using the installer in silent or response file mode.



Note:

You must complete all required preinstallation tasks on a system before running the installer in silent or response file mode.

1. Verify the Windows Registry key `HKEY_LOCAL_MACHINE\Software\Oracle` exists and that the value for `inst_loc` is the location of the Oracle Inventory directory on the local node.

To install Oracle RAC, you must first have installed Oracle Grid Infrastructure on your cluster nodes. The Oracle Inventory directory was created and added to the Windows registry during the installation of Oracle Grid Infrastructure. If the `inst_loc` key does not exist in the Windows registry, then install Oracle Grid Infrastructure for a cluster before continuing.



Note:

Changing the value for `inst_loc` in the Windows registry is not supported after the installation of Oracle software

2. Prepare a response file.
3. Run the installer in silent or response file mode.
4. If you completed a software-only installation, then run Oracle Database Configuration Assistant (DBCA) in silent or response file mode.

Preparing Response Files

There are two methods you can use to prepare response files for silent mode or response file mode installations.

- [About Response File Templates](#)
Oracle provides response file templates for each product and installation type and for each configuration tool.
- [Editing a Response File Template](#)
You can copy and modify a response file template for each product and installation type and for each configuration tool.
- [Recording Response Files](#)
You can use the installer in interactive mode to record response files, which you can then edit and use to complete silent mode or response file mode installations.

About Response File Templates

Oracle provides response file templates for each product and installation type and for each configuration tool.

For Oracle Database, the response file templates are located in the `database\response` directory on the installation media and in the `Oracle_home\inventory\response` directory. For Oracle Grid Infrastructure, the response file templates are located in the `Grid_home\install\response` directory after the software is installed.

All response file templates contain comment entries, sample formats, examples, and other useful instructions. Read the response file instructions to understand how to specify values for the response file variables, so that you can customize your installation.

The following response files are provided with this software:

Table A-2 Response Files for Oracle Database and Oracle Grid Infrastructure

Response File	Used For
<code>db_install.rsp</code>	Silent configuration of Oracle Database software
<code>dbca.rsp</code>	Silent creation and configuration of an Oracle Database using Oracle Database Configuration Assistant (DBCA)
<code>netca.rsp</code>	Silent configuration of Oracle Net using NETCA
<code>grid_install.rsp</code>	Silent configuration of Oracle Grid Infrastructure installations

 Caution:

When you modify a response file template and save a file for use, the response file may contain plain text passwords. Ownership of the response file must be given to the Oracle software installation owner only, and access restricted to the response file. Oracle strongly recommends that database administrators or other administrators delete or secure response files when they are not in use.

Editing a Response File Template

You can copy and modify a response file template for each product and installation type and for each configuration tool.

To copy and modify a response file, perform the following steps:

1. Copy the response file from the response file directory to a directory on your system.

```
copy Oracle_home\install\response\product_timestamp.rsp local_directory
```

2. Open the response file in a text editor.
3. Follow the instructions in the file to edit it.

 Note:

The installer or configuration assistant fails if you do not correctly configure the response file. Also, ensure that your response file name has the `.rsp` suffix.

4. Secure the response file.

Ensure that only the user that installed the Oracle software can view or modify response files. Consider deleting the modified response file after the installation succeeds.

 Note:

A fully specified response file for an Oracle Grid Infrastructure installation or an Oracle Database installation can contain the passwords for:

- Oracle Automatic Storage Management (Oracle ASM) administrative accounts
- Database administrative accounts
- A user who is a member of the operating system group `ORA_DBA` (required for automated backups)

Recording Response Files

You can use the installer in interactive mode to record response files, which you can then edit and use to complete silent mode or response file mode installations.

This method is useful for Advanced or software-only installations. You can save all the installation steps into a response file during installation by clicking **Save Response File** on the Summary page. You can use the generated response file for a silent installation later.

When you record the response file, you can either complete the installation, or you can exit from the installer on the Summary page, before the installer starts to copy the software to the local disk.

If you use record mode during a response file mode installation, then the installer records the variable values that were specified in the original source response file into the new response file.

 Note:

You cannot save passwords while recording the response file.

1. Complete preinstallation tasks for a standard installation.

When you run the installer to record a response file, it checks the system to verify that it meets the requirements to install the software. For this reason, Oracle recommends that you complete all of the required preinstallation tasks and record the response file while completing an installation.

2. Log in as the Oracle Installation User. Ensure that the Oracle Installation User has permissions to create or write to the Oracle home path that you specify during installation.
3. Start the installer. On each installation screen, specify the required information.
4. When the installer displays the Summary screen, perform the following steps:
 - a. Click **Save Response File**. In the pop up window, specify a file name and location to save the values for the response file, then click **Save**.
 - b. Click **Finish** to continue with the installation.

Click **Cancel** if you do not want to continue with the installation. The installation stops, but the recorded response file is retained.

 Note:

Your response file name must end with the `.rsp` suffix.

5. If you do not complete the installation, then delete the Oracle home directory that the installer created using the path you specified in the Specify File Locations screen.
6. Before you use the saved response file on another system, edit the file and make any required changes. Use the instructions in the file as a guide when editing it.

Running Oracle Universal Installer Using a Response File

After creating the response file, run Oracle Universal Installer at the command line, specifying the response file you created, to perform the installation.

The Oracle Universal Installer executables, `setup.bat` and `gridSetup.bat`, provide several options. For help information on the full set of these options, run the `gridSetup.bat` or `setup.bat` command with the `-help` option. For example:

- For Oracle Database:

```
db_home> setup.bat -help
```

- For Oracle Grid Infrastructure:

```
Grid_home> gridSetup.bat -help
```

The help information appears in your session window after a short period of time.

To run the installer using a response file, perform the following steps:

1. Complete the preinstallation tasks for a normal installation.
2. Log in as an Administrator user or the user that installed the software.
3. To start the installer in silent or response file mode, enter a command similar to the following:
 - For Oracle Database:

```
C:\> directory_path\setup.bat [-silent] [-noconfig] \  
-responseFile response_filename
```

- For Oracle Grid Infrastructure:

```
C:\> directory_path\gridSetup.bat [-silent] [-noconfig] \  
-responseFile response_filename
```

 Note:

Do not specify a relative path to the response file. If you specify a relative path, then the installer fails.

In this example:

- `directory_path` is the path of the DVD or the path of the directory on the hard drive where you have copied the installation software.
- `-silent` runs the installer in silent mode.
- `-noconfig` suppresses running the configuration assistants during installation, and a software-only installation is performed instead.
- `response_filename` is the full path and file name of the installation response file that you configured.

Running Configuration Assistants Using Response Files

You can run configuration assistants in response file mode to configure and start Oracle software after it is installed on your system.

To run configuration assistants in response file mode, you must first copy and edit a response file template.



Note:

If you copied the software to a hard disk, then the response file template is located in the `\response` directory.

- [Silent Mode of Database Configuration Assistant](#)
Use the `-silent` option in combination with the `-responseFile` option to set the mode to silent.
- [Running Oracle DBCA Using Response Files](#)
You can run Oracle DBCA in response file mode to configure and start an Oracle database on the system.
- [Running Oracle Net Configuration Assistant Using Response Files](#)
You can run Oracle Net Configuration Assistant (NETCA) in silent mode to configure and start an Oracle Net listener on the system, configure naming methods, and configure Oracle Net service names.

Silent Mode of Database Configuration Assistant

Use the `-silent` option in combination with the `-responseFile` option to set the mode to silent.

In the silent mode, Database Configuration Assistant uses values that you specify, in the response file or as command-line options, to create a database. No window or user interface is displayed in the silent mode.

Running Oracle DBCA Using Response Files

You can run Oracle DBCA in response file mode to configure and start an Oracle database on the system.

To run Oracle DBCA in response file mode, you must copy and edit a response file template.

1. Oracle provides a response file template named `dbca.rsp` in the response directory with the installation. Copy the `dbca.rsp` response file template from the response file directory to a directory on your system.

If you have copied the software to a hard drive, then the response files are located in the `\response` directory.

 Note:

As an alternative to editing the response file template, you can also create a database by specifying all required information as command-line options when you run Oracle DBCA. For information about the list of options supported, enter the following command:

```
C:\> %ORACLE_HOME%\bin\dbca -help
```

2. Log in as the Oracle Home user. Set the `%ORACLE_HOME%` environment variable to the correct Oracle home directory.
3. Open the response file in a text editor.
4. Edit the file, following the instructions in the file.

In response file mode, Oracle DBCA uses values that you specify in the response file or as command-line options, to create a database.

 Note:

Oracle DBCA fails if you do not correctly configure the response file.

5. Open a command-line window. Change directories to the location of the Oracle home directory.
6. Use a command similar to the following example to run Oracle DBCA in silent or response file mode using a response file:

```
C:\> %ORACLE_HOME%\bin\dbca [-silent] -createDatabase -responseFile  
  \local_dir\dbca.rsp
```

In this example:

- The `-silent` option runs Oracle DBCA in silent mode, suppressing user prompts.
- `-createDatabase` creates the database.
- `local_dir` is the full path of the directory where the `dbca.rsp` response file is located.

As Oracle DBCA configures and starts the database, it displays a window that contains status messages and a progress bar. The window that Oracle DBCA displays is the same window that is displayed when you choose to create a preconfigured database during an Oracle Database Enterprise Edition, Standard Edition, or Standard Edition 2 (SE2) installation.

 See Also:

- ["Multiple Oracle Home Directories on Windows"](#) for more information about changing the current setting for Oracle Home.
- *Oracle Database Administrator's Guide* for information about using Oracle DBCA in noninteractive, or silent mode, to create a database
- *Oracle Automatic Storage Management Administrator's Guide* for information about running Oracle Automatic Storage Management Configuration Assistant (ASMCA) in noninteractive mode

Running Oracle Net Configuration Assistant Using Response Files

You can run Oracle Net Configuration Assistant (NETCA) in silent mode to configure and start an Oracle Net listener on the system, configure naming methods, and configure Oracle Net service names.

To run NETCA in silent mode, you must copy and edit a response file template. Oracle provides a response file template named `netca.rsp` in the `%ORACLE_HOME%\assistants\netca` directory.

To run NETCA using a response file:

1. Copy the `netca.rsp` response file template from the response file directory to a directory on your system.

```
copy \directory_path\assistants\netca\netca.rsp local_directory
```

In this example, `directory_path` is the path of the directory where you have copied the installation binaries.

If the software is staged on a hard drive, or has already been installed, then you can edit the file in the `response` directory located on the local disk instead.

2. Open the response file in a text editor.
3. Follow the instructions in the file to edit it.

 Note:

NETCA fails if you do not correctly configure the response file.

4. Log in as the Oracle Home user. Set the `%ORACLE_HOME%` environment variable to the correct Oracle home directory.
5. Enter a command similar to the following to run NETCA in silent mode:

```
C:\> Oracle_home\bin\netca -silent -responsefile X:\local_dir\netca.rsp
```

In this command:

- The `-silent` option runs NETCA in silent mode.

- `X:\local_dir` is the full path of the directory where you copied the `netca.rsp` response file template, where `X` represents the drive on which the file is located, and `local_dir` the path on that drive.

Related Topics

- [Multiple Oracle Home Directories on Windows](#)
Install each Oracle product in its own Oracle home.
- [Changing the Current Setting for Oracle Home](#)
Use Oracle Universal Installer (OUI) to change the current Oracle home.

Postinstallation Configuration Using Response File Created During Installation

Use response files to configure Oracle software after installation. You can use the same response file created during installation to also complete postinstallation configuration.

- [Using the Installation Response File for Postinstallation Configuration](#)
Starting with Oracle Database 12c release 2 (12.2), you can use the response file created during installation to also complete postinstallation configuration.
- [Running Postinstallation Configuration Using a Response File](#)
Complete this procedure to run configuration assistants configuration with the `-executeConfigTools` command.

Using the Installation Response File for Postinstallation Configuration

Starting with Oracle Database 12c release 2 (12.2), you can use the response file created during installation to also complete postinstallation configuration.

Run the installer with the `-executeConfigTools` option to configure configuration assistants after installing Oracle Grid Infrastructure or Oracle Database. You can use the response file located at `%ORACLE_HOME%\install\response\grid_timestamp.rsp` to obtain the passwords required to run the configuration tools. You must update the response file with the required passwords before running the `-executeConfigTools` command.

Oracle strongly recommends that you maintain security with a password response file. The owner of the response file must be the installation owner user.

Example A-2 Response File Passwords for Oracle Grid Infrastructure

```
oracle.install.crs.config.ipmi.bmcPassword=password
oracle.install.asm.SYSASMPassword=GRID_HOME\gridSetup.bat -executeConfigTools
-responseFile %ORACLE_HOME%
\install\response\grid_time_stamp.rsporacle.install.asm.monitorPassword=password
rd
oracle.install.config.emAdminPassword=password
oracle.install.OracleHomeUserPassword=password
```

If you do not have a BMC card, or you do not want to enable IPMI, then leave the `ipmi.bmcPassword` input field blank.

If you do not want to enable Oracle Enterprise Manager for management, then leave the `emAdminPassword` password field blank.

If you did not specify an Oracle Home user for the Oracle Grid Infrastructure installation, then leave the `OracleHomeUserPassword` field blank.

Example A-3 Response File Passwords for Oracle Grid Infrastructure for a Standalone Server (Oracle Restart)

```
oracle.install.asm.SYSASMPassword=password
oracle.install.asm.monitorPassword=password
oracle.install.config.emAdminPassword=password
oracle.install.OracleHomeUserPassword=password
```

If you do not want to enable Oracle Enterprise Manager for management, then leave the `emAdminPassword` password field blank.

If you did not specify an Oracle Home user for the Oracle Grid Infrastructure for a Standalone Server (Oracle Restart) installation, then leave the `OracleHomeUserPassword` field blank.

Example A-4 Response File Passwords for Oracle Database

This example illustrates the passwords to specify for use with the database configuration assistants.

```
oracle.install.db.config.starterdb.password.SYS=password
oracle.install.db.config.starterdb.password.SYSTEM=password
oracle.install.db.config.starterdb.password.DBSNMP=password
oracle.install.db.config.starterdb.password.PDBADMIN=password
oracle.install.db.config.starterdb.emAdminPassword=password
oracle.install.db.config.asm.ASMNMPPassword=password
oracle.install.OracleHomeUserPassword=password
```

You can also specify `oracle.install.db.config.starterdb.password.ALL=password` to use the same password for all database users.

Oracle Database configuration assistants require the SYS, SYSTEM, and DBSNMP passwords for use with Oracle Database Configuration Assistant (DBCA). Specify the following passwords, depending on your system configuration:

- If the database uses Oracle ASM for storage, then you must specify a password for the `ASMSNMPPassword` variable. If you are not using Oracle ASM, then leave the value for this password variable blank.
- If you create a multitenant container database (CDB) with one or more pluggable databases (PDBs), then you must specify a password for the `PDBADMIN` variable. If you are not using Oracle ASM, then leave the value for this password variable blank.
- If you did not specify an Oracle Home user for the Oracle Database installation, then leave the `OracleHomeUserPassword` field blank.

Running Postinstallation Configuration Using a Response File

Complete this procedure to run configuration assistants configuration with the `-executeConfigTools` command.

1. Edit the response file and specify the required passwords for your configuration. You can use the response file created during installation, located at

`Oracle_home\install\response\product_timestamp.rsp`. For example, for Oracle Grid Infrastructure:

```
oracle.install.asm.SYSASMPassword=password
oracle.install.config.emAdminPassword=password
```

2. Change directory to the Oracle home containing the installation software. For example, for Oracle Grid Infrastructure:

```
cd Grid_home
```

3. Run the configuration script using the following syntax:

For Oracle Grid Infrastructure:

```
setup.exe -executeConfigTools -responsefile
Grid_home\install\response\product_timestamp.rsp
```

For Oracle Database:

```
setup.exe -executeConfigTools -responseFile
Oracle_home\install\response\product_timestamp.rsp
```

For Oracle Database, you can also edit and use the response file located in the directory `Oracle_home\inventory\response\`:

```
setup.exe -executeConfigTools -responseFile
Oracle_home\inventory\response\db_install.rsp
```

The postinstallation configuration tool runs the installer in the graphical user interface mode, displaying the progress of the postinstallation configuration.

Specify the `[-silent]` option to run the postinstallation configuration in the silent mode.

For example, for Oracle Grid Infrastructure:

```
setup.exe -executeConfigTools -responseFile
Grid_home\install\response\grid_2016-09-09_01-03-36PM.rsp -silent
```

For Oracle Database:

```
setup.exe -executeConfigTools -responseFile
Oracle_home\inventory\response\db_2016-09-09_01-03-36PM.rsp -silent
```

Postinstallation Configuration Using the ConfigToolAllCommands Script

You can create and run a response file configuration after installing Oracle software.

The `configToolAllCommands` script requires users to create a second response file, of a different format than the one used for installing the product. Starting with Oracle Database 12c

Release 2 (12.2), the `configToolAllCommands` script is deprecated and may be desupported in a future release.

Starting with Oracle Database 18c Release, use the `executeConfigTools` script to complete the postinstall configuration.

- [About the Postinstallation Configuration File](#)
The configuration assistants are started with a script called `configToolAllCommands`.
- [Creating a Password Response File](#)
Use these steps to create a password response file for use with the configuration assistants.
- [Performing Postinstallation Configuration Using a Response File](#)
To run configuration assistants with the `configToolAllCommands` script in silent mode or response file mode, perform these steps.



See Also:

"[Postinstallation Configuration Using Response File Created During Installation](#)" for an alternate method of postinstallation configuration of Oracle software using the same response file created at the time of installation.

About the Postinstallation Configuration File

The configuration assistants are started with a script called `configToolAllCommands`.

When you perform an installation using silent mode or response file mode, you provide information about your servers in a response file that you otherwise provide manually using a graphical user interface. However, the response file does not contain passwords for user accounts that configuration assistants require after software installation is complete. To run the configuration assistants after the installation completes in silent mode, you must run the `configToolAllCommands` script and provide the passwords used by the assistants in a password file.

You can run the `configToolAllCommands` script in silent mode by using a password response file. The script uses the passwords in the file to run the configuration tools in succession to complete the software configuration. If you keep the password file to use when cloning installations, then Oracle strongly recommends that you store the password file in a secure location.

You can also use the password file to restart a failed installation. If you stop an installation to fix an error, then you can rerun the configuration assistants using `configToolAllCommands` and a password response file.

The `configToolAllCommands` password response file has the following options:

- `oracle.crs` for Oracle Grid Infrastructure components or `oracle.server` for Oracle Database components that the configuration assistants configure.
- `variable_name` is the name of the configuration file variable.
- `value` is the desired value to use for configuration.

The command syntax is as follows:

```
internal_component_name|variable_name=value
```

For example, to set the password for the SYS user of Oracle ASM:

```
oracle.crs|S_ASMPASSWORD=PassWord
```

Creating a Password Response File

Use these steps to create a password response file for use with the configuration assistants.

1. Create a response file that has a name of the format *filename.properties*.
2. Open the file with a text editor, and cut and paste the sample password file contents, as shown in the example below, modifying as needed.
3. If the file is stored on a volume formatted for Windows New Technology File System (NTFS), then modify the security permissions to secure the file.

Example A-5 Sample Password Response File for Oracle RAC Databases

This example provides a template for a password response file to use with the database configuration assistants.

```
oracle.install.db.config.starterdb.password.SYS=password
oracle.install.db.config.starterdb.password.SYSTEM=password
oracle.install.db.config.starterdb.password.DBSNMP=password
oracle.install.db.config.starterdb.password.PDBADMIN=password
oracle.install.db.config.starterdb.emAdminPassword=password
oracle.install.db.config.asm.ASMSNMPPassword=password
oracle.install.db.config.OracleHomeUserPassword=password
```

If you do not want to enable access for Oracle Enterprise Manager or Oracle ASM, then leave those password fields blank.

Performing Postinstallation Configuration Using a Response File

To run configuration assistants with the `configToolAllCommands` script in silent mode or response file mode, perform these steps.

1. Change directory to `%ORACLE_HOME%\cfgtoollogs`.
2. Run the configuration script using the following syntax:

```
configToolAllCommands RESPONSE_FILE=\path\name.properties
```

or

```
setup.exe -executeConfigTools -responseFile responsefile_location -silent -
debug
```

Example A-6 Running Configuration Assistants in Response File Mode

Assume you created a password response file in the `C:\users\oracle\db` directory with a name of `cfg_db.properties`. To run the configuration assistants in response file mode to configure the Oracle software after installation, enter commands similar to the following:

```
C:\> cd %ORACLE_HOME%\cfgtoollogs  
C:\..\cfgtoollogs> configToolAllCommands RESPONSE_FILE=C:\users\oracle\db\cfg  
_db.properties
```

B

Directory Structure for Oracle RAC Environments

After you install the software, there are several directory structures for Oracle Real Application Clusters (Oracle RAC).

- [Understanding the Oracle RAC Directory Structure](#)
When you install Oracle Database 18c with Oracle RAC, several directory structures are created.
- [Directory Structures for Oracle RAC](#)
The sample Optimal Flexible Architecture (OFA)-compliant database has a hierarchical directory tree of folders.

Understanding the Oracle RAC Directory Structure

When you install Oracle Database 18c with Oracle RAC, several directory structures are created.

All subdirectories except for the Oracle Inventory directory, the Oracle Automatic Storage Management (Oracle ASM) home (if applicable), and the Oracle Clusterware home, are under a top-level Oracle base directory. The Oracle home and `admin` directories are also located under the Oracle base directory.



See Also:

Oracle Database Installation Guide for Microsoft Windows for more information about the Oracle home and `admin` directories

Directory Structures for Oracle RAC

The sample Optimal Flexible Architecture (OFA)-compliant database has a hierarchical directory tree of folders.

Table B-1 Directory Structure for a Sample OFA-Compliant Environment

Directory	Description
<code>%ORACLE_BASE%</code>	<code>C:\app\oracle</code> The default <code>ORACLE_BASE</code> directory, where the software was installed by the Oracle Installation user
<code>%ORACLE_BASE%\installation_type</code>	<code>C:\app\oracle\product\19.0.0</code> The type of installation under the Oracle base directory. For example, when installing Oracle Database 19c, the value for installation type is <code>product\19.0.0\db</code> .

Table B-1 (Cont.) Directory Structure for a Sample OFA-Compliant Environment

Directory	Description
%ORACLE_HOME%	The location in which the Oracle Database software is installed. You can also choose to add a counter, if you have multiple installations of the software. For example, if you are creating a second Oracle home for the Oracle Database 19c software, then the path is as follows: C:\app\oracle\product\19.0.0\dbhome_2
(%ORACLE_BASE%\installation_type\Home_name)	Note that the Oracle database home is under the Oracle base directory path. An Oracle Clusterware directory must <i>not</i> be under the Oracle base directory path where the database executable files are located.
%ORACLE_HOME%\database	The directory in which the local initialization parameter file is stored for the database.
%ORACLE_BASE%\admin	C:\app\oracle\admin The administrative directory. Note that with Oracle Database 11g and later releases, bdump, cdump, and udump files are relocated to the directory associated with %ADR_BASE%.
%ORACLE_BASE%\admin\db_unique_name	The database unique name; this is the same as dbname when the database name is 8 or fewer characters in length. For example, if your database name is sales, the directory path is as follows: C:\app\oracle\admin\sales
%ORACLE_BASE%\admin\db_unique_name\hdump	The dump destinations for the database server.
%ORACLE_BASE%\admin\db_unique_name\pfile	
%ADR_BASE%	This directory path is set by the initialization parameter DIAGNOSTIC_DEST, and the path for the Automatic Diagnostic Repository must be located on the same location available to all the nodes. By default, this path is a subset of the Oracle base directory, in the following path: %ORACLE_BASE%\diag\
%ADR_BASE%\bdump	Automatic Diagnostic Repository dump destination trace files.
%ADR_BASE%\cdump	
%ADR_BASE%\udump	
Oracle Grid Infrastructure for a cluster home (Grid home)	An OFA-compliant path for the Oracle Clusterware home. The default value is C:\app\19.0.0\grid. During the Oracle Grid Infrastructure for a cluster installation, Oracle Clusterware and Oracle Automatic Storage Management (Oracle ASM) software is installed.

Table B-1 (Cont.) Directory Structure for a Sample OFA-Compliant Environment

Directory	Description
<i>Grid_home\bin</i>	The subtree for Oracle Clusterware and Oracle ASM executable files.
<i>Grid_home\network</i>	The subtree for Oracle Net Services configuration files and utilities.

C

Preparing to Upgrade an Existing Oracle RAC Database

Understand how you can prepare an Oracle Real Application Clusters (Oracle RAC) database for patch updates or upgrade.

- [Backing Up the Oracle RAC Database](#)
Make a backup of the Oracle software installation before modifying the installed software.
- [Using CVU to Validate Readiness for Oracle RAC Upgrades](#)
Review the contents in this section to validate that your Oracle RAC cluster is ready for upgrades.



See Also:

Oracle Database Upgrade Guide for information about how to prepare for upgrading an existing database

Backing Up the Oracle RAC Database

Make a backup of the Oracle software installation before modifying the installed software.

- Before you make any changes to the Oracle software, Oracle recommends that you create a backup of the Oracle Database installation.



See Also:

- *Oracle Database Upgrade Guide* for information about creating a backup strategy
- *Oracle Database Backup and Recovery User's Guide* for information about backing a database using RMAN
- *Oracle Database Administrator's Reference for Microsoft Windows* for information about backing up a database using VSS

Using CVU to Validate Readiness for Oracle RAC Upgrades

Review the contents in this section to validate that your Oracle RAC cluster is ready for upgrades.

- [Using the CVU Database Upgrade Validation Command Options](#)
Use the Cluster Verification Utility (CVU) to check the readiness of your Oracle RAC installation for upgrades.

- [Example of Verifying System Upgrade Readiness for Oracle RAC Infrastructure](#)
You can verify that the permissions required for installing Oracle RAC have been configured on the nodes `node1` and `node2` using `cluvfy`.
- [Verifying System Readiness for Oracle Database Upgrades](#)
To avoid interruptions during the upgrade process, you can use Cluster Verification Utility to ensure your system is ready.

Using the CVU Database Upgrade Validation Command Options

Use the Cluster Verification Utility (CVU) to check the readiness of your Oracle RAC installation for upgrades.

Purpose

Running `cluvfy` with the `-pre dbinst` and `-upgrade` options performs system checks to confirm if the cluster is in a correct state for upgrading from an existing Oracle RAC installation.

Command Syntax

```
cluvfy stage -pre dbinst -upgrade -src_dbhome src_RAChome [-dbname
db_names_list] -dest_dbhome dest_RAChome -dest_version dest_version
[-dest_serviceuser username [-dest_servicepasswd]] [-verbose]
```

Command Options

Table C-1 Command Options for CLUVFY Pre-upgrade Check

Command Option	Description
<code>-src_dbhome src_RAChome</code>	The location of the source Oracle RAC home that you are upgrading, where <code>src_RAChome</code> is the path to the home that you want to upgrade.
<code>-dbname db_names_list</code>	Optional: List of unique names of the databases being upgraded.
<code>-dest_dbhome dest_RAChome</code>	The location of the upgraded Oracle RAC home, where <code>dest_RAChome</code> is the path to the Oracle RAC home.
<code>-dest_version dest_version</code>	Use the <code>dest_version</code> option to indicate the release number of the upgrade, including any patchset. The release number must include the five digits designating the release to the level of the platform-specific patch, for example: 12.2.0.1.0.
<code>-dest_serviceuser username</code>	Optional: The Oracle Home user for the destination Oracle home.
<code>-dest_servicepasswd</code>	Optional: Prompt for the Oracle Home user password
<code>-verbose</code>	Use the <code>-verbose</code> option to produce detailed output of individual checks.



See Also:

Oracle Database Administrator's Guide for information about release number format.

Example of Verifying System Upgrade Readiness for Oracle RAC Infrastructure

You can verify that the permissions required for installing Oracle RAC have been configured on the nodes `node1` and `node2` using `cluvfy`.

```
C:\..bin> cluvfy stage -pre dbinst -upgrade  
-src_dbhome C:\app\oracle\product\12.2.0\dbhome_1  
-dest_dbhome C:\app\oracle\product\19.0.0\dbhome_1  
-dest_version 19.0.0.0.0 -verbose
```

Verifying System Readiness for Oracle Database Upgrades

To avoid interruptions during the upgrade process, you can use Cluster Verification Utility to ensure your system is ready.

- Use Cluster Verification Utility to assist you with system checks in preparation for starting a database upgrade.

The installer runs the appropriate CVU checks automatically, and prompts you to fix problems before proceeding with the upgrade.

D

Configuring Read-Only Oracle Homes

Understand how read-only Oracle homes work and how you can configure read-only Oracle homes.

- [Evolution of Oracle Homes](#)
Learn about read-only Oracle home concepts like `ORACLE_BASE_HOME` and `ORACLE_BASE_CONFIG`.
- [Enabling a Read-Only Oracle Home](#)
Configure your Oracle home as a read-only Oracle home after you have performed a software-only Oracle Database installation.
- [Determining if an Oracle Home is Read-Only](#)
You can use the Registry Editor to determine if your Oracle home is a read/write or read-only Oracle home.
- [File Path and Directory Changes in Read-Only Oracle Homes](#)
Examples of hierarchical file mappings in a read-only Oracle home as compared to a read/write Oracle home.

Evolution of Oracle Homes

Learn about read-only Oracle home concepts like `ORACLE_BASE_HOME` and `ORACLE_BASE_CONFIG`.

- [About Read-Only Oracle Homes](#)
Starting with Oracle Database 18c, you can configure an Oracle home in read-only mode.
- [About Oracle Base Homes](#)
Both, in a read-only `ORACLE_HOME` and read/write `ORACLE_HOME`, the user-specific files, instance-specific files, and log files reside in a location known as the `ORACLE_BASE_HOME`.
- [About Oracle Base Config](#)
Both, in a read-only `ORACLE_HOME` and read/write `ORACLE_HOME`, the configuration files reside in a location known as `ORACLE_BASE_CONFIG`.

About Read-Only Oracle Homes

Starting with Oracle Database 18c, you can configure an Oracle home in read-only mode.

A read-only Oracle home simplifies provisioning by implementing separation of installation and configuration.

In a read-only Oracle home, all the configuration data and log files reside outside of the read-only Oracle home.

Apart from the traditional `ORACLE_BASE` and `ORACLE_HOME` directories, the following directories contain files that used to be in `ORACLE_HOME`:

- `ORACLE_BASE_HOME`
- `ORACLE_BASE_CONFIG`

**Note:**

This feature does not affect how database administrators monitor, diagnose, and tune their system performance.

About Oracle Base Homes

Both, in a read-only ORACLE_HOME and read/write ORACLE_HOME, the user-specific files, instance-specific files, and log files reside in a location known as the ORACLE_BASE_HOME.

In a read/write ORACLE_HOME, the ORACLE_BASE_HOME path is the same as the ORACLE_HOME directory. However, in a read-only ORACLE_HOME, the ORACLE_BASE_HOME directory is not co-located with ORACLE_HOME but is located at ORACLE_BASE\homes\HOME_NAME.

Where, HOME_NAME is the internal name for ORACLE_HOME.

For example, the networking directories `network\admin`, `network\trace`, and `network\log` are located in the ORACLE_BASE_HOME directory. In a read/write ORACLE_HOME the networking directories appear to be in ORACLE_HOME because ORACLE_BASE_HOME is co-located with ORACLE_HOME, whereas in a read-only ORACLE_HOME the networking directories are located in ORACLE_BASE\homes\HOME_NAME.

To print the ORACLE_BASE_HOME path, run the `orabasehome` command from the `%ORACLE_HOME%\bin` directory:

```
set ORACLE_HOME C:\app\oracle\product\19.0.0\dbhome_1
cd %ORACLE_HOME%\bin
orabasehome
```

For example:

```
orabasehome
C:\app\oracle\homes\OraDB19Home1
```

Where, `C:\app\oracle` is ORACLE_BASE and `OraDB19Home1` is HOME_NAME

About Oracle Base Config

Both, in a read-only ORACLE_HOME and read/write ORACLE_HOME, the configuration files reside in a location known as ORACLE_BASE_CONFIG.

In a read/write ORACLE_HOME, the ORACLE_BASE_CONFIG path is the same as the ORACLE_HOME path because it is located at ORACLE_HOME. However, in a read-only ORACLE_HOME, the ORACLE_BASE_CONFIG path is the same as ORACLE_BASE.

`ORACLE_BASE_CONFIG\dfs` contains the configuration files for ORACLE_HOME. Each file in the `dfs` directory contains ORACLE_SID so that the directory can be shared by many different ORACLE_SIDs.

To print the `ORACLE_BASE_CONFIG` path, run the `orabaseconfig` command from the `%ORACLE_HOME%\bin` directory:

```
C:\> set ORACLE_HOME C:\app\oracle\product\19.0.0\dbhome_1
C:\> cd %ORACLE_HOME%\bin
orabaseconfig
```

For example:

```
orabaseconfig
C:\> C:\app\oracle
```

Where, `C:\app\oracle` is `ORACLE_BASE`.

Enabling a Read-Only Oracle Home

Configure your Oracle home as a read-only Oracle home after you have performed a software-only Oracle Database installation.

To enable a read-only Oracle home:

1. Perform a software-only Oracle Database installation.
2. Run the `roohctl -enable` script.

Note:

Do not use the `-disable` flag with the `roohctl` command, as it is not supported.

3. Run Oracle Database Configuration Assistant (Oracle DBCA) to create a database.

Software-Only Database Installation

1. Log in as the Oracle installation owner user account (`oracle`) that you want to own the software binaries.
2. Download the Oracle Database installation image files (`db_home.zip`) to a directory of your choice. For example, you can download the image files to the `\tmp` directory.
3. Create the Oracle home directory and extract the image files that you have downloaded into this Oracle home directory.

Note:

Ensure that the Oracle home directory path you create is in compliance with the Oracle Optimal Flexible Architecture recommendations. Also, unzip the installation image files only in this Oracle home directory that you created.

4. From the Oracle home directory, run the `setup.exe` command to start the Oracle Database installer.
5. In the Select Configuration Option screen, select **Set Up Software Only**.

6. Select your installation type. Installation screens vary depending on the installation option you select. Respond to the configuration prompts as needed.

**Note:**

Click **Help** if you have any questions about the information you are asked to submit during installation.

Run the roohctl Script

1. Go to the `bin` directory

```
cd C:\app\oracle\product\19.0.0\dbhome_1\bin
```

2. Run the `roohctl` script to enable read-only Oracle home.

```
roohctl.bat -enable
```

Run Oracle Database Configuration Assistant

1. Ensure that you are still in the `bin` directory and run Oracle DBCA.

```
dbca
```

2. In the Select Database Operation screen, select **Create a Database**.
3. The configuration screens vary depending on the options you select. Respond to the prompts as needed.

**Note:**

Click **Help** if you have any questions about the information you are asked to submit during database creation.

Related Topics

- *Oracle Database 2 Day DBA*

Determining if an Oracle Home is Read-Only

You can use the Registry Editor to determine if your Oracle home is a read/write or read-only Oracle home.

Start Registry Editor and navigate to the `ORACLE_HOME_READONLY` entry in the `HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE\KEY_OracleHomeName` Windows Registry key.

The default value for the `ORACLE_HOME_READONLY` parameter appears as `N`. The `ORACLE_HOME_READONLY` parameter value appears as `Y` if your Oracle home is in a read-only mode. Do not modify this value.

File Path and Directory Changes in Read-Only Oracle Homes

Examples of hierarchical file mappings in a read-only Oracle home as compared to a read/write Oracle home.

This example shows an Optimal Flexible Architecture-compliant Oracle Database installation, for the user `oracle`, with the `ORACLE_HOME`, `ORACLE_BASE`, `ORACLE_BASE_HOME`, and `ORACLE_BASE_CONFIG` logical locations. The database files are under `oraclebase\oradata`.

This example also shows the changes in the Oracle Database software defined paths of configuration files, log files, and other directories in a read-only Oracle home when compared to a read/write Oracle home.

Table D-1 read/write and Read-Only Oracle Home File Path Examples

Directory	Read/Write Oracle Home File Path	Read-Only Oracle Home File Path
ORACLE_HOME	C:\app\oracle\product\19.0.0\dbhome_1	C:\app\oracle\product\19.0.0\dbhome_1
ORACLE_BASE	C:\app\oracle\	C:\app\oracle\
ORACLE_BASE_HOME	ORACLE_HOME (or) C:\app\oracle\product\19.0.0\dbhome_1	ORACLE_BASE\homes\HOME_NAME. (or) C:\app\oracle\homes\OracleDB19Home1
ORACLE_BASE_CONFIG	ORACLE_HOME (or) C:\app\oracle\product\19.0.0\dbhome_1	ORACLE_BASE (or) C:\app\oracle\
network	ORACLE_HOME\network (or) C:\app\oracle\product\19.0.0\dbhome_1\network	ORACLE_BASE_HOME\network (or) C:\app\oracle\homes\OracleDB19Home1\network
database	ORACLE_HOME\database (or) C:\app\oracle\product\19.0.0\dbhome_1\database	ORACLE_BASE\database (or) C:\app\oracle\database

E

Managing Oracle Database Port Numbers

Review the default port numbers. If needed, use these steps to change assigned ports after installation.

- [About Managing Ports](#)
During installation, Oracle Universal Installer (OUI) assigns port numbers to components from a set of default port numbers. Many Oracle Real Application Clusters (Oracle RAC) components and services use ports.
- [About Viewing Port Numbers and Access URLs](#)
In most cases, the Oracle Database component's port number is listed in the tool used to configure the port.
- [Setting UDP and TCP Dynamic Port Range for Oracle RAC Installations](#)
For certain configurations of Oracle RAC in high load environments it is possible for the system to exhaust the available number of sockets. To avoid this problem, expand the dynamic port range for both UDP and TCP.
- [Port Numbers and Protocols of Oracle Components](#)
Review this information for port numbers and protocols used by components that are configured during the installation. By default, the first port in the range is assigned to the component, if it is available.
- [Changing the Oracle Services for Microsoft Transaction Server Port](#)
In most cases, you are not required to reconfigure the port number for the Oracle Services for Microsoft Transaction Server.

About Managing Ports

During installation, Oracle Universal Installer (OUI) assigns port numbers to components from a set of default port numbers. Many Oracle Real Application Clusters (Oracle RAC) components and services use ports.

As an administrator, it is important to know the port numbers used by these services, and to ensure that the same port number is not used by two services on your system.

Most port numbers are assigned during installation. Every component and service has an allotted port range, which is the set of port numbers Oracle RAC attempts to use when assigning a port. Oracle RAC starts with the lowest number in the range and performs the following checks:

- Is the port used by another Oracle Database installation on the system?
The installation can be either active or inactive at the time; Oracle Database can still detect if the port is used.
- Is the port used by a process that is currently running?
This could be any process on the host, including processes other than Oracle Database processes.

If the answer to any of the preceding questions is yes, then Oracle RAC moves to the next highest port in the allotted port range and continues checking until it finds a free port.

About Viewing Port Numbers and Access URLs

In most cases, the Oracle Database component's port number is listed in the tool used to configure the port.

- Ports for some Oracle Database applications are listed in the `portlist.ini` file. This file is located in the directory `%ORACLE_HOME%\install`.
- If you change a port number after installation, then it is not updated in the `portlist.ini` file, so you can rely on this file only immediately after installation.
- To find or change a port number, use the methods described in this appendix.

Setting UDP and TCP Dynamic Port Range for Oracle RAC Installations

For certain configurations of Oracle RAC in high load environments it is possible for the system to exhaust the available number of sockets. To avoid this problem, expand the dynamic port range for both UDP and TCP.

1. Open a command line window as an Administrator user.
2. Run the following commands to set the dynamic port range:

```
netsh int ipv4 set dynamicport udp start=9000 num=56000
netsh int ipv4 set dynamicport tcp start=9000 num=56000
```

3. Run the following commands to verify that the dynamic port range was set:

```
netsh int ipv4 show dynamicport udp
netsh int ipv4 show dynamicport tcp
```

For IPv6 network, replace IPv4 with IPv6 in the above examples.

Port Numbers and Protocols of Oracle Components

Review this information for port numbers and protocols used by components that are configured during the installation. By default, the first port in the range is assigned to the component, if it is available.

Table E-1 Ports Used in Oracle Components

Component and Description	Default Port Number	Port Range	Protocol	Used Only On Interconnect
Cluster Manager The port number is assigned automatically during installation. You cannot view or modify it afterward.	Dynamic	Dynamic	TCP	Yes

Table E-1 (Cont.) Ports Used in Oracle Components

Component and Description	Default Port Number	Port Range	Protocol	Used Only On Interconnect
Cluster Synchronization Service daemon (CSSD) The Cluster Synchronization Service (CSS) daemon uses a fixed port for node restart advisory messages. This port is used on all interfaces that have broadcast capability. Broadcast occurs only when a node eviction restart is imminent.	42424	Dynamic	TCP	Yes
Grid Plug and Play (GPNPD) GPNPD provides access to the Grid Plug and Play profile, and coordinates updates to the profile among the nodes of the cluster to ensure that all of the nodes have the most recent profile.	Dynamic	Dynamic	TCP	No
Multicast Domain Name Service (MDNSD) The mDNS process is a background process on Linux and UNIX, and a service on Windows, and is necessary for Grid Plug and Play and GNS.	5353	Dynamic	TCP	No
Oracle Cluster Registry The port number is assigned automatically during installation. You cannot view or modify it afterward.	Dynamic	Dynamic	TCP	Yes
Oracle Clusterware Daemon (CRSD) Oracle Clusterware daemon internode connection. The port number is assigned automatically during installation. You cannot view or modify it afterward.	Dynamic	Dynamic	TCP	Yes
Oracle Connection Manager Listening port for Oracle client connections to Oracle Connection Manager. You can configure Oracle Connection Manager after installation using NETCA.	1630	1630	TCP	No
Quality of Management Service (QOMS) Server The CRS Agent uses port 8888 locally to manage the lifecycle of the container.	8888	8888	HTTP	Not applicable
Quality of Management Service (QOMS) Server Port for the Quality of Management Service server.	23792	23792	JMX/RMI	No

Table E-1 (Cont.) Ports Used in Oracle Components

Component and Description	Default Port Number	Port Range	Protocol	Used Only On Interconnect
Oracle Data Guard Shares the Oracle Net listener port and is configured during installation. To reconfigure this port, use Oracle Net Configuration Assistant (NETCA) to reconfigure the listener.	1521 (same value as the listener)	modifiable manually to any available port	TCP	No
Oracle Event Manager (EVM) Generates events for Oracle Clusterware. The port number is assigned automatically during installation. You cannot view or modify it afterward.	Dynamic	Dynamic	TCP	Yes
Oracle Grid Interprocess Communication (GIPCD) A support daemon that enables Redundant Interconnect Usage.	42424	Dynamic	TCP	Yes
Oracle Grid Naming Service (GNSD) The Oracle Grid Naming Service daemon performs name resolution for the cluster.	53	53	UDP	No
Oracle Grid Naming Service (GNSD) The Oracle Grid Naming Service daemon performs name resolution for the cluster.	Dynamic	Dynamic	TCP	No
Oracle HA Services daemon (OHASD) The Oracle High Availability Services (OHAS) daemon starts the Oracle Clusterware stack.	42424	Dynamic	TCP	Yes
Oracle Net Listener Allows Oracle clients to connect to the database by using Oracle Net Services. You can configure this port during installation. To reconfigure this port, use NETCA.	1521	Port number changes to the next available port. Modifiable manually to any available port.	TCP	No
Oracle Notification Services (ONS) Port for ONS, used for the publish and subscribe service for communicating information about Fast Application Notification (FAN) events. The FAN notification process uses system events that Oracle Database publishes when cluster servers become unreachable or if network interfaces fail. Use <code>srvctl</code> to modify ONS ports.	6100 (local) 6200 (remote)	Configured manually	TCP	No
Oracle Real Application Clusters The port number is assigned automatically during installation. You cannot view or modify it afterward.	Dynamic	9000 to 64999	UDP	Yes

Table E-1 (Cont.) Ports Used in Oracle Components

Component and Description	Default Port Number	Port Range	Protocol	Used Only On Interconnect
Oracle Services for Microsoft Transaction Server The port number for Microsoft Transaction Server is configured automatically by Oracle Universal Installer (OUI) the first time you install the software on a particular server. If you install the software in multiple Oracle homes on the same server, then OUI uses the same port number for all installations. In most cases, you do not have to reconfigure the port number.	Dynamic	49152 to 65535	TCP	No
Oracle XML DB - FTP The Oracle XML DB FTP port is used when applications must access an Oracle database from an FTP listener. The port is configured during installation and you <i>cannot</i> view it afterward.	0	Configured manually	FTP	No
Oracle XML DB - HTTP The Oracle XML DB HTTP port is used if web-based applications must access an Oracle database from an HTTP listener. The port is configured during installation, and you <i>cannot</i> view it afterward.	0	Configured manually	HTTP	No

 See Also:

- *Oracle XML DB Developer's Guide* for information about changing Oracle XML DB FTP port number.
- *Oracle XML DB Developer's Guide* for information about changing Oracle XML DB HTTP port number.
- *Oracle Enterprise Manager Cloud Control Advanced Installation and Configuration Guide* for information on Oracle Management Agent ports.
- "[Changing the Oracle Services for Microsoft Transaction Server Port](#)" for information about changing Oracle Services for Microsoft Transaction Server port number.

Changing the Oracle Services for Microsoft Transaction Server Port

In most cases, you are not required to reconfigure the port number for the Oracle Services for Microsoft Transaction Server.

- If you must change the port number, then you can use the Registry Editor to edit its value in the
HKEY_LOCAL_MACHINE\SOFTWARE\ORACLE\OracleMTSRecoveryService\Protid_0 Windows Registry key to any available port within the range 1024 to 65535.

During installation, Oracle Universal Installer takes the value for the port from the key, if it exists. Otherwise, a free port ranging from 49152 to 65535 is chosen automatically.

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